

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]
```

```

cols = list(dict.fromkeys(target_col + features_X +
                          feats_to_balance + feats_grouping +
                          feats_reference+ feats_postproc))

df = df[cols]
#df['campaign'] = "ALL"

return df

```

```

def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/switched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")

    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4

```

```
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)
```

```
return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	7988
Nombre de lignes df clean	7988

Nombre de features	77
% de NaN	0.0
Taille dataset entraînement	5591
Taille dataset test	2396
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_rms, r_NH3_st_8h_median, r_NH3_st_8h_min, r_NH3_st_8h_max, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_NH3_st_12h_mean, r_NH3_st_12h_std, r_NH3_st_12h_rms, r_NH3_st_12h_median, r_NH3_st_12h_min, r_NH3_st_12h_max, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_NH3_st_24h_mean, r_NH3_st_24h_std, r_NH3_st_24h_rms, r_NH3_st_24h_median, r_NH3_st_24h_min, r_NH3_st_24h_max, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_temperature_st_8h_mean, r_temperature_st_8h_std, r_temperature_st_8h_rms, r_temperature_st_8h_median, r_temperature_st_8h_min, r_temperature_st_8h_max, r_temperature_st_8h_diff_last, r_temperature_st_8h_slope, r_temperature_st_12h_mean, r_temperature_st_12h_std, r_temperature_st_12h_rms, r_temperature_st_12h_median, r_temperature_st_12h_min, r_temperature_st_12h_max, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_temperature_st_24h_mean, r_temperature_st_24h_std, r_temperature_st_24h_rms, r_temperature_st_24h_median, r_temperature_st_24h_min, r_temperature_st_24h_max, r_temperature_st_24h_diff_last, r_temperature_st_24h_slope, r_umidity_st_8h_mean, r_umidity_st_8h_std, r_umidity_st_8h_rms, r_umidity_st_8h_median, r_umidity_st_8h_min, r_umidity_st_8h_max, r_umidity_st_8h_diff_last, r_umidity_st_8h_slope, r_umidity_st_12h_mean, r_umidity_st_12h_std, r_umidity_st_12h_rms, r_umidity_st_12h_median, r_umidity_st_12h_min, r_umidity_st_12h_max, r_umidity_st_12h_diff_last, r_umidity_st_12h_slope, r_umidity_st_24h_mean, r_umidity_st_24h_std, r_umidity_st_24h_rms, r_umidity_st_24h_median, r_umidity_st_24h_min, r_umidity_st_24h_max, r_umidity_st_24h_diff_last, r_umidity_st_24h_slope

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Data correlation and clustering

2.1 Clustermap and cluster list

The following section has been done with a threshold choosen of : 0.8

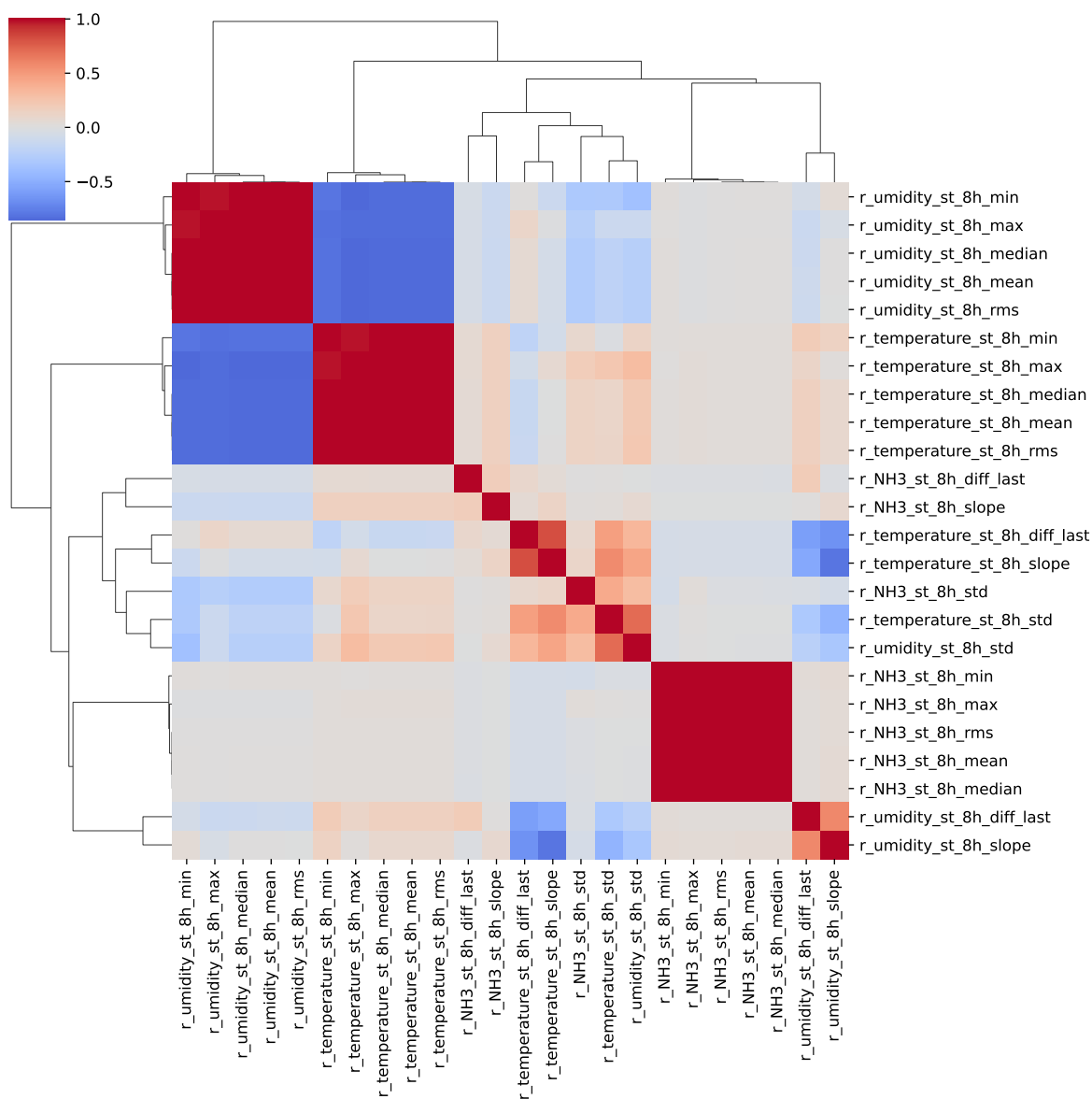
```
Index(['r_NH3_st_8h_mean', 'r_NH3_st_8h_std', 'r_NH3_st_8h_rms', 'r_NH3_st_8h_me-  
dian', 'r_NH3_st_8h_min', 'r_NH3_st_8h_max', 'r_NH3_st_8h_diff_last', 'r_NH3_st_8h_slope',  
'r_temperature_st_8h_mean', 'r_temperature_st_8h_std', 'r_temperature_st_8h_rms',  
'r_temperature_st_8h_median', 'r_temperature_st_8h_min', 'r_temperature_st_8h_max',  
'r_temperature_st_8h_diff_last', 'r_temperature_st_8h_slope', 'r_umidity_st_8h_mean',  
'r_umidity_st_8h_std', 'r_umidity_st_8h_rms', 'r_umidity_st_8h_median', 'r_umid-  
ity_st_8h_min', 'r_umidity_st_8h_max', 'r_umidity_st_8h_diff_last', 'r_umid-  
ity_st_8h_slope'], dtype='object') Index(['r_NH3_st_12h_mean', 'r_NH3_st_12h_std',  
'r_NH3_st_12h_rms', 'r_NH3_st_12h_median', 'r_NH3_st_12h_min', 'r_NH3_st_12h_max',  
'r_NH3_st_12h_diff_last', 'r_NH3_st_12h_slope', 'r_temperature_st_12h_mean', 'r_tem-  
perature_st_12h_std', 'r_temperature_st_12h_rms', 'r_temperature_st_12h_me-  
dian', 'r_temperature_st_12h_min', 'r_temperature_st_12h_max', 'r tempera-  
ture_st_12h_diff_last', 'r_temperature_st_12h_slope', 'r_umidity_st_12h_mean',  
'r_umidity_st_12h_std', 'r_umidity_st_12h_rms', 'r_umidity_st_12h_median', 'r_umid-  
ity_st_12h_min', 'r_umidity_st_12h_max', 'r_umidity_st_12h_diff_last', 'r_umid-  
ity_st_12h_slope'], dtype='object') Index(['r_NH3_st_24h_mean', 'r_NH3_st_24h_std',  
'r_NH3_st_24h_rms', 'r_NH3_st_24h_median', 'r_NH3_st_24h_min', 'r_NH3_st_24h_max',  
'r_NH3_st_24h_diff_last', 'r_NH3_st_24h_slope', 'r_temperature_st_24h_mean', 'r_tem-  
perature_st_24h_std', 'r_temperature_st_24h_rms', 'r_temperature_st_24h me-  
dian', 'r_temperature_st_24h_min', 'r_temperature_st_24h_max', 'r tempera-  
ture_st_24h_diff_last', 'r_temperature_st_24h_slope', 'r_umidity_st_24h_mean',  
'r_umidity_st_24h_std', 'r_umidity_st_24h_rms', 'r_umidity_st_24h_median', 'r_umid-  
ity_st_24h_min', 'r_umidity_st_24h_max', 'r_umidity_st_24h_diff_last', 'r_umid-  
ity_st_24h_slope'], dtype='object') cluster 1: r, , N, H, 3 ; cluster 2: r, , t, e, m, p, e,  
r, a, t, u, r, e ; cluster 3: r, , u, m, i, d, i, t, y ; cluster 4: r, , C, O, 2 ; cluster 5: r,  
_, T, H, I ; cluster 6: r_NH3_st_8h_mean, r_NH3_st_8h_max, r_NH3_st_8h_rms,  
r_NH3_st_8h_min, r_NH3_st_8h_median ; cluster 7: r_NH3_st_8h_std ; clus-  
ter 8: r_NH3_st_8h_diff_last ; cluster 9: r_NH3_st_8h_slope ; cluster 10:  
r_temperature_st_8h_min, r_temperature_st_8h_max, r_umidity_st_8h_rms,  
r_umidity_st_8h_max, r_temperature_st_8h_median, r_temperature_st_8h_mean,  
r_umidity_st_8h_min, r_umidity_st_8h_median, r_umidity_st_8h_mean, r tem-  
perature_st_8h_rms ; cluster 11: r_temperature_st_8h_std ; cluster 12: r_tem-  
perature_st_8h_diff_last, r_temperature_st_8h_slope, r_umidity_st_8h_slope  
; cluster 13: r_umidity_st_8h_std ; cluster 14: r_umidity_st_8h_diff_last  
; cluster 15: r_NH3_st_12h_rms, r_NH3_st_12h_min, r_NH3_st_12h_max,  
r_NH3_st_12h_median, r_NH3_st_12h_mean ; cluster 16: r_NH3_st_12h_std ;  
cluster 17: r_NH3_st_12h_diff_last ; cluster 18: r_NH3_st_12h_slope ; cluster 19:
```

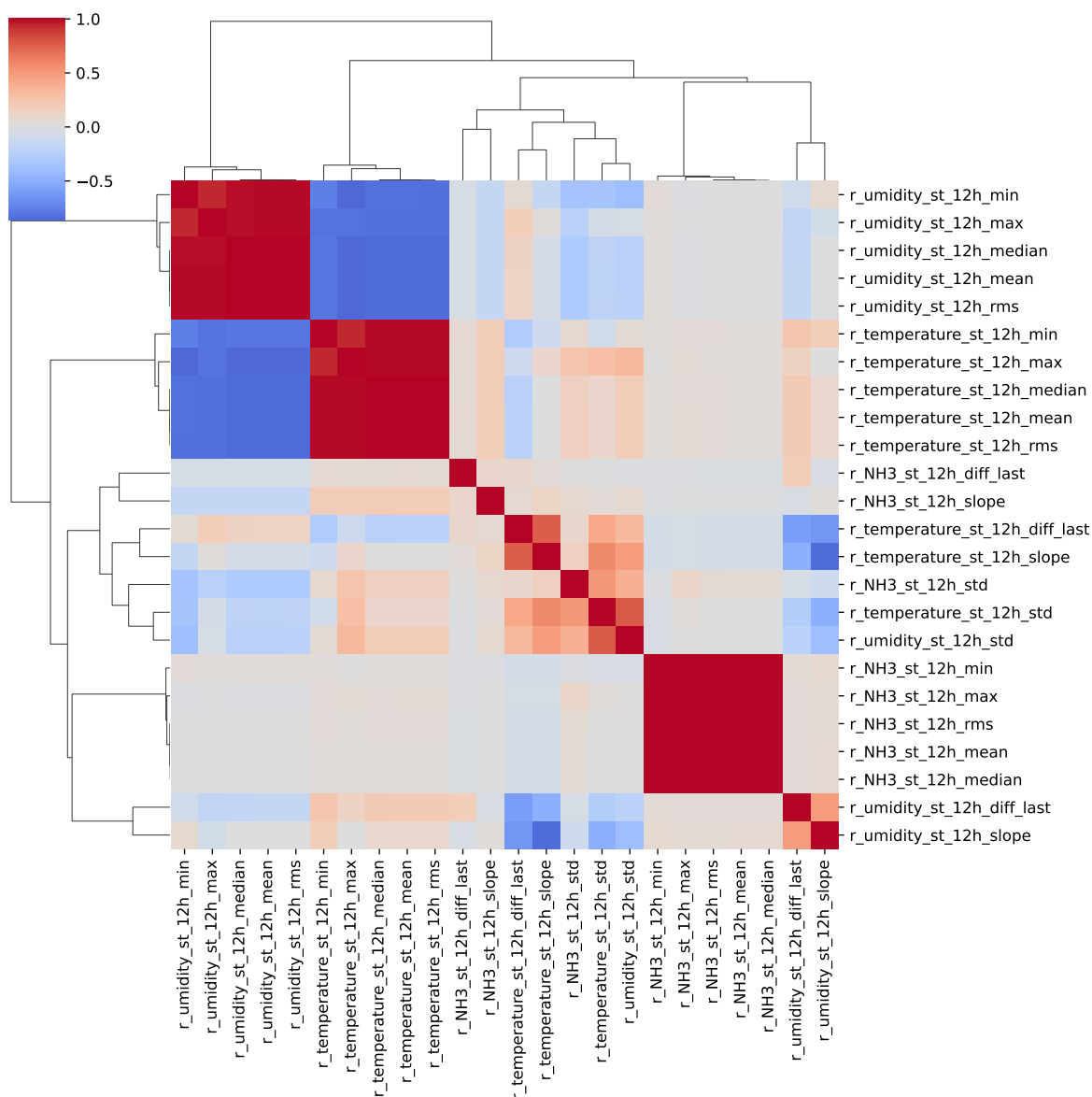
r_temperature_st_12h_median, r_umidity_st_12h_median, r_umidity_st_12h_mean, r_umidity_st_12h_min, r_umidity_st_12h_rms, r_umidity_st_12h_max, r_temperature_st_12h_rms, r_temperature_st_12h_min, r_temperature_st_12h_mean, r_temperature_st_12h_max ; **cluster 20:** r_temperature_st_12h_std ; **cluster 21:** r_temperature_st_12h_diff_last ; **cluster 22:** r_temperature_st_12h_slope, r_umidity_st_12h_slope ; **cluster 23:** r_umidity_st_12h_std ; **cluster 24:** r_umidity_st_12h_diff_last ; **cluster 25:** r_NH3_st_24h_min, r_NH3_st_24h_mean, r_NH3_st_24h_max, r_NH3_st_24h_rms, r_NH3_st_24h_median ; **cluster 26:** r_NH3_st_24h_std ; **cluster 27:** r_NH3_st_24h_diff_last ; **cluster 28:** r_NH3_st_24h_slope ; **cluster 29:** r_temperature_st_24h_mean, r_umidity_st_24h_min, r_umidity_st_24h_max, r_temperature_st_24h_max, r_temperature_st_24h_rms, r_temperature_st_24h_median, r_umidity_st_24h_mean, r_umidity_st_24h_rms, r_umidity_st_24h_median, r_temperature_st_24h_min ; **cluster 30:** r_umidity_st_24h_std, r_temperature_st_24h_std ; **cluster 31:** r_temperature_st_24h_diff_last ; **cluster 32:** r_temperature_st_24h_slope, r_umidity_st_24h_slope ; **cluster 33:** r_umidity_st_24h_diff_last ;

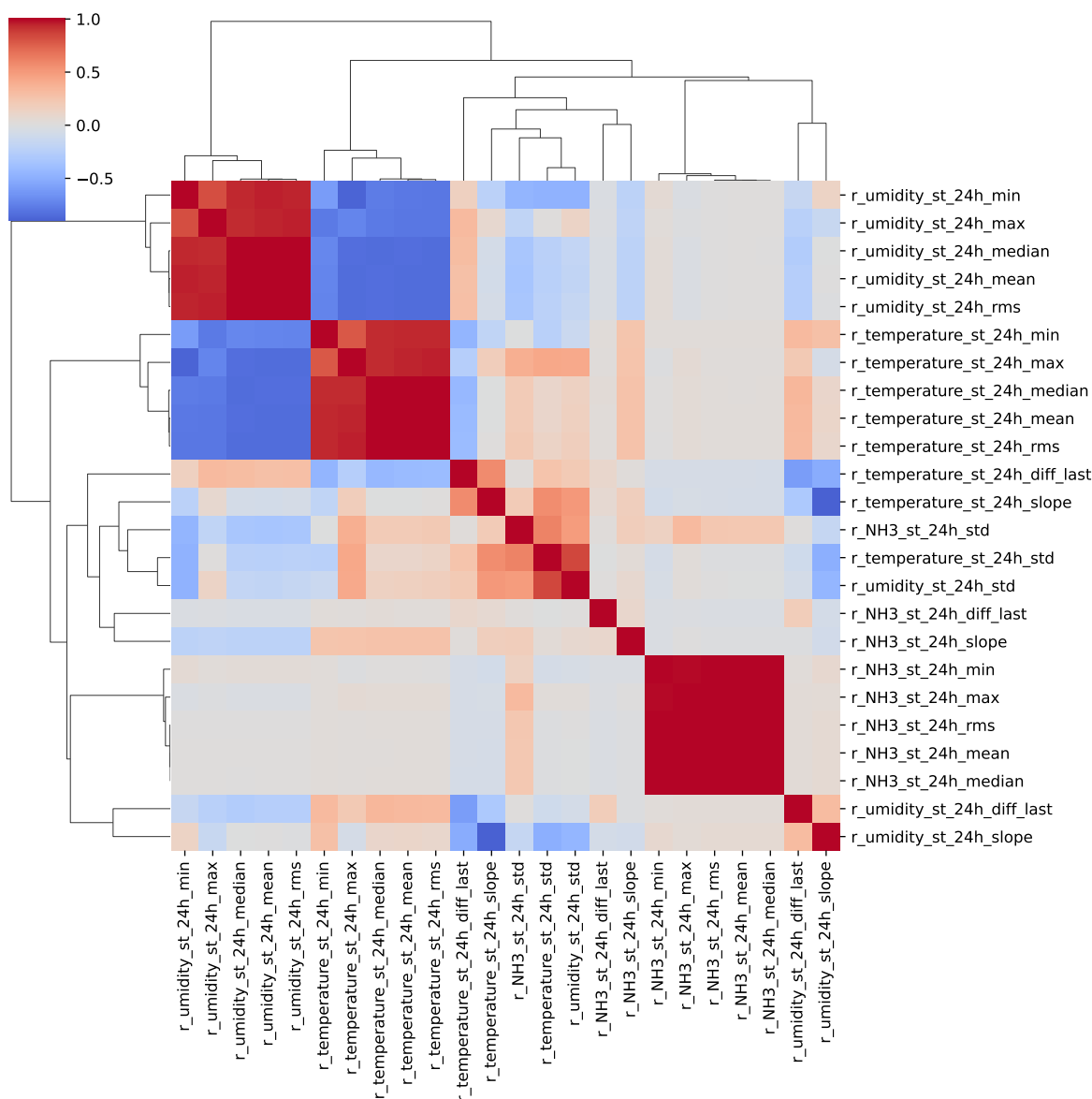
2.2 Features selection

Only the first items of the clusters is choosen for the rest of the study

Chooosen features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_8h_min, r_temperature_st_8h_std, r_temperature_st_8h_diff_last, r_umidity_st_8h_std, r_umidity_st_8h_diff_last, r_NH3_st_12h_rms, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_12h_median, r_temperature_st_12h_std, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last, r_NH3_st_24h_min, r_NH3_st_24h_std, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_temperature_st_24h_mean, r_umidity_st_24h_std, r_temperature_st_24h_diff_last, r_temperature_st_24h_slope, r_umidity_st_24h_diff_last







3 Model(s) analysis

3.1 Metadata

Features_X: r_NH3, r_temperature, r_umidity, r_CO2, r_THI, r_NH3_st_8h_mean, r_NH3_st_8h_std, r_NH3_st_8h_diff_last, r_NH3_st_8h_slope, r_temperature_st_8h_min, r_temperature_st_8h_std, r_temperature_st_8h_diff_last, r_umid-

ity_st_8h_std, r_umidity_st_8h_diff_last, r_NH3_st_12h_rms, r_NH3_st_12h_std, r_NH3_st_12h_diff_last, r_NH3_st_12h_slope, r_temperature_st_12h_median, r_temperature_st_12h_std, r_temperature_st_12h_diff_last, r_temperature_st_12h_slope, r_umidity_st_12h_std, r_umidity_st_12h_diff_last, r_NH3_st_24h_min, r_NH3_st_24h_std, r_NH3_st_24h_diff_last, r_NH3_st_24h_slope, r_temperature_st_24h_mean, r_umidity_st_24h_std, r_temperature_st_24h_diff_last, r_temperature_st_24h_slope, r_umidity_st_24h_diff_last

Len(Features_X): 33

Config Gridsearch:{‘show’: True, ‘search_type’: ‘optuna’, ‘model_name’: None, ‘model’: None, ‘param_grid’: None, ‘best_param’: None, ‘cv_func’: ShuffleSplit(n_splits=5, random_state=42, test_size=0.3, train_size=None), ‘save_models’: True, ‘sumup-table’: True}

3.2 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_{\text{test}} < 0.95$ (under performance of the test)

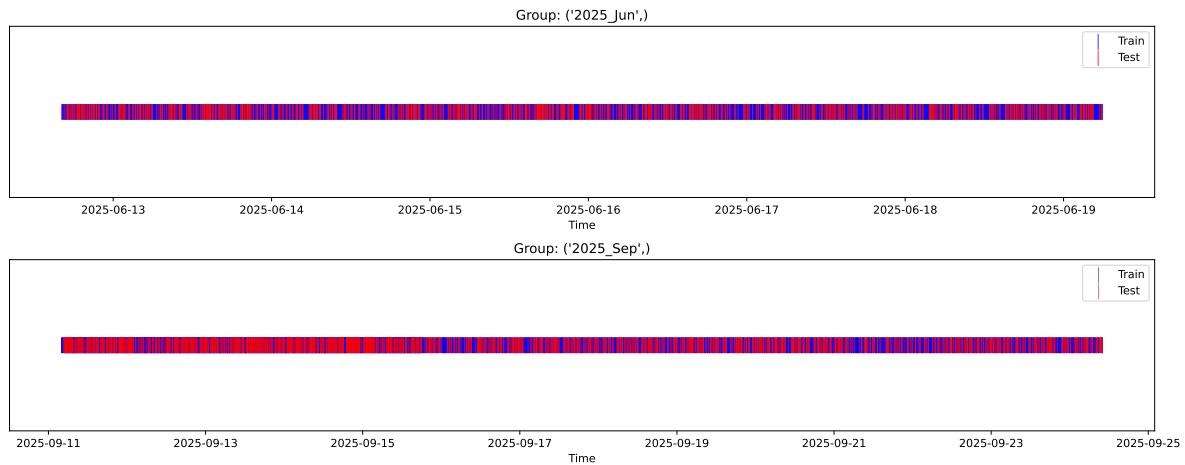
[Green] $r_{\text{test}} > 1.05$ (over performance of the test)

[Red] r_{train} not in $[0.95, 1.05]$ (gap between train/test)

3.3 Repartition of train and test

3.3.1 External Split: X_{train} vs X_{test}

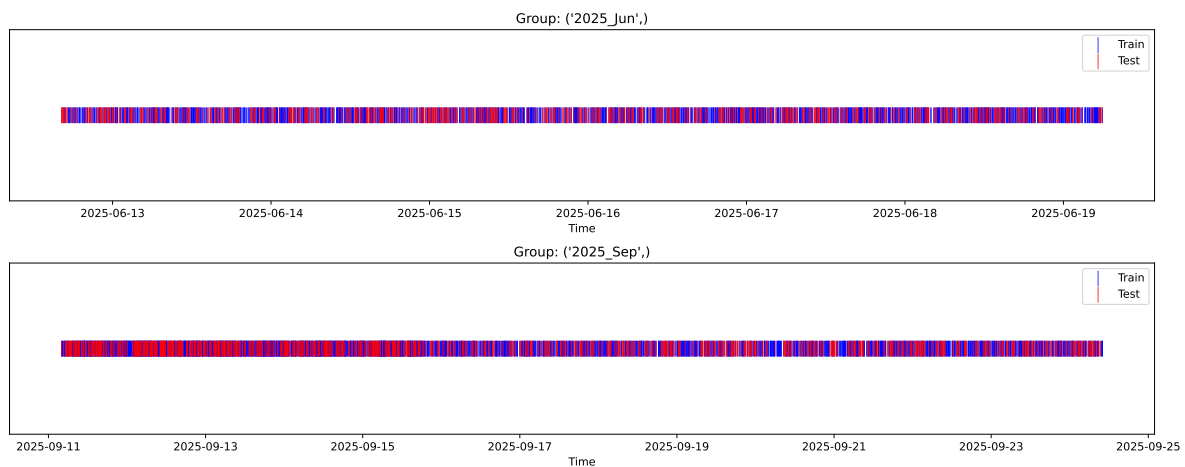
External Split: Train: 5591 samples Test: 2397 samples



3.3.2 Internal CV Folds

Fold 1/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 2/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 3/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



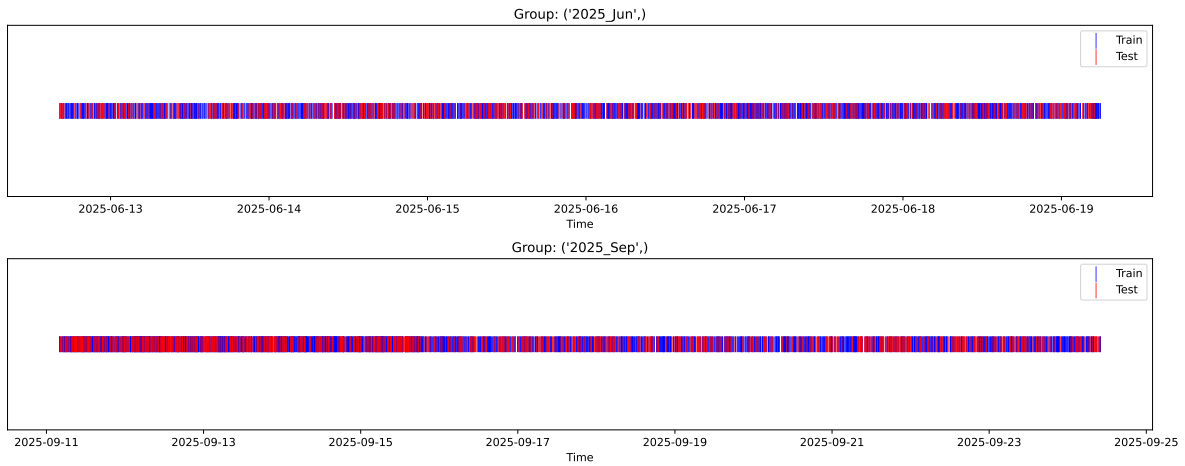
Fold 4/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 3913 samples Test: 1678 samples Excluded: 0 samples Window exclusion: 0h



3.4 Model : Linear

3.4.1 Gridsearch

Distribution y_{train} vs y_{test} : y_{train} : mean=2.350, std=0.793, min=0.959, max=8.682
 y_{test} : mean=2.359, std=0.815, min=0.973, max=13.845

Fold 0 — y_{test} : mean=2.36, std=0.79, y_{train} : mean=2.35, std=0.79

Fold 1 — y_{test} : mean=2.35, std=0.78, y_{train} : mean=2.35, std=0.80

Fold 2 — y_{test} : mean=2.34, std=0.79, y_{train} : mean=2.35, std=0.79

Fold 3 — y_test: mean=2.37, std=0.83, y_train: mean=2.34, std=0.77

Fold 4 — y_test: mean=2.37, std=0.79, y_train: mean=2.34, std=0.79

train scores CV: [0.56770758 0.55540685 0.55731225 0.56149127 0.56692844]

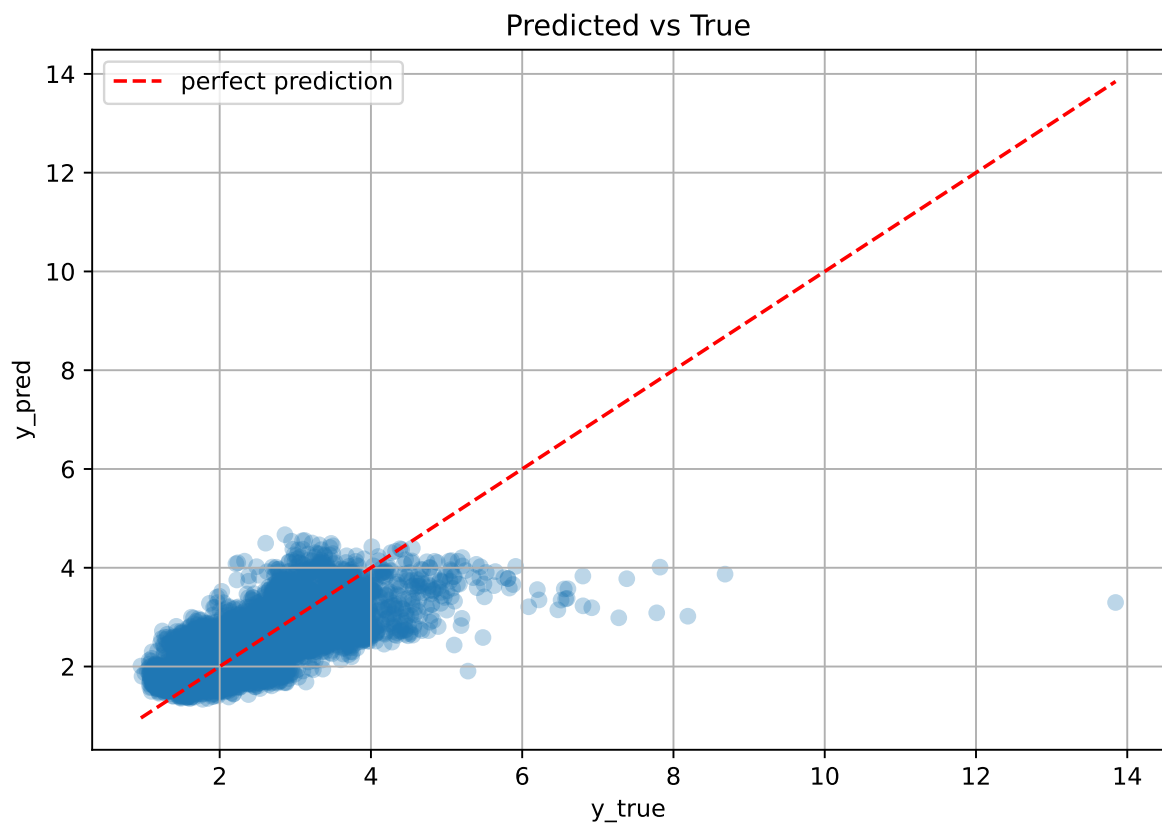
test scores CV: [0.54071218 0.56892219 0.56607344 0.55433866 0.54257019]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.5618	0.5545	0.0116	0.5254	0.5258	-0.0287	0.9482	1.0131

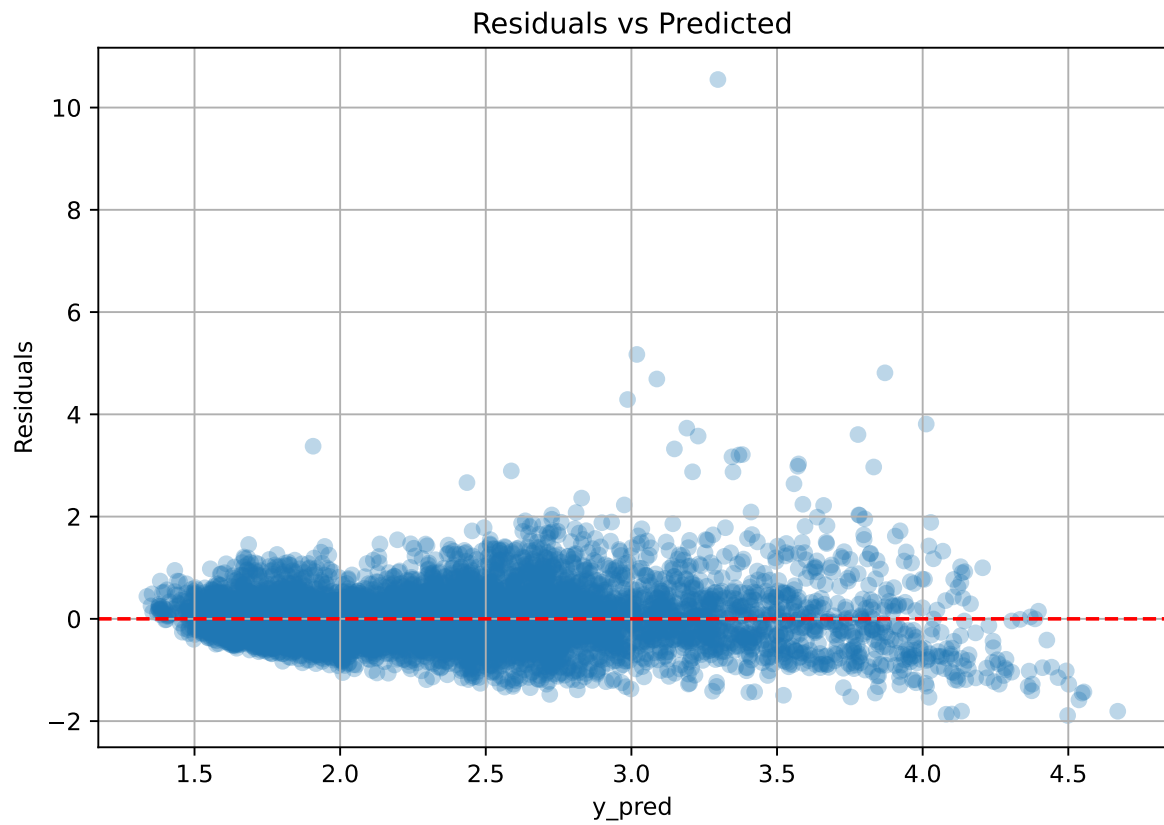
Coefficients:

coef_r_CO2 = -0.0475, coef_r_NH3 = 0.2487, coef_r_NH3_st_12h_diff_last = 0.0029, coef_r_NH3_st_12h_rms = 4.0095, coef_r_NH3_st_12h_slope = 0.1382, coef_r_NH3_st_12h_std = 0.0152, coef_r_NH3_st_24h_diff_last = 0.0029, coef_r_NH3_st_24h_min = -0.4324, coef_r_NH3_st_24h_slope = 0.0257, coef_r_NH3_st_24h_std = 0.0083, coef_r_NH3_st_8h_diff_last = 0.0029, coef_r_NH3_st_8h_mean = -3.8738, coef_r_NH3_st_8h_slope = -0.0381, coef_r_NH3_st_8h_std = -0.0077, coef_r_THI = 0.6419, coef_r_temperature = -0.0679, coef_r_temperature_st_12h_diff_last = -0.0049, coef_r_temperature_st_12h_median = 0.5968, coef_r_temperature_st_12h_slope = -0.0381, coef_r_temperature_st_12h_std = 0.1843, coef_r_temperature_st_24h_diff_last = -0.0049, coef_r_temperature_st_24h_mean = -1.6805, coef_r_temperature_st_24h_slope = -0.5533, coef_r_temperature_st_8h_diff_last = -0.0049, coef_r_temperature_st_8h_min = 0.4919, coef_r_temperature_st_8h_std = -0.0146, coef_r_umidity = -0.5023, coef_r_umidity_st_12h_diff_last = -0.0205, coef_r_umidity_st_12h_std = 0.0027, coef_r_umidity_st_24h_diff_last = -0.0205, coef_r_umidity_st_24h_std = -0.0900, coef_r_umidity_st_8h_diff_last = -0.0205, coef_r_umidity_st_8h_std = 0.0503, intercept = 2.3500,

3.4.2 Scatter



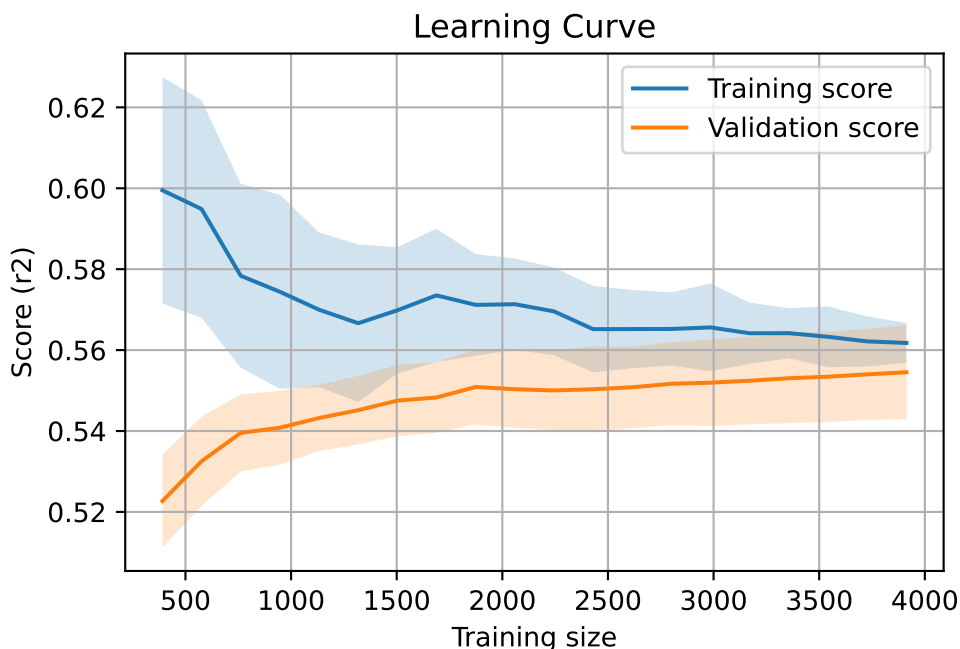
3.4.3 Scatter residuals



3.4.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396

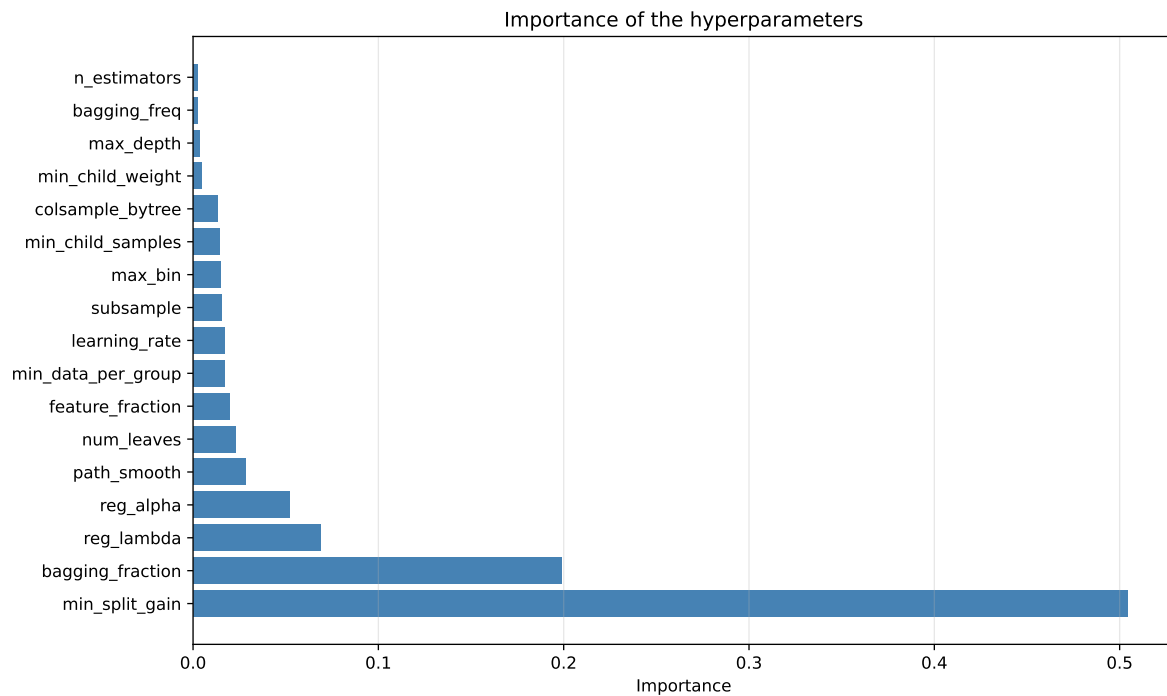
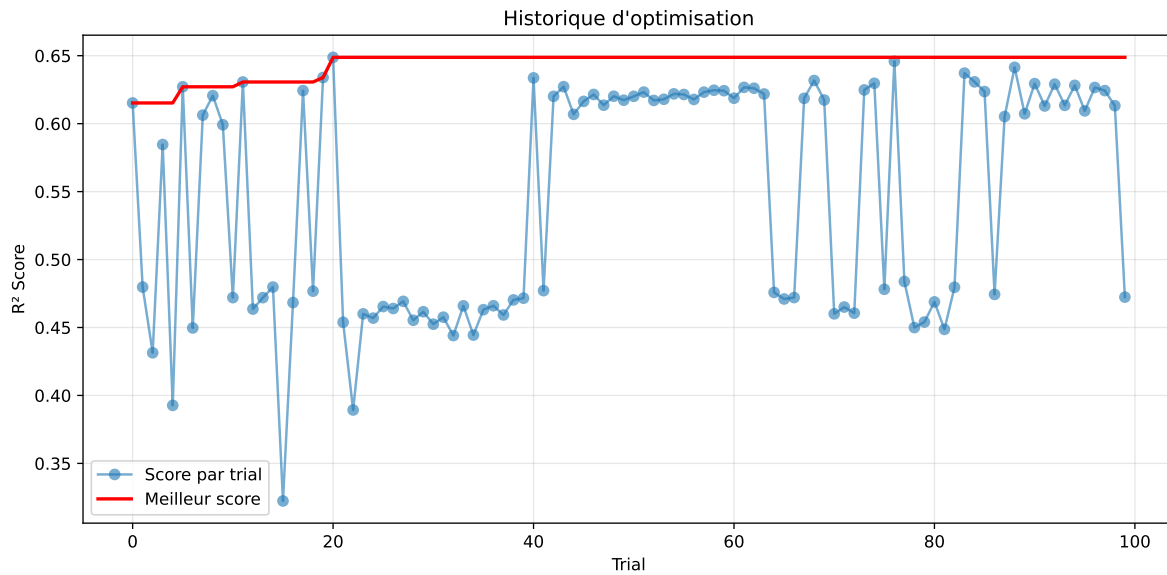


3.5 Model : LGBM

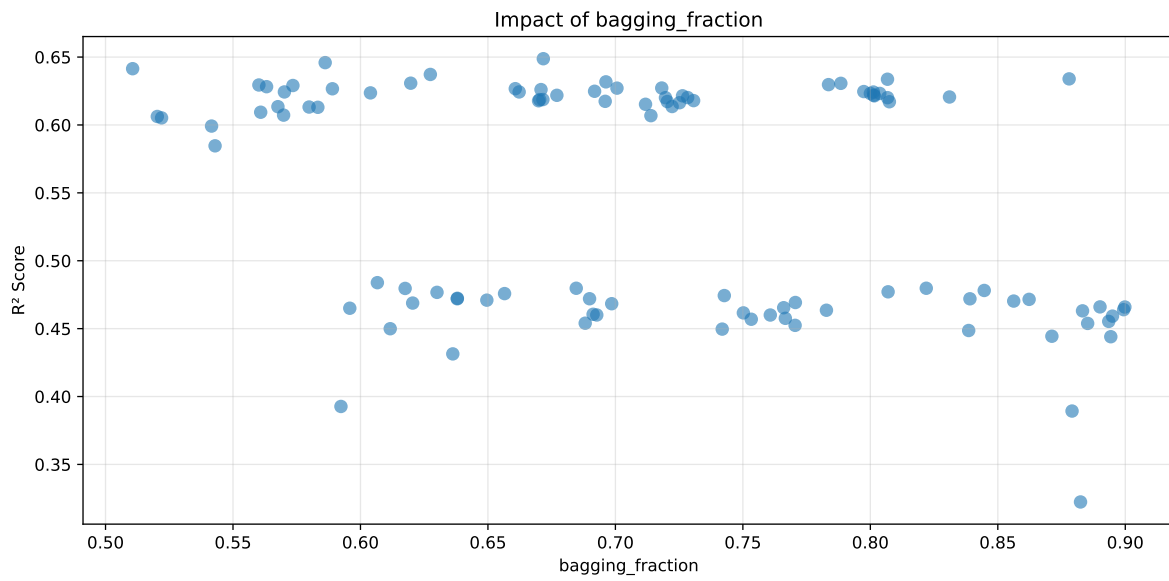
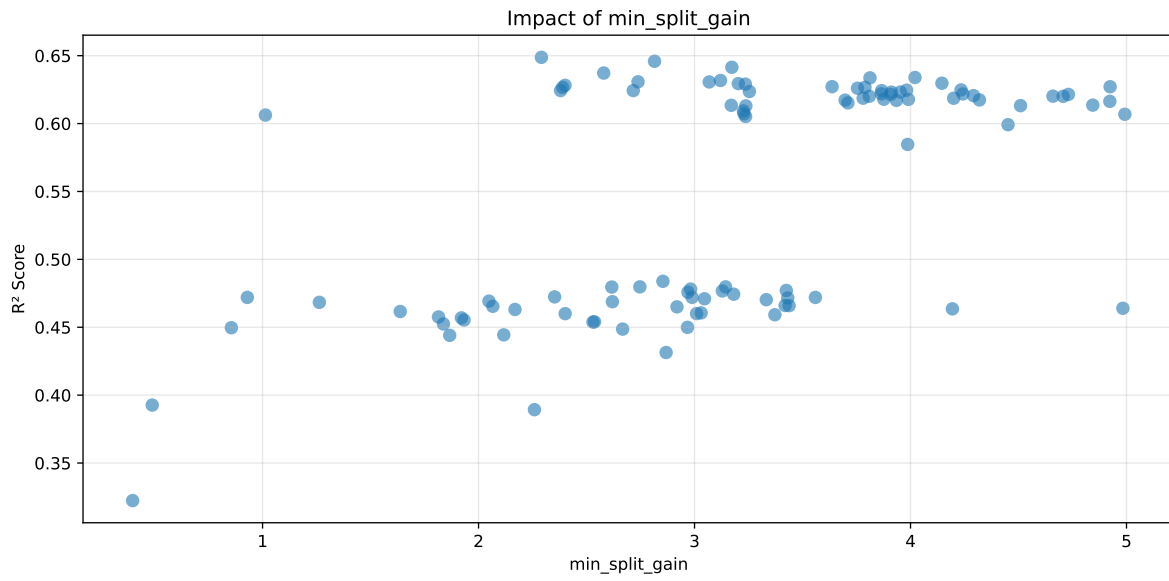
3.5.1 Gridsearch

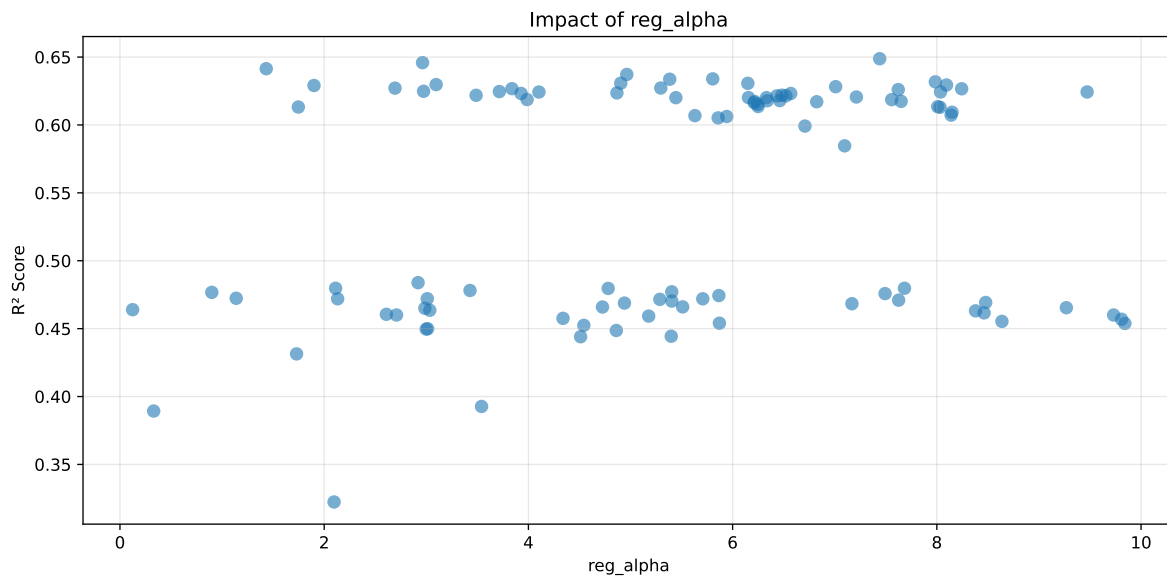
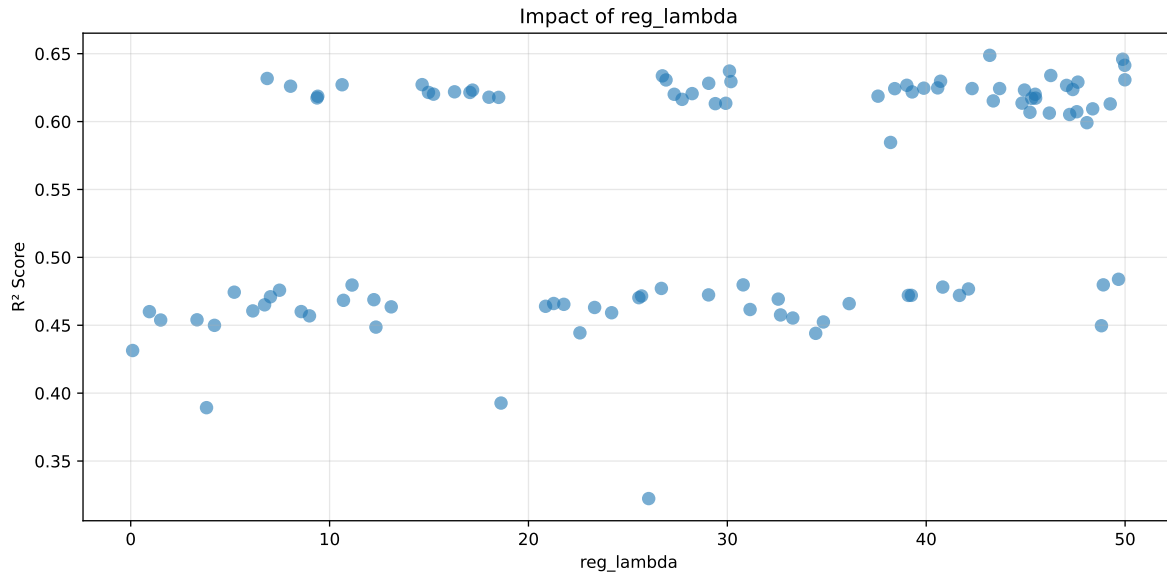
Distribution y_train vs y_test: y_train: mean=2.350, std=0.793, min=0.959, max=8.682
y_test: mean=2.359, std=0.815, min=0.973, max=13.845

```
===== TOP Importance FEATURES =====
feature importance r_CO2 260 r_umidity 156 r_NH3_st_24h_min 134 r_tempera-
ture_st_12h_slope 87 r_NH3_st_8h_std 85 r_temperature_st_8h_diff_last 56 r_NH3
53 r_temperature_st_24h_slope 47 r_NH3_st_24h_std 43 r_NH3_st_12h_std 41
r_NH3_st_8h_diff_last 39 r_umidity_st_8h_diff_last 36 r_NH3_st_24h_slope 35
r_temperature 34 r_temperature_st_24h_mean 32 r_temperature_st_8h_std 29 r_umid-
ity_st_12h_std 27 r_temperature_st_12h_std 25 r_temperature_st_12h_median 22
r_temperature_st_8h_min 20 r_NH3_st_8h_slope 20 r_NH3_st_12h_rms 18 r_THI
18 r_NH3_st_12h_slope 16 r_umidity_st_24h_std 12 r_umidity_st_12h_diff_last 10
r_umidity_st_24h_diff_last 5 r_temperature_st_12h_diff_last 5 r_umidity_st_8h_std
5 r_NH3_st_8h_mean 5 r_NH3_st_12h_diff_last 4 r_temperature_st_24h_diff_last 3
r_NH3_st_24h_diff_last 0
```



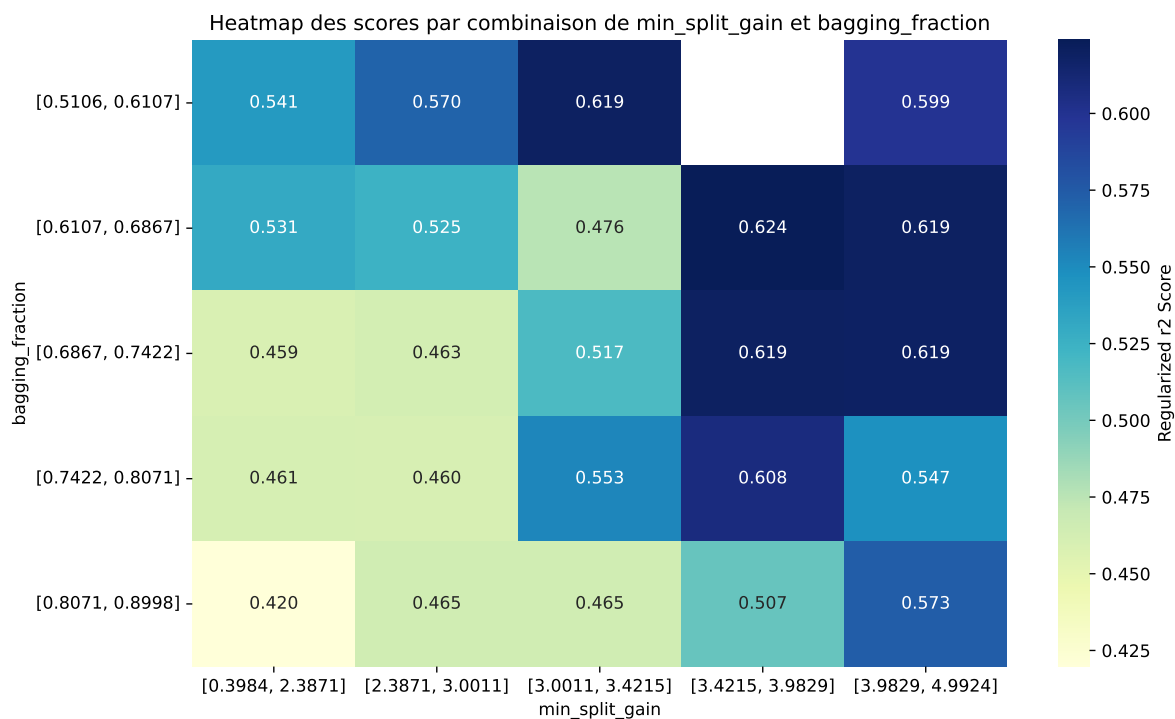
Paramètres avec >5% d'importance : ['min_split_gain', 'bagging_fraction', 'reg_lambda', 'reg_alpha']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6811	0.6488	0.6326	0.0137	0.6488	0.6536	0.0048	1.0074	1.0498
2	0.6766	0.6459	0.6311	0.0143	0.6459	0.6505	0.0046	1.0071	1.0475
3	0.6721	0.6414	0.6284	0.0139	0.6414	0.6504	0.009	1.014	1.0479
4	0.6666	0.6372	0.6206	0.0146	0.6372	0.6322	-0.005	0.9921	1.0461
5	0.6655	0.6339	0.6169	0.0144	0.6339	0.6366	0.0027	1.0042	1.0498
6	0.6649	0.6337	0.6158	0.0145	0.6337	0.633	-0.0007	0.9989	1.0493
7	0.6631	0.6317	0.6134	0.0136	0.6317	0.6333	0.0015	1.0024	1.0496
8	0.6592	0.6308	0.6121	0.0136	0.6308	0.63	-0.0008	0.9988	1.0451
9	0.6618	0.6307	0.6123	0.0134	0.6307	0.6283	-0.0024	0.9962	1.0493
10	0.6573	0.6297	0.6106	0.0129	0.6297	0.6347	0.0049	1.0078	1.0438
---	---	---	---	---	---	---	---	---	---
100	0.8155	0.7333	0.7183	0.0127	0.3224	0.7417	0.0083	1.0113	1.1121

Hyperparameters:

Rank 1: {'reg__n_estimators': 400, 'reg__max_depth': 8, 'reg__learning_rate': 0.030166617424212967, 'reg__num_leaves': 24, 'reg__subsample': 0.6338575646507988, 'reg__colsample_bytree': 0.6240217519815728, 'reg__min_child_samples': 24, 'reg__reg_alpha': 7.440159232078022, 'reg__reg_lambda': 43.19461424546416, 'reg__min_split_gain': 2.291115768444195, 'reg__path_smooth': 4.960184480806823, 'reg__min_child_weight': 3.486527597482765, 'reg__feature_fraction': 0.7812641645486846, 'reg__bagging_fraction':

0.6717240983411557, 'reg__bagging_freq': 1, 'reg__max_bin': 85, 'reg__min_data_per_group': 80}

Rank 2: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.040546651514775704, 'reg__num_leaves': 33, 'reg__subsample': 0.8226501548132003, 'reg__colsample_bytree': 0.5764780908076104, 'reg__min_child_samples': 50, 'reg__reg_alpha': 2.9633669448567064, 'reg__reg_lambda': 49.87908720086455, 'reg__min_split_gain': 2.814761809754091, 'reg__path_smooth': 2.8737602483134435, 'reg__min_child_weight': 0.03425558688141986, 'reg__feature_fraction': 0.7572791540915529, 'reg__bagging_fraction': 0.5861575518616862, 'reg__bagging_freq': 1, 'reg__max_bin': 107, 'reg__min_data_per_group': 155}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.04813292013682678, 'reg__num_leaves': 17, 'reg__subsample': 0.8703700048868928, 'reg__colsample_bytree': 0.5175774705739138, 'reg__min_child_samples': 48, 'reg__reg_alpha': 1.4324456927386566, 'reg__reg_lambda': 49.98288994133154, 'reg__min_split_gain': 3.172807098944704, 'reg__path_smooth': 3.016304810241083, 'reg__min_child_weight': 0.03629135250977384, 'reg__feature_fraction': 0.7573701899821277, 'reg__bagging_fraction': 0.510632272293719, 'reg__bagging_freq': 1, 'reg__max_bin': 107, 'reg__min_data_per_group': 166}

Rank 4: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.033565194875236196, 'reg__num_leaves': 25, 'reg__subsample': 0.8519211007311531, 'reg__colsample_bytree': 0.5136672868312576, 'reg__min_child_samples': 42, 'reg__reg_alpha': 4.9635674231663955, 'reg__reg_lambda': 30.106420888141038, 'reg__min_split_gain': 2.579423578631915, 'reg__path_smooth': 2.776201845643464, 'reg__min_child_weight': 0.02637539152889755, 'reg__feature_fraction': 0.7602629272966572, 'reg__bagging_fraction': 0.6274108685327477, 'reg__bagging_freq': 6, 'reg__max_bin': 63, 'reg__min_data_per_group': 159}

Rank 5: {'reg__n_estimators': 600, 'reg__max_depth': 4, 'reg__learning_rate': 0.013941341338477672, 'reg__num_leaves': 27, 'reg__subsample': 0.6649843991508563, 'reg__colsample_bytree': 0.5321925575129853, 'reg__min_child_samples': 26, 'reg__reg_alpha': 5.8051680475559255, 'reg__reg_lambda': 46.26371317573277, 'reg__min_split_gain': 4.020728344305542, 'reg__path_smooth': 1.925540400384872, 'reg__min_child_weight': 0.002115901416889657, 'reg__feature_fraction': 0.7721516499453102, 'reg__bagging_fraction': 0.8779736591355085, 'reg__bagging_freq': 1, 'reg__max_bin': 162, 'reg__min_data_per_group': 103}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 4, 'reg__learning_rate': 0.01086868617849551, 'reg__num_leaves': 41, 'reg__subsample': 0.5485573681495454, 'reg__colsample_bytree': 0.6120610456523293, 'reg__min_child_samples': 22, 'reg__reg_alpha': 5.384017992517651, 'reg__reg_lambda': 26.733562956339195, 'reg__min_split_gain': 3.812304550969358, 'reg__path_smooth': 2.193215487569007, 'reg__min_child_weight': 0.9340514271319212, 'reg__feature_fraction': 0.6837317227487169, 'reg__bagging_fraction':

0.8067289790976957, 'reg__bagging_freq': 2, 'reg__max_bin': 171, 'reg__min_data_per_group': 117}

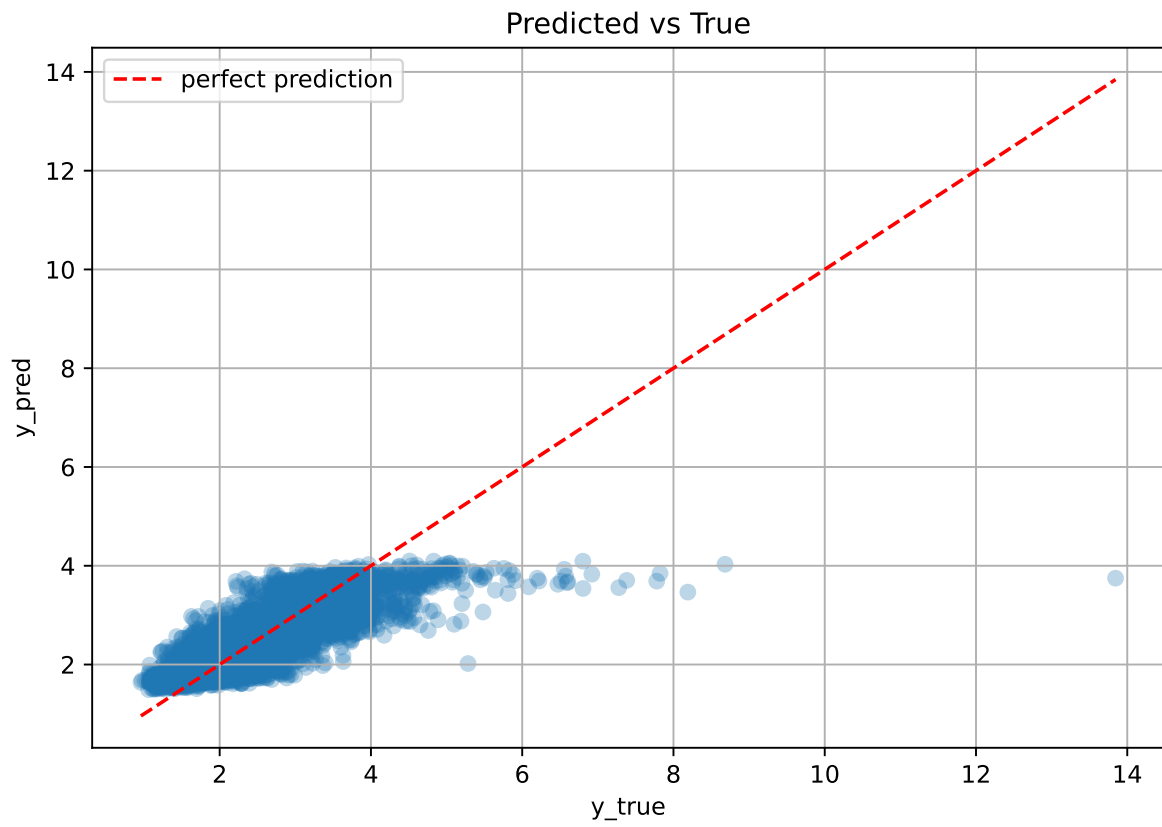
Rank 7: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.012169936511314277, 'reg__num_leaves': 48, 'reg__subsample': 0.8756274795671789, 'reg__colsample_bytree': 0.5893534545309397, 'reg__min_child_samples': 48, 'reg__reg_alpha': 7.985127091970099, 'reg__reg_lambda': 6.8615211517473504, 'reg__min_split_gain': 3.120299095302849, 'reg__path_smooth': 4.550362685699387, 'reg__min_child_weight': 2.871969251548714, 'reg__feature_fraction': 0.8347012471139738, 'reg__bagging_fraction': 0.6962762917912387, 'reg__bagging_freq': 5, 'reg__max_bin': 206, 'reg__min_data_per_group': 163}

Rank 8: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.03565658594742754, 'reg__num_leaves': 27, 'reg__subsample': 0.8767334668869863, 'reg__colsample_bytree': 0.5167230065702985, 'reg__min_child_samples': 44, 'reg__reg_alpha': 4.90433119228536, 'reg__reg_lambda': 49.98918718982668, 'reg__min_split_gain': 2.738415597681322, 'reg__path_smooth': 2.7599213388316746, 'reg__min_child_weight': 0.032025533864598665, 'reg__feature_fraction': 0.7568931339045237, 'reg__bagging_fraction': 0.6197058124977811, 'reg__bagging_freq': 6, 'reg__max_bin': 98, 'reg__min_data_per_group': 107}

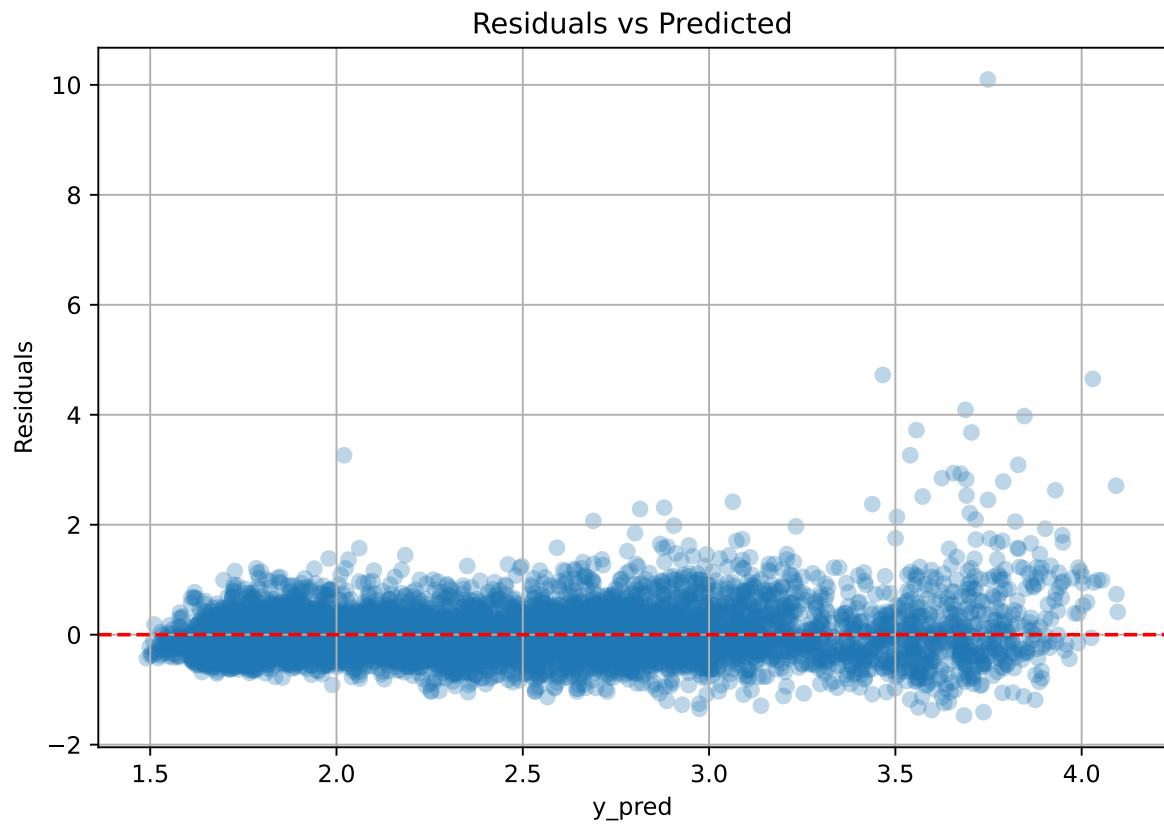
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 3, 'reg__learning_rate': 0.03904542170982645, 'reg__num_leaves': 46, 'reg__subsample': 0.5930482267141849, 'reg__colsample_bytree': 0.8151805752164649, 'reg__min_child_samples': 41, 'reg__reg_alpha': 6.148792541066086, 'reg__reg_lambda': 26.91809632817016, 'reg__min_split_gain': 3.0679642995844274, 'reg__path_smooth': 1.9809507906776709, 'reg__min_child_weight': 0.09346402875785262, 'reg__feature_fraction': 0.8518354377763651, 'reg__bagging_fraction': 0.7884254580406915, 'reg__bagging_freq': 5, 'reg__max_bin': 220, 'reg__min_data_per_group': 181}

Rank 10: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.01198238775223765, 'reg__num_leaves': 26, 'reg__subsample': 0.5970460874721655, 'reg__colsample_bytree': 0.5879810555435087, 'reg__min_child_samples': 50, 'reg__reg_alpha': 3.096184751470137, 'reg__reg_lambda': 40.72165134834493, 'reg__min_split_gain': 4.145310364521447, 'reg__path_smooth': 3.262193610747252, 'reg__min_child_weight': 5.150605955552051, 'reg__feature_fraction': 0.7087977308556651, 'reg__bagging_fraction': 0.7835958945988784, 'reg__bagging_freq': 5, 'reg__max_bin': 108, 'reg__min_data_per_group': 109}

3.5.2 Scatter



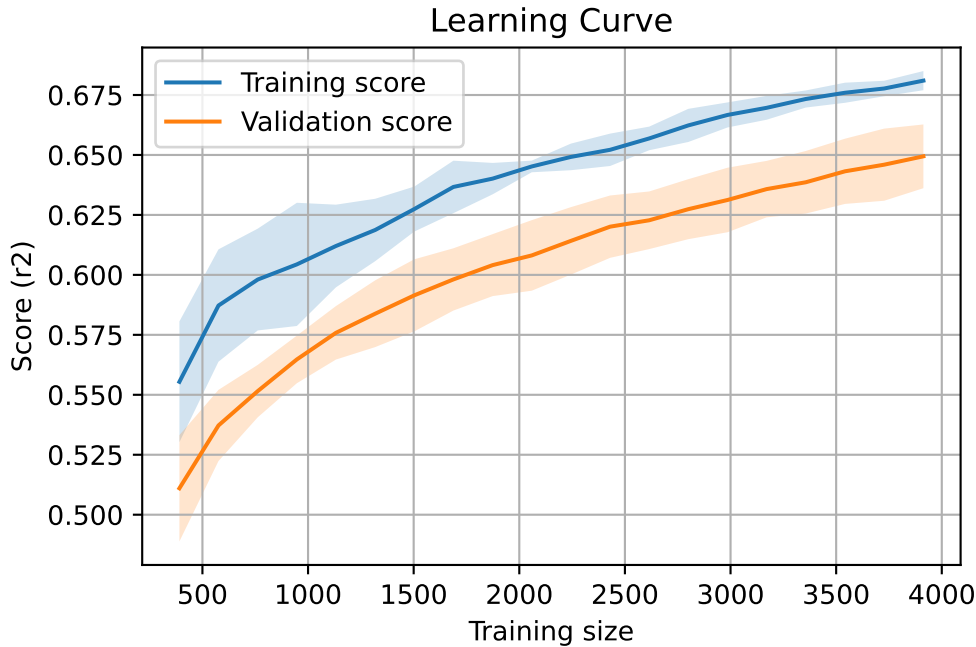
3.5.3 Scatter residuals



3.5.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396



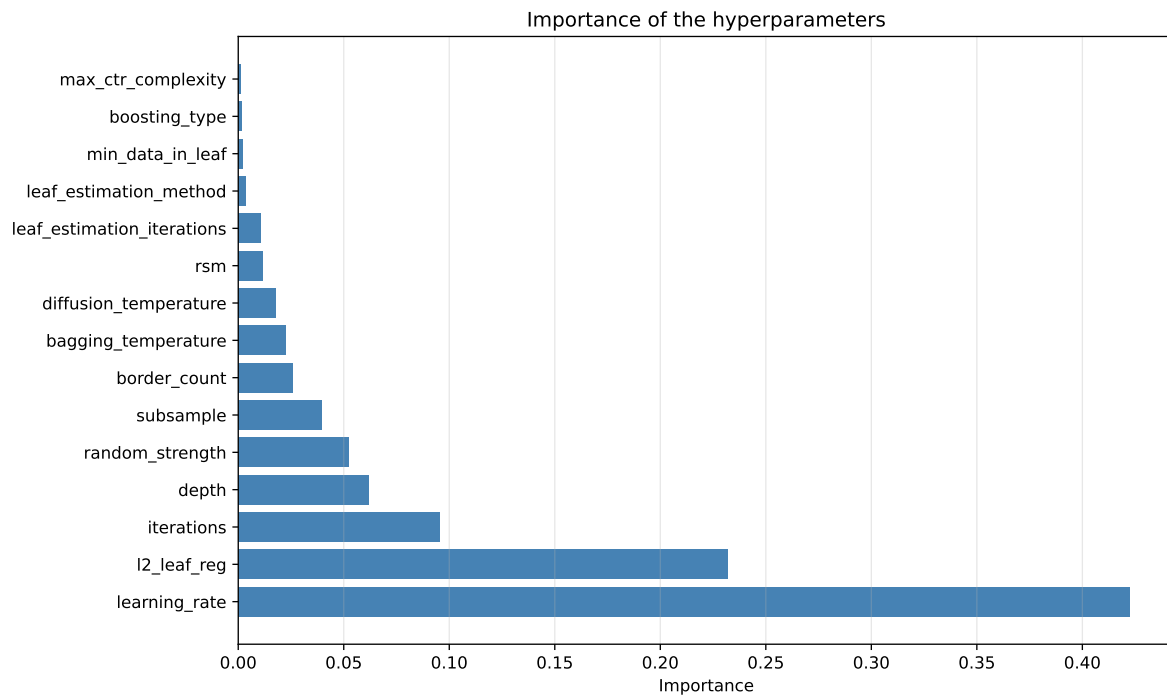
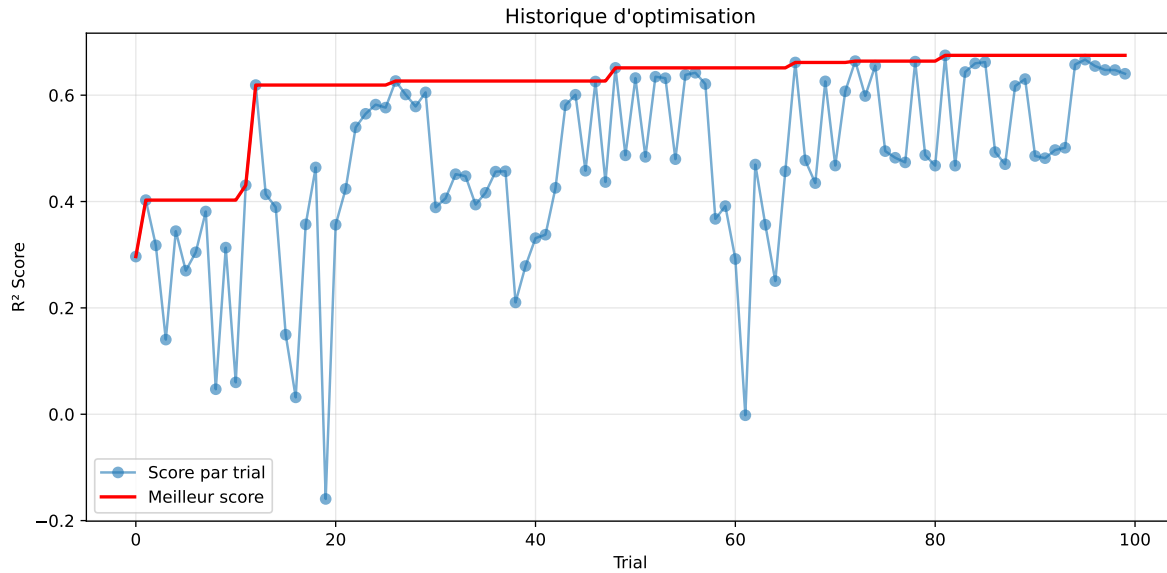
3.6 Model : CatBoost

3.6.1 Gridsearch

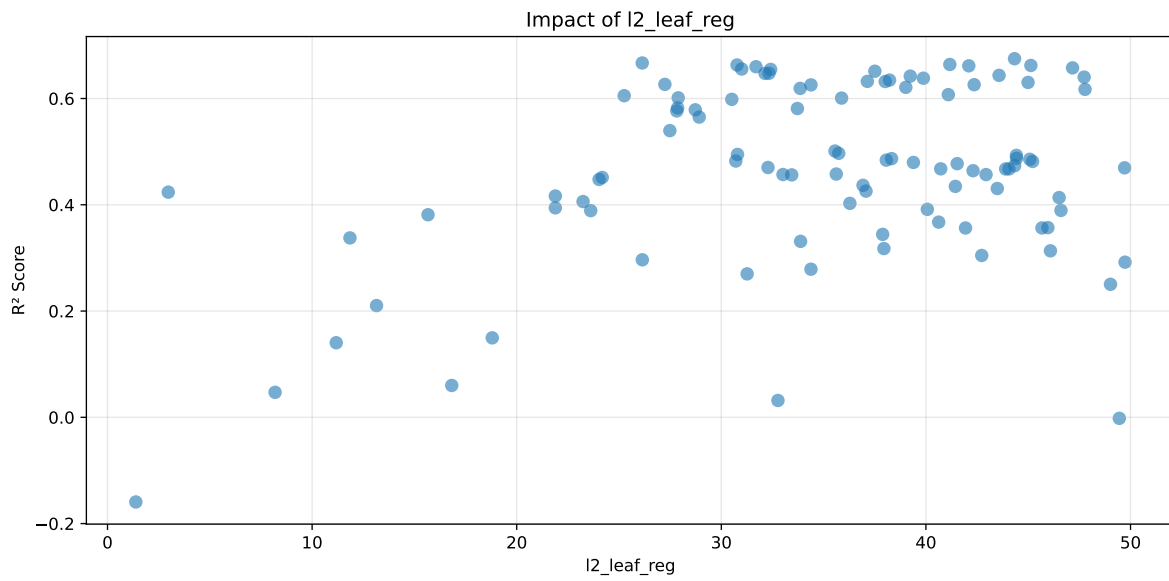
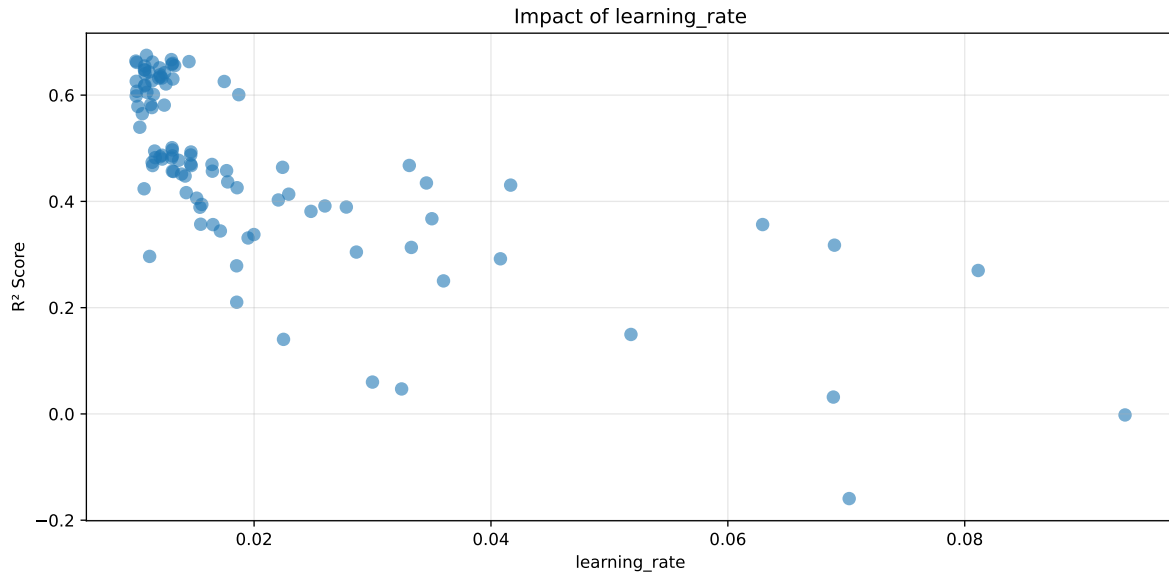
Distribution y_train vs y_test: y_train: mean=2.350, std=0.793, min=0.959, max=8.682
y_test: mean=2.359, std=0.815, min=0.973, max=13.845

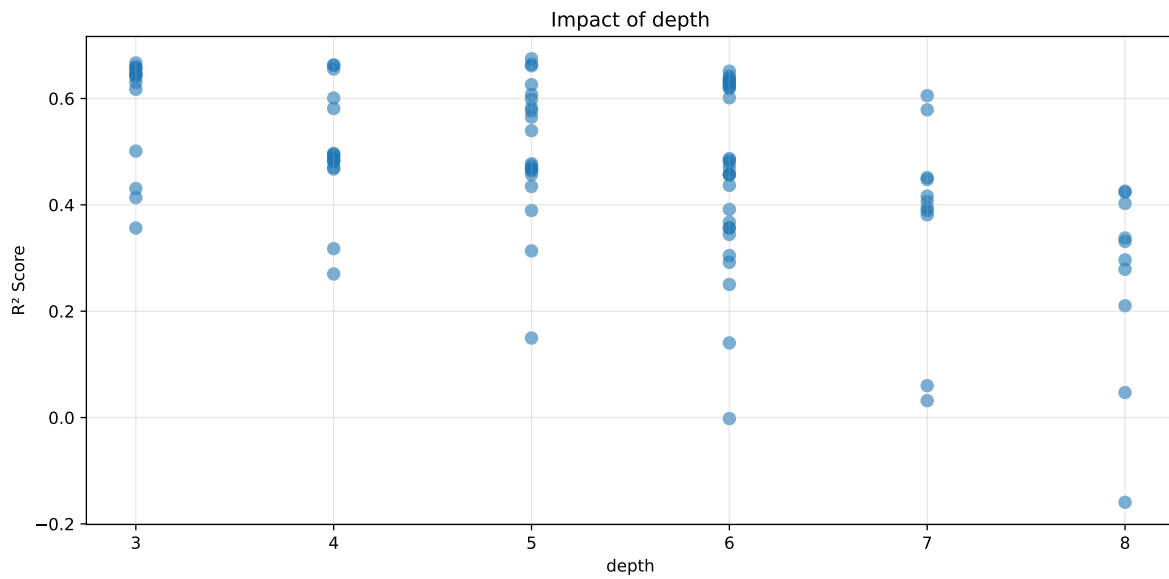
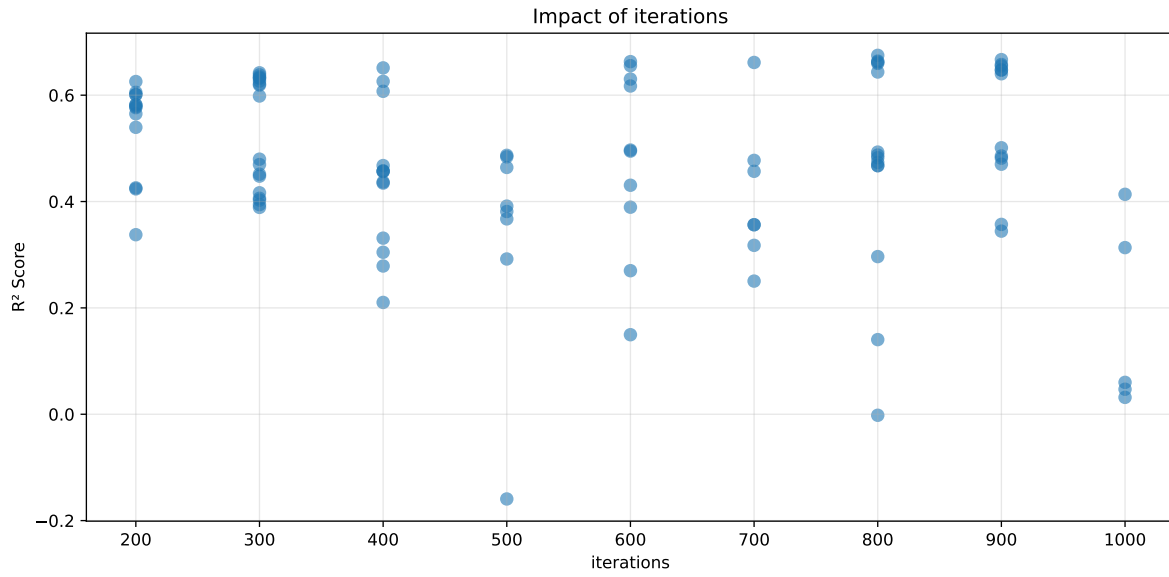
===== TOP Importance FEATURES =====

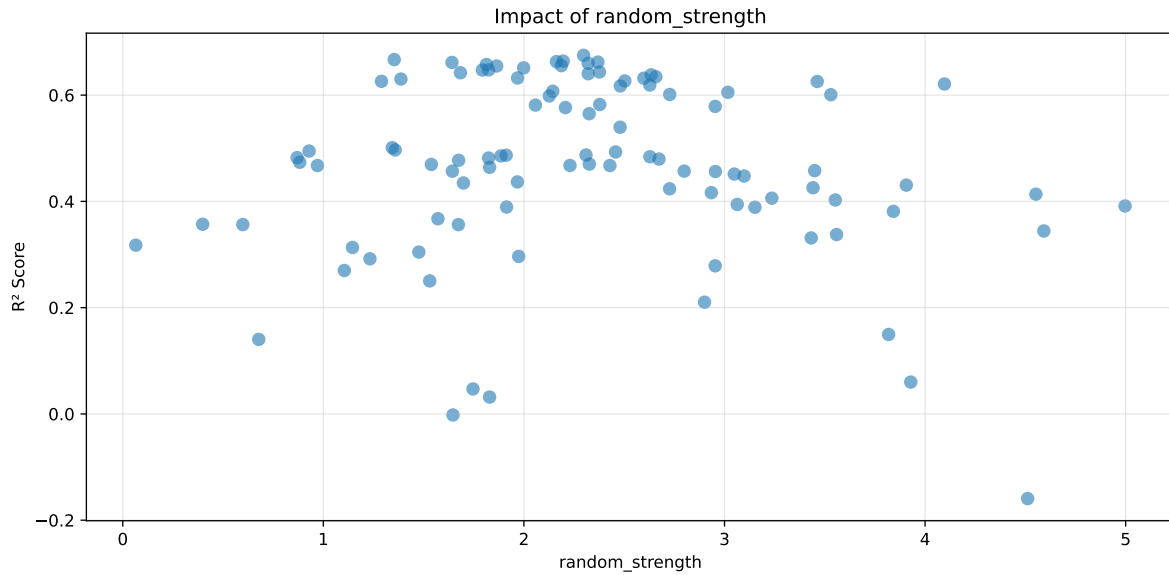
feature	importance
r_umidity	20.291345
r_temperature_st_24h_slope	10.694033
r_temperature_st_12h_slope	10.413307
r_CO2	9.388962
r_temperature_st_12h_std	3.590449
r_temperature_st_8h_std	3.231973
r_NH3_st_24h_min	3.229359
r_temperature_st_12h_diff_last	2.898815
r_NH3_st_8h_std	2.686449
r_temperature_st_24h_mean	2.636398
r_NH3_st_12h_std	2.627894
r_NH3	2.355809
r_temperature	2.236667
r_temperature_st_12h_median	2.217848
r_NH3_st_24h_std	2.014006
r_NH3_st_12h_rms	2.002490
r_temperature_st_8h_min	1.802710
r_temperature_st_8h_diff_last	1.792151
r_temperature_st_24h_diff_last	1.573032
r_NH3_st_8h_mean	1.439974
r_NH3_st_24h_slope	1.436792
r_umidity_st_12h_std	1.257562
r_NH3_st_8h_slope	1.211133
r_THI	1.116567
r_NH3_st_12h_slope	1.115265
r_umidity_st_24h_std	0.754771
r_umidity_st_12h_diff_last	0.720970
r_umidity_st_24h_diff_last	0.666267
r_umidity_st_8h_std	0.657025
r_NH3_st_8h_diff_last	0.604712
r_umidity_st_8h_diff_last	0.491921
r_NH3_st_24h_diff_last	0.429529
r_NH3_st_12h_diff_last	0.413818



Paramètres avec >5% d'importance : ['learning_rate', 'l2_leaf_reg', 'iterations', 'depth', 'random_strength']

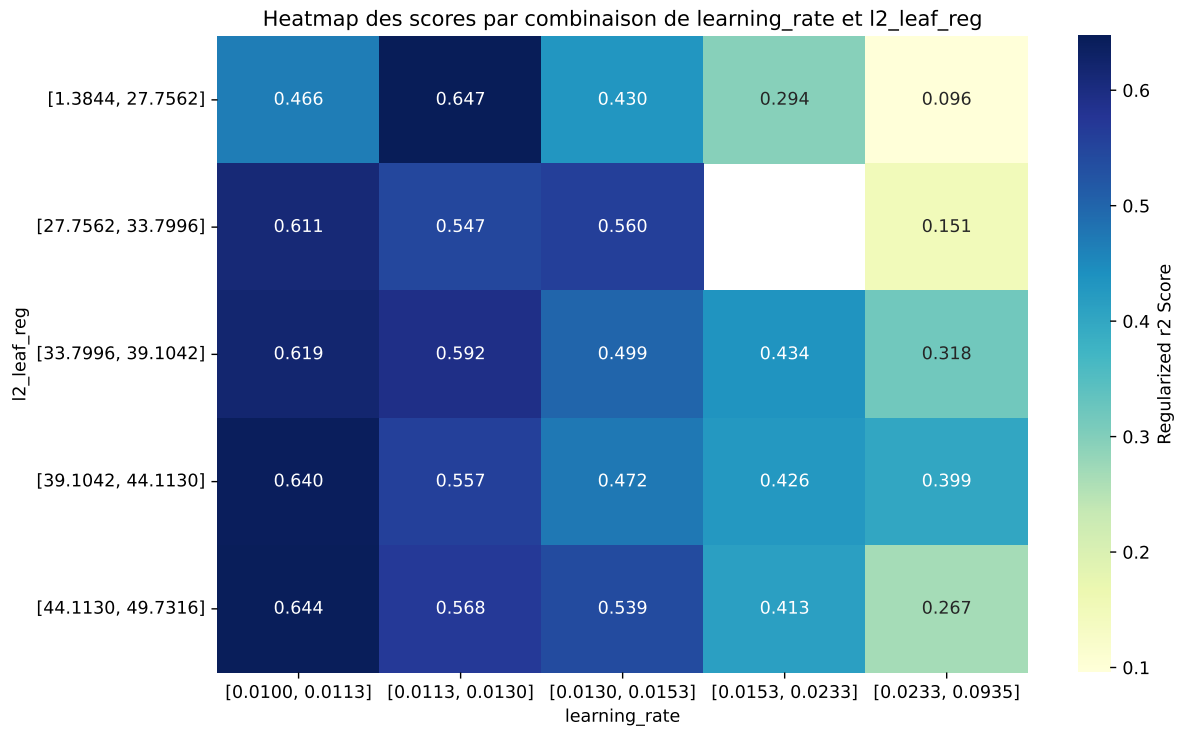






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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.7086	0.6749	0.659	0.0125	0.6749	0.6712	-0.0037	0.9945	1.0499
2	0.699	0.6669	0.6535	0.0123	0.6669	0.6645	-0.0024	0.9964	1.0482
3	0.6964	0.6641	0.6452	0.0115	0.6641	0.6566	-0.0075	0.9887	1.0486
4	0.693	0.6631	0.6472	0.0128	0.6631	0.6616	-0.0015	0.9977	1.0451
5	0.6914	0.6622	0.6463	0.012	0.6622	0.6593	-0.0029	0.9957	1.0441
6	0.6936	0.6615	0.6434	0.0122	0.6615	0.6565	-0.005	0.9924	1.0485
7	0.6892	0.6599	0.6455	0.0127	0.6599	0.6548	-0.0051	0.9923	1.0444
8	0.6859	0.6577	0.6437	0.0139	0.6577	0.6577	0.0	1.0	1.0429
9	0.6836	0.6555	0.6393	0.0125	0.6555	0.6497	-0.0058	0.9912	1.0429
10	0.6816	0.6548	0.6403	0.013	0.6548	0.6504	-0.0044	0.9932	1.0409
---	---	---	---	---	---	---	---	---	---
100	0.9829	0.7925	0.7623	0.0089	-0.1593	0.7852	-0.0073	0.9908	1.2402

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 5, 'reg__learning_rate': 0.010913215996310312, 'reg__l2_leaf_reg': 44.330941513400624, 'reg__bagging_temperature': 0.7960983494398761, 'reg__random_strength': 2.2975226063004226, 'reg__subsample': 0.8798645595681472, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.8247564435056035, 'reg__border_count': 77, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8825.209695405707}

Rank 2: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.013016451157226265, 'reg__l2_leaf_reg': 26.134343136959597, 'reg__bagging_temperature': 2.0480077522616753, 'reg__random_strength': 1.3533140243709791, 'reg__subsample': 0.8670258214744911, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8010365443351347, 'reg__border_count': 73, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3343.229924185714}

Rank 3: {'reg__iterations': 800, 'reg__depth': 5, 'reg__learning_rate': 0.010015253784620875, 'reg__l2_leaf_reg': 41.16854762482597, 'reg__bagging_temperature': 1.4493518374505638, 'reg__random_strength': 2.1957878959612644, 'reg__subsample': 0.8846502117527817, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.7581577634121589, 'reg__border_count': 81, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 7740.862388532721}

Rank 4: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.014505103127577175, 'reg__l2_leaf_reg': 30.768825805685942, 'reg__bagging_temperature': 1.9578132857314494, 'reg__random_strength': 2.1620881468489777, 'reg__subsample': 0.806900425076886, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':

0.6782623294231085, 'reg__border_count': 82, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3173.457004850563}

Rank 5: {'reg__iterations': 800, 'reg__depth': 4, 'reg__learning_rate': 0.011417442861418072, 'reg__l2_leaf_reg': 45.129774997006756, 'reg__bagging_temperature': 1.8652400463274459, 'reg__random_strength': 2.368736402165992, 'reg__subsample': 0.8858797318631132, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6720678373621314, 'reg__border_count': 100, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 8545.292950393748}

Rank 6: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.010086164472214776, 'reg__l2_leaf_reg': 42.0948875192242, 'reg__bagging_temperature': 1.5043157848250996, 'reg__random_strength': 1.6411234342354952, 'reg__subsample': 0.8532123137530543, 'reg__min_data_in_leaf': 11, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6607259271922359, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 9513.233965976742}

Rank 7: {'reg__iterations': 800, 'reg__depth': 3, 'reg__learning_rate': 0.01312663017183308, 'reg__l2_leaf_reg': 31.6948622138516, 'reg__bagging_temperature': 1.9738056545121687, 'reg__random_strength': 2.320441172360723, 'reg__subsample': 0.8753083103976107, 'reg__min_data_in_leaf': 18, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7500795771456764, 'reg__border_count': 101, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8508.691845590767}

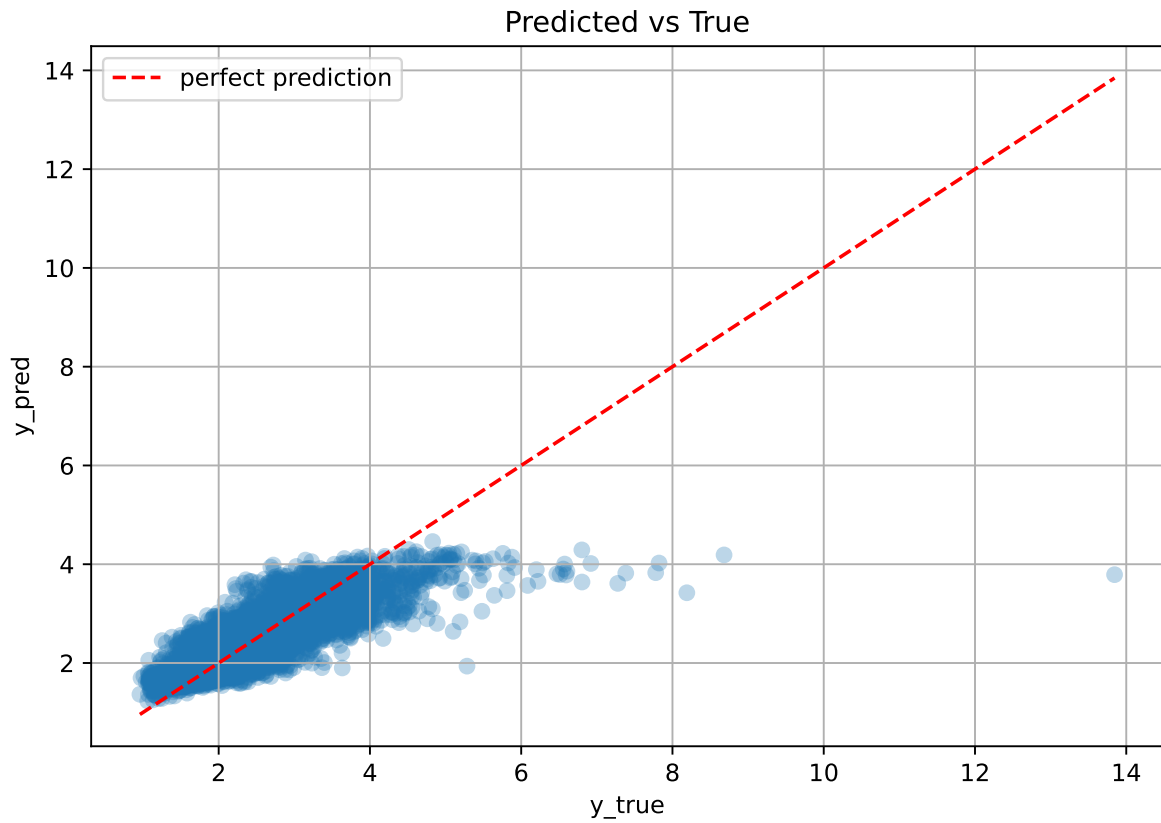
Rank 8: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.013060462823860128, 'reg__l2_leaf_reg': 47.16657017196917, 'reg__bagging_temperature': 1.7054465075260261, 'reg__random_strength': 1.8128156746139075, 'reg__subsample': 0.8638560185732612, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8016648348631035, 'reg__border_count': 87, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3332.3444372873273}

Rank 9: {'reg__iterations': 600, 'reg__depth': 4, 'reg__learning_rate': 0.013294063950604515, 'reg__l2_leaf_reg': 30.99708189044061, 'reg__bagging_temperature': 1.7786148479326638, 'reg__random_strength': 2.1867546709969043, 'reg__subsample': 0.8935330524905053, 'reg__min_data_in_leaf': 14, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6076175290487826, 'reg__border_count': 99, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8678.703970116654}

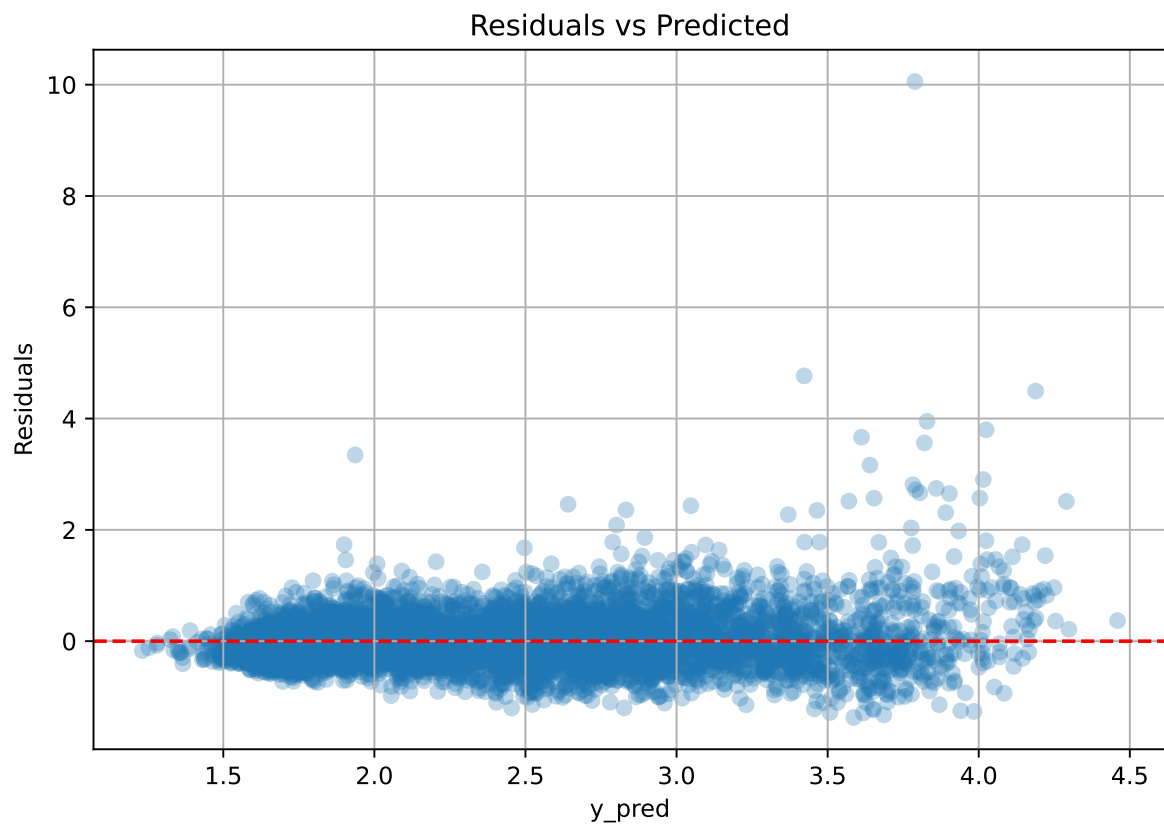
Rank 10: {'reg__iterations': 900, 'reg__depth': 3, 'reg__learning_rate': 0.010762520911870944, 'reg__l2_leaf_reg': 32.41045899397506, 'reg__bagging_temperature': 2.4355992266684074,

'reg__random_strength': 1.8642200207024746, 'reg__subsample': 0.8636770430675674,
'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':
0.6903225438752979, 'reg__border_count': 72, 'reg__leaf_estimation_method': 'Gradient',
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_tempera-
ture': 3349.7130483707924}

3.6.2 Scatter



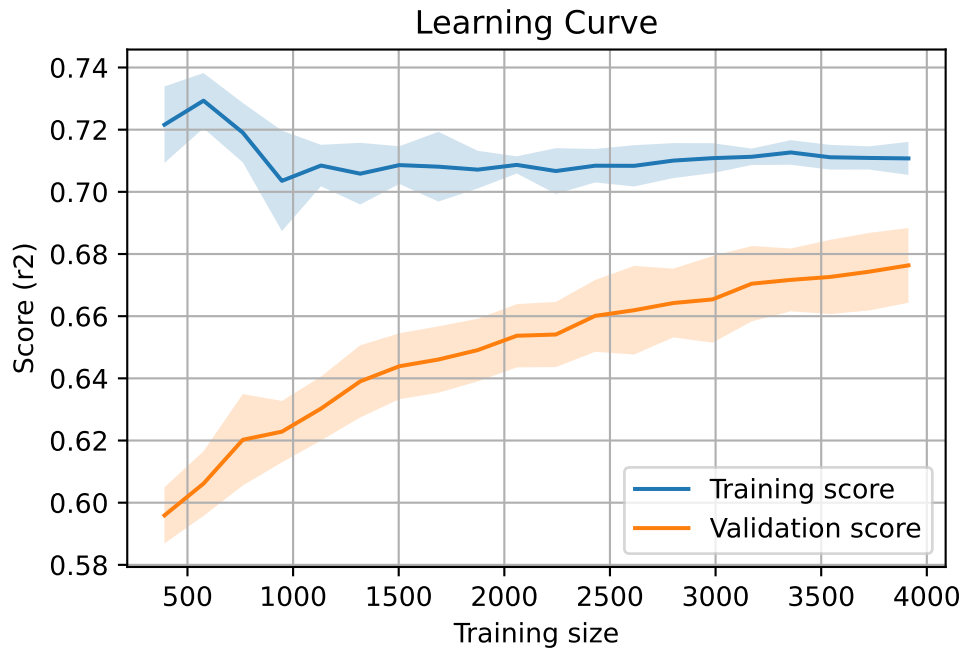
3.6.3 Scatter residuals



3.6.4 Learning curve

`Len(group)` : 7988, `Len(training)` : 5592,

`Len(training_real)` : 3915, `Len(test_of_training)` : 1677, `Len(test)` : 2396



3.7 Sumup-table

model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.562	0.554	0.012	0.525	0.526	-0.029	0.948	1.013	nan
LGBM	0.681	0.649	0.014	0.633	0.654	0.005	1.007	1.05	0.6488
CatBoost	0.709	0.675	0.012	0.659	0.671	-0.004	0.994	1.05	0.6749

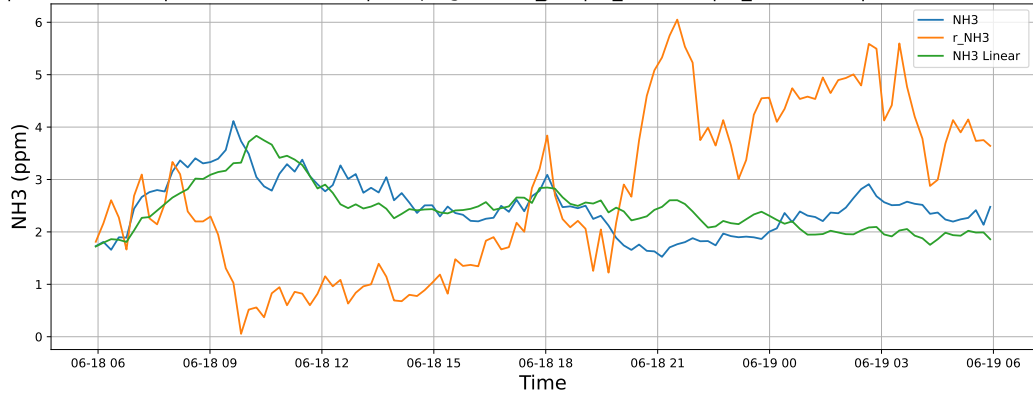
4 Plot

4.1 Standalone plots

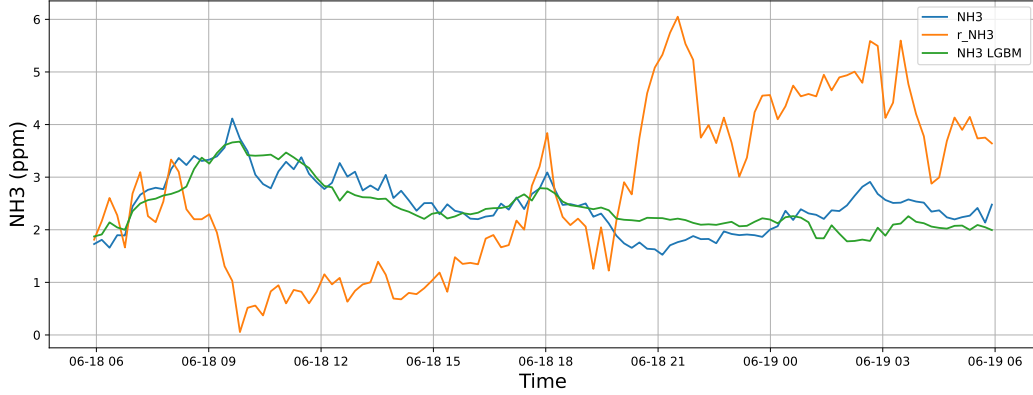
4.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

4.1.1.1 Time window : 24

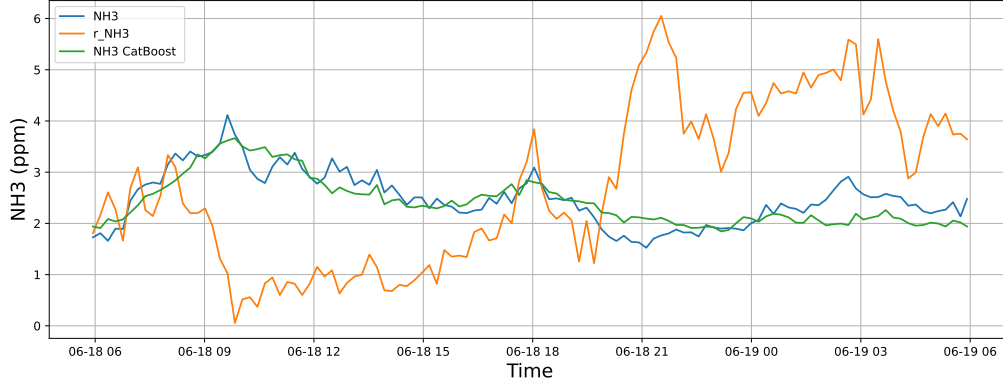
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

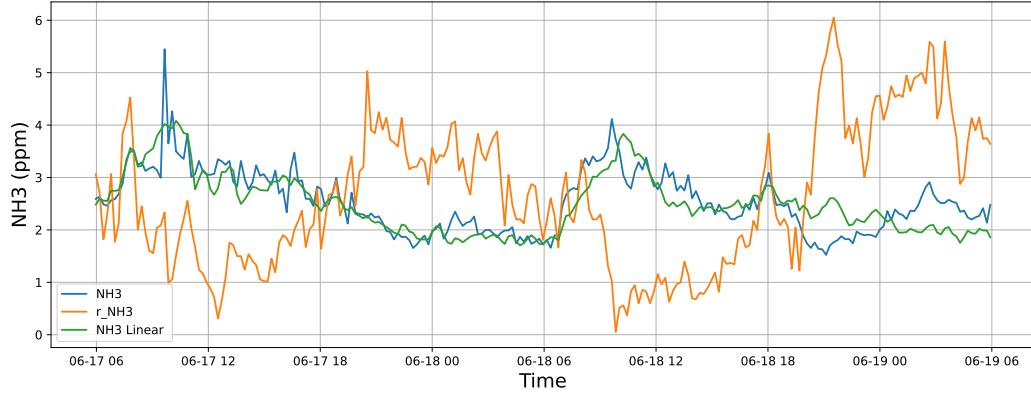


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

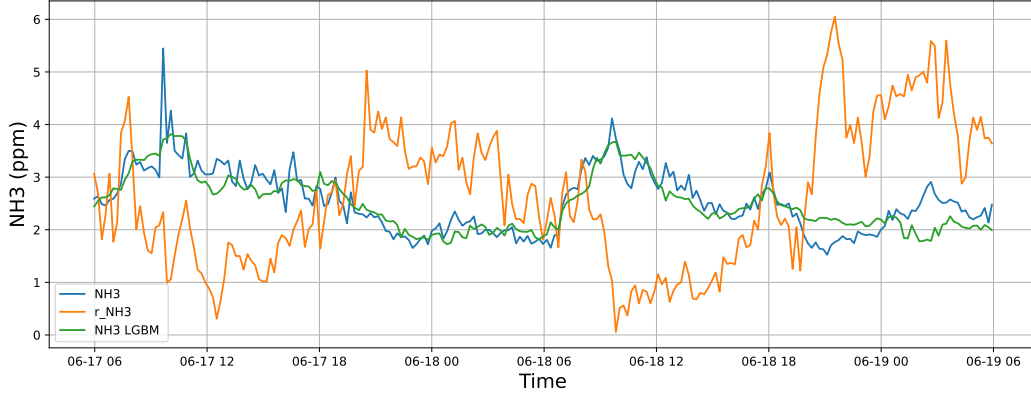


4.1.1.2 Time window : 48

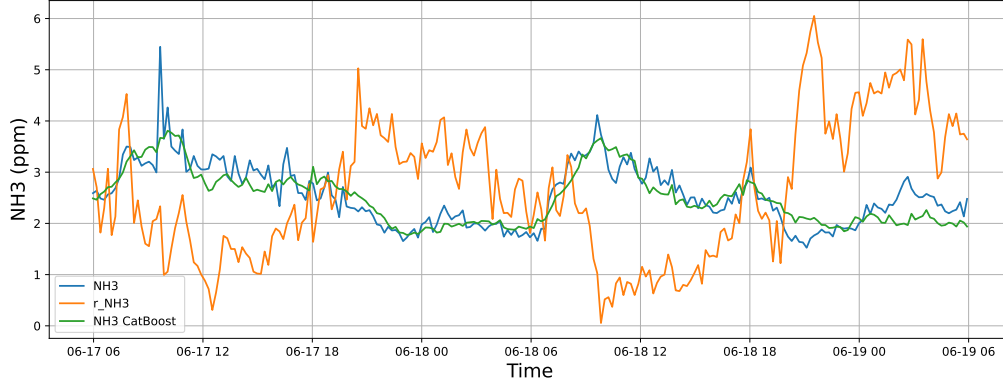
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

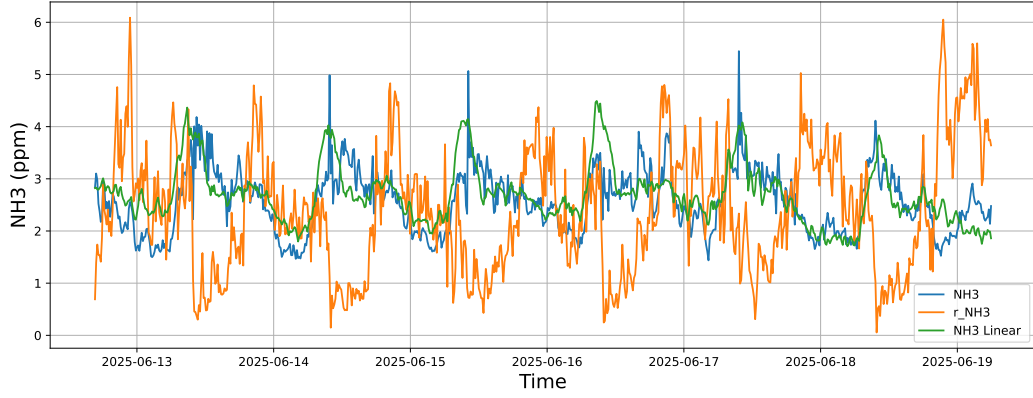


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

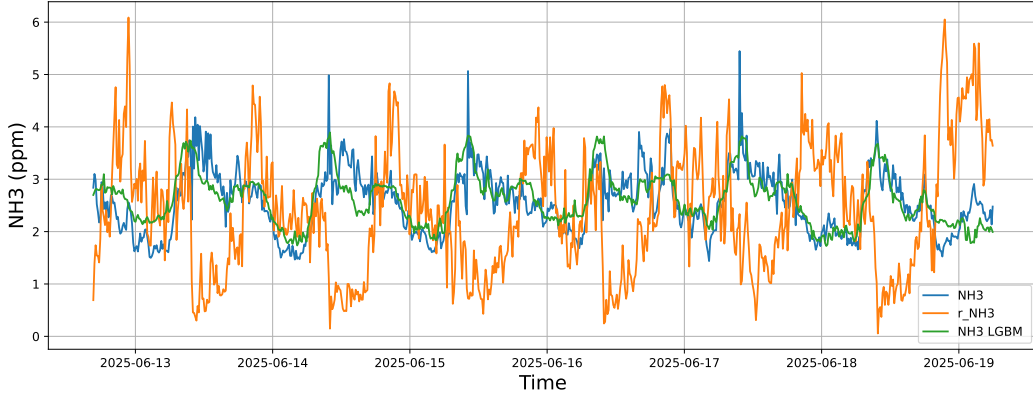


4.1.1.3 Time window : ALL

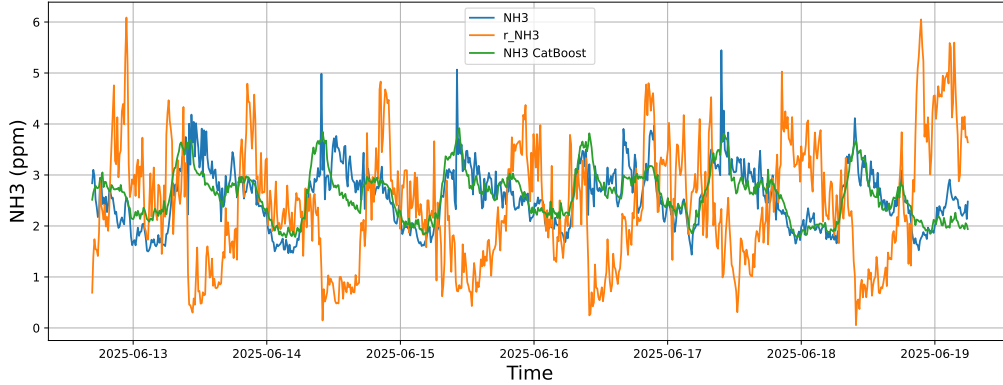
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



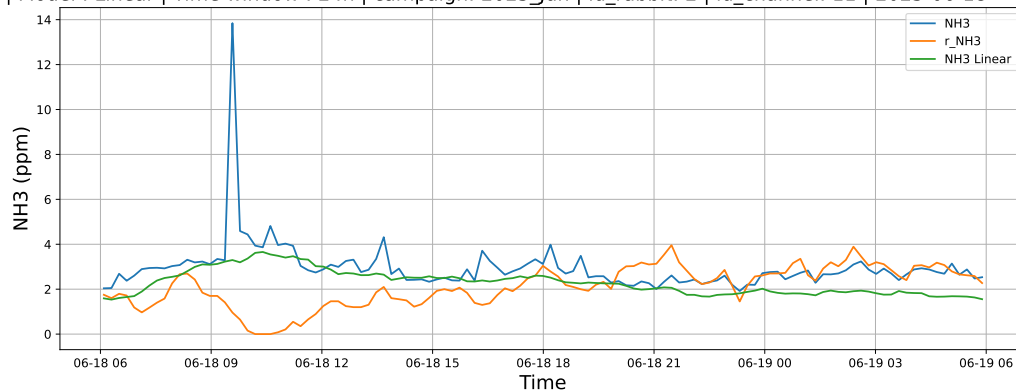
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



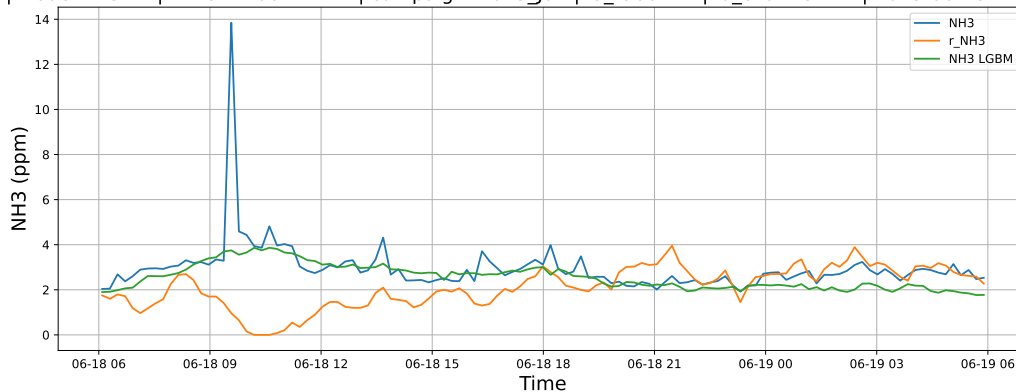
4.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

4.1.2.1 Time window : 24

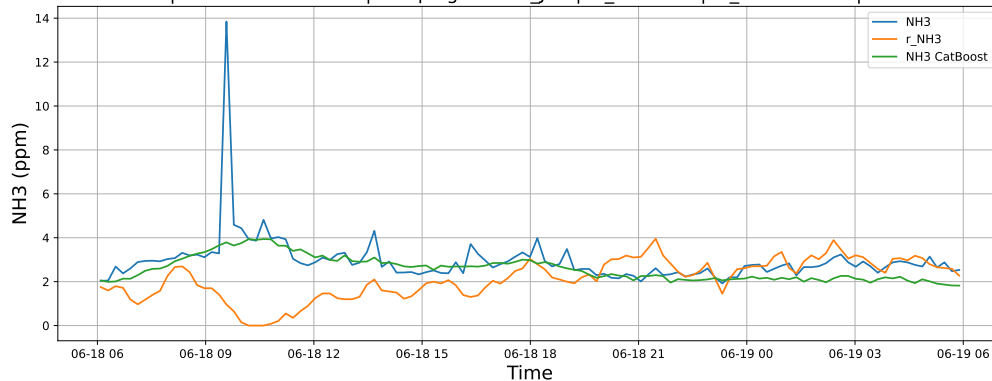
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

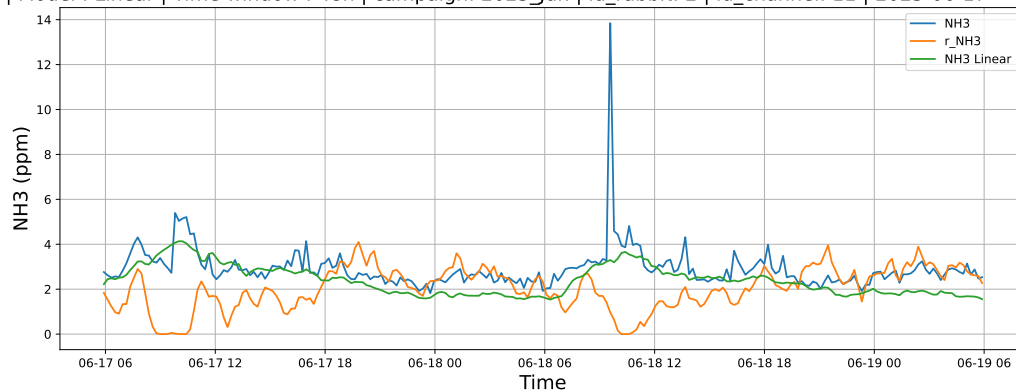


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

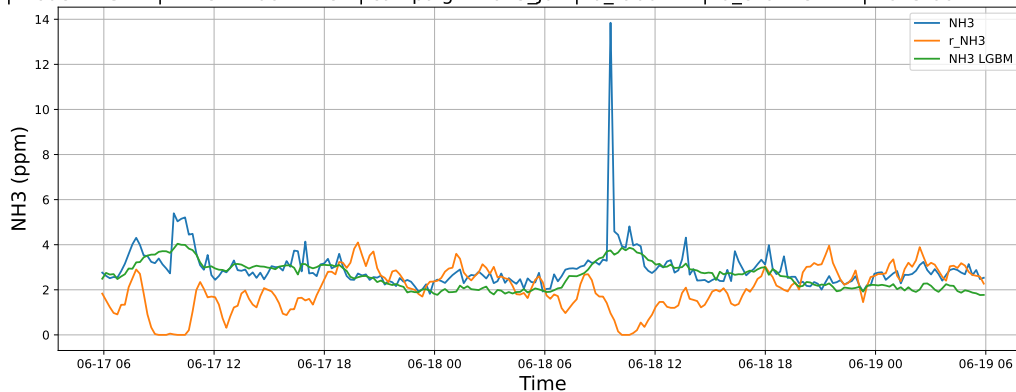


4.1.2.2 Time window : 48

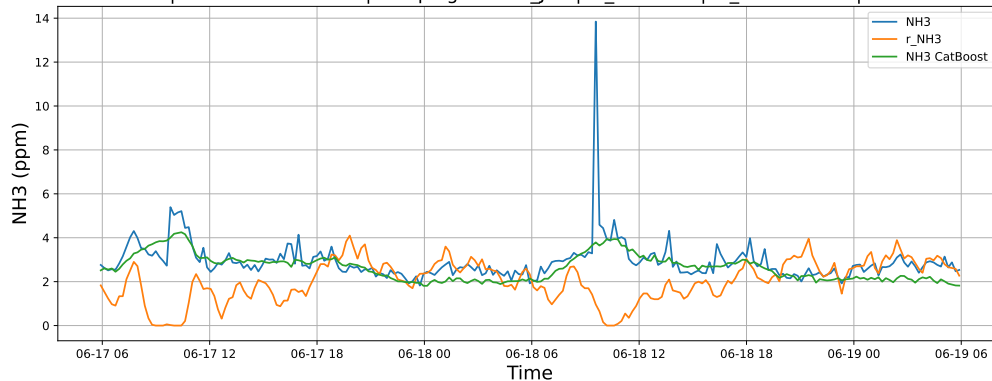
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

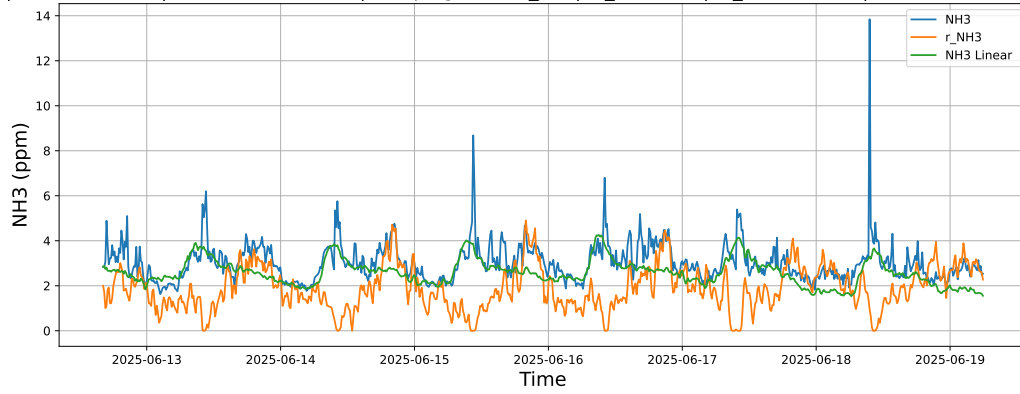


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

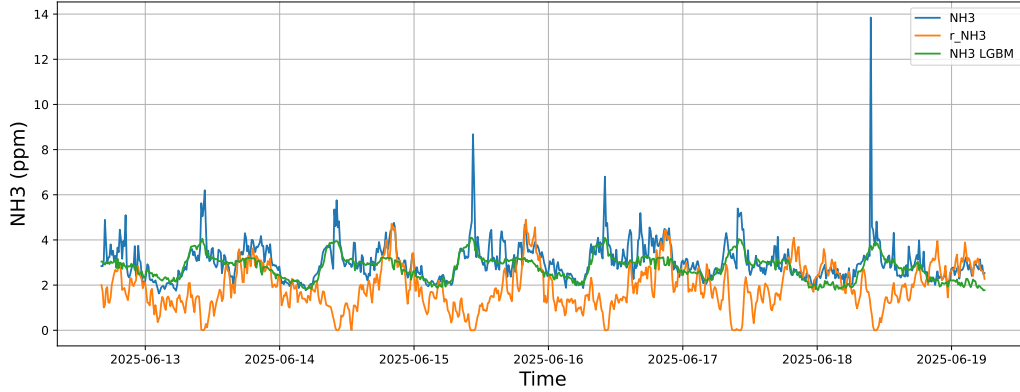


4.1.2.3 Time window : ALL

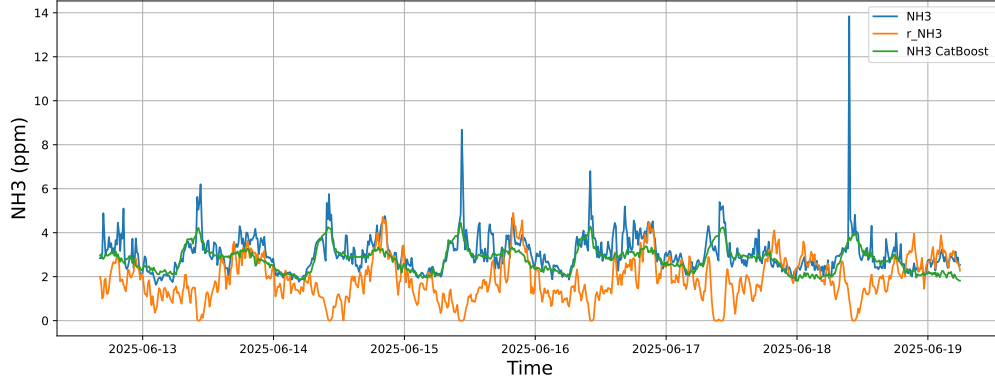
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



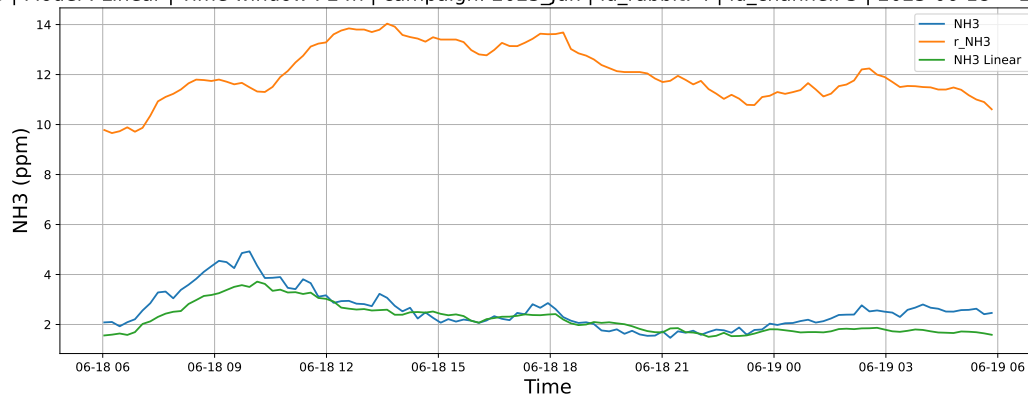
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



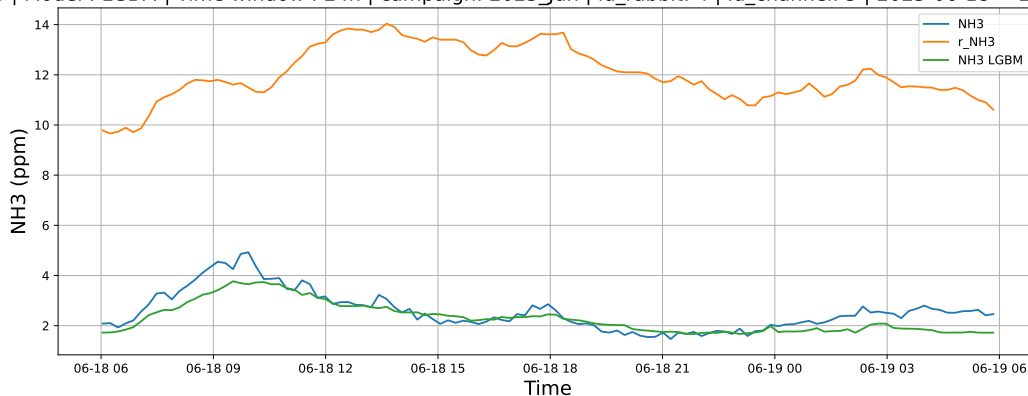
4.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

4.1.3.1 Time window : 24

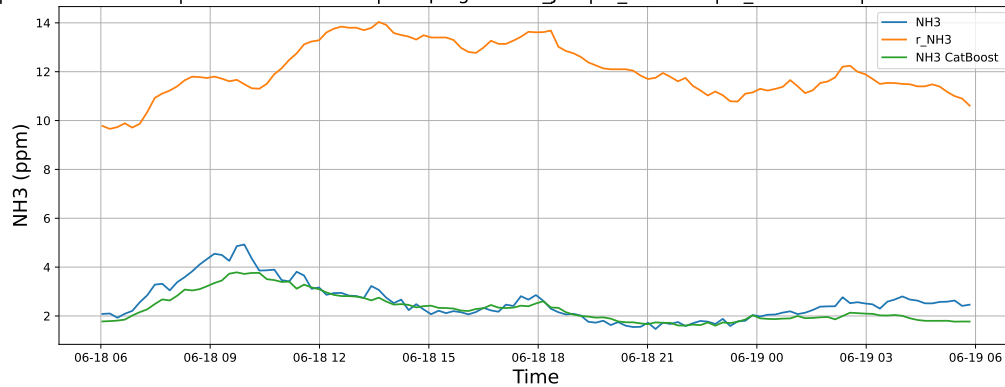
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

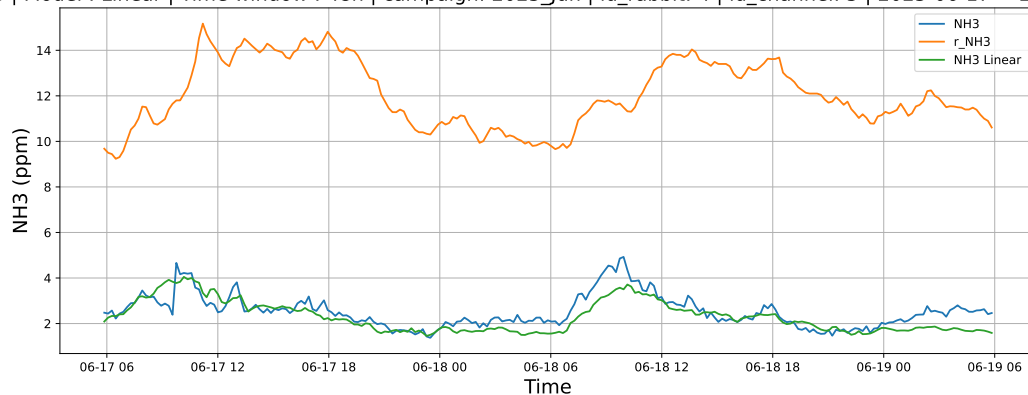


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

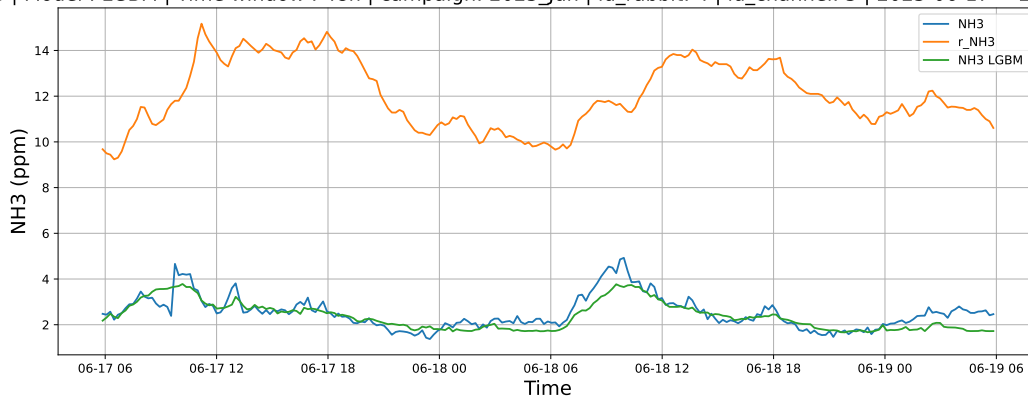


4.1.3.2 Time window : 48

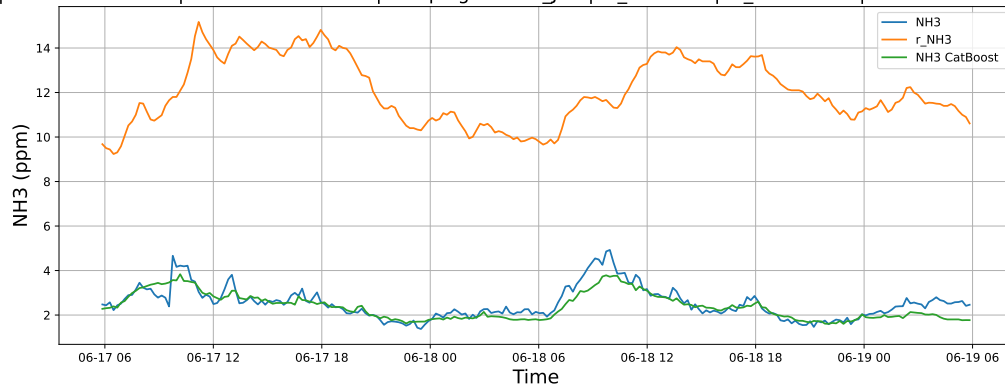
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

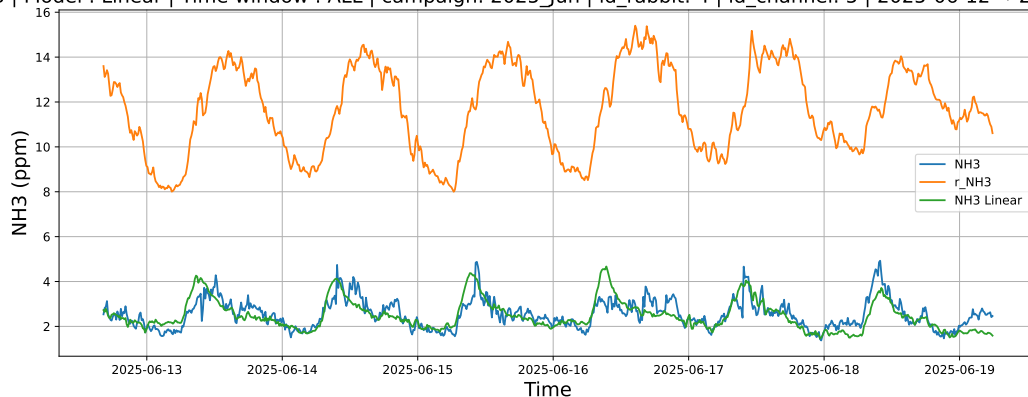


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

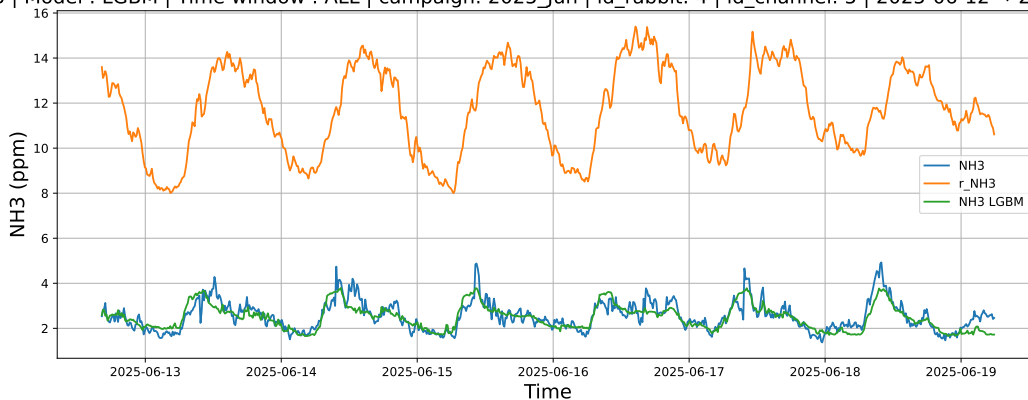


4.1.3.3 Time window : ALL

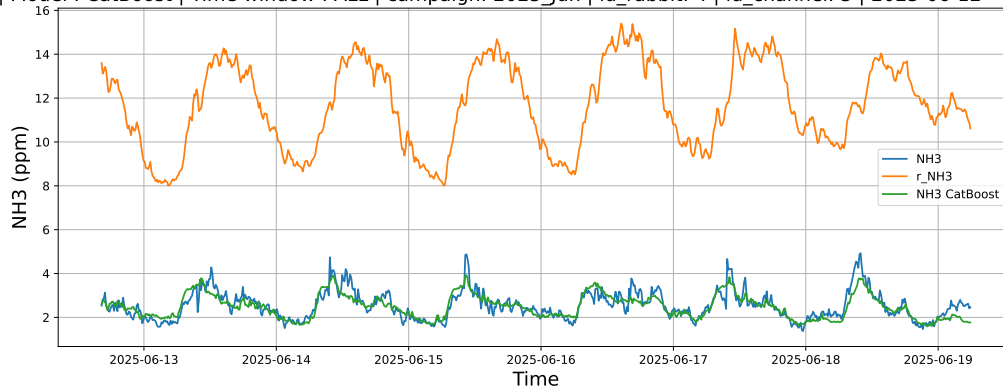
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



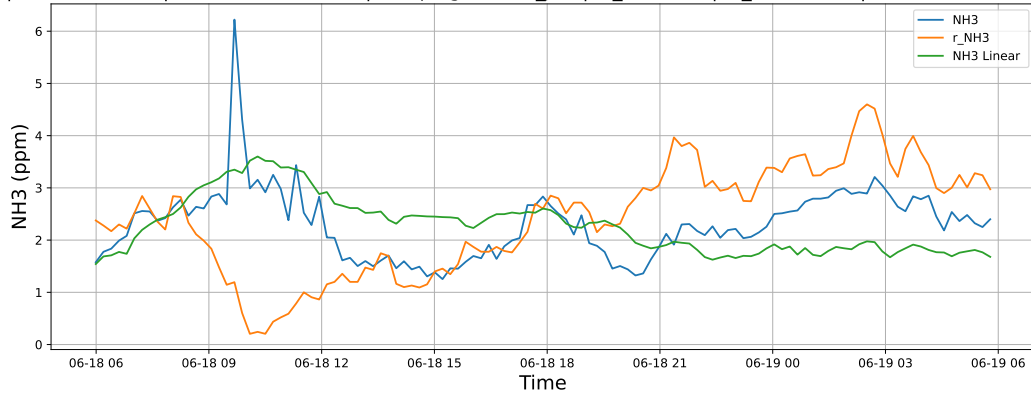
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



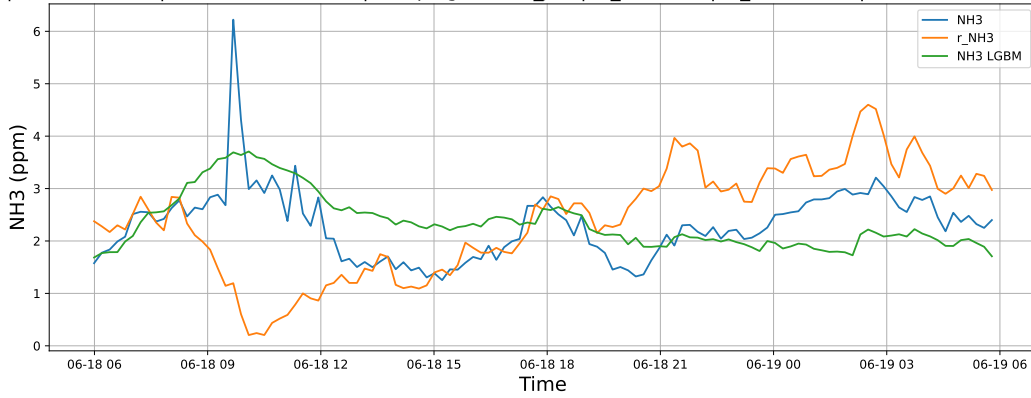
4.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

4.1.4.1 Time window : 24

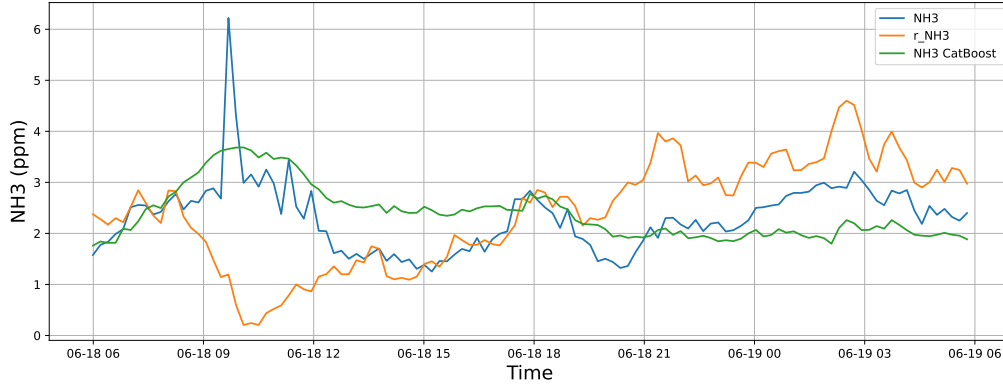
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

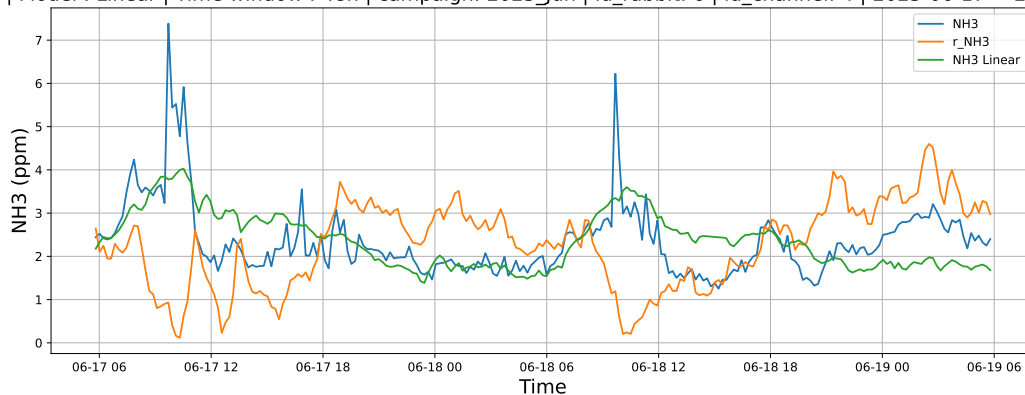


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

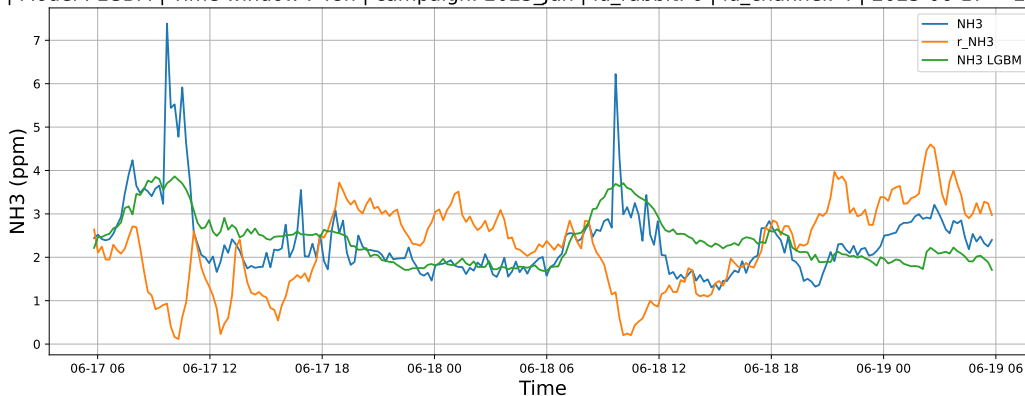


4.1.4.2 Time window : 48

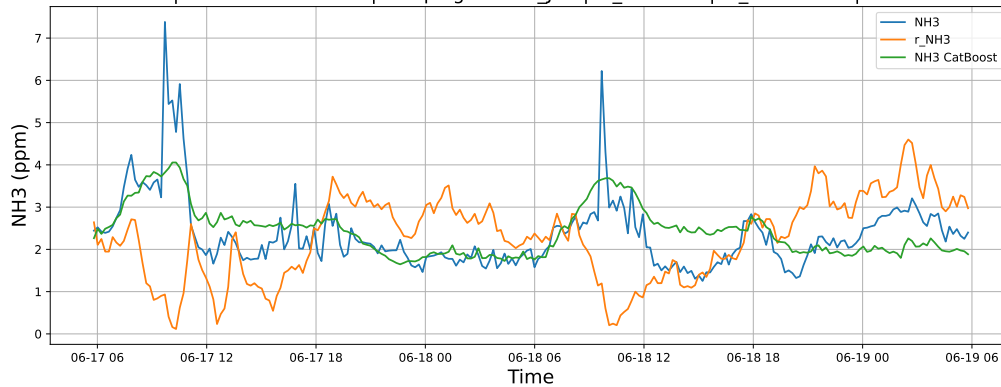
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

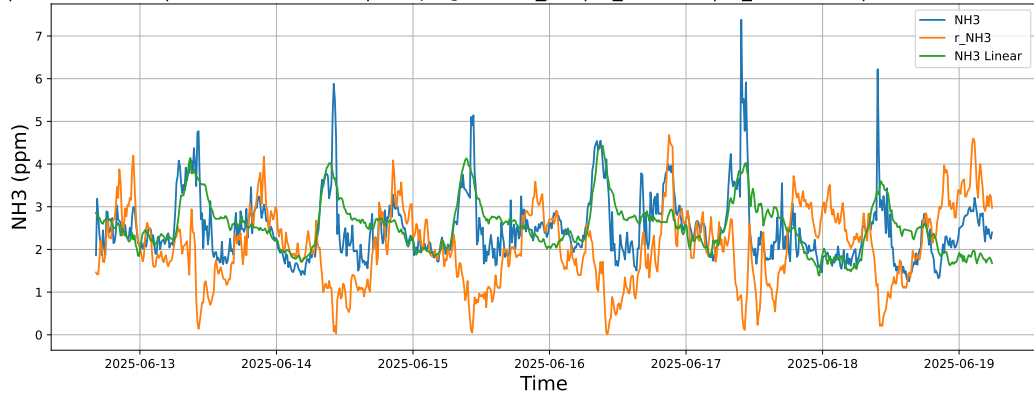


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

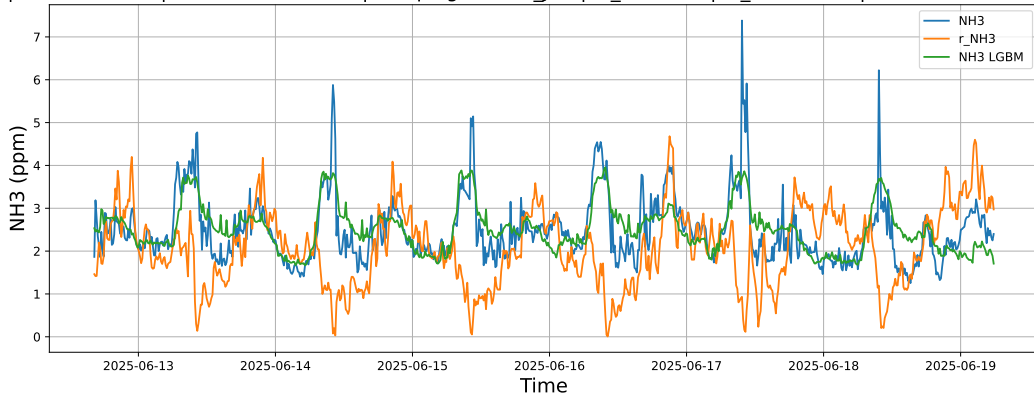


4.1.4.3 Time window : ALL

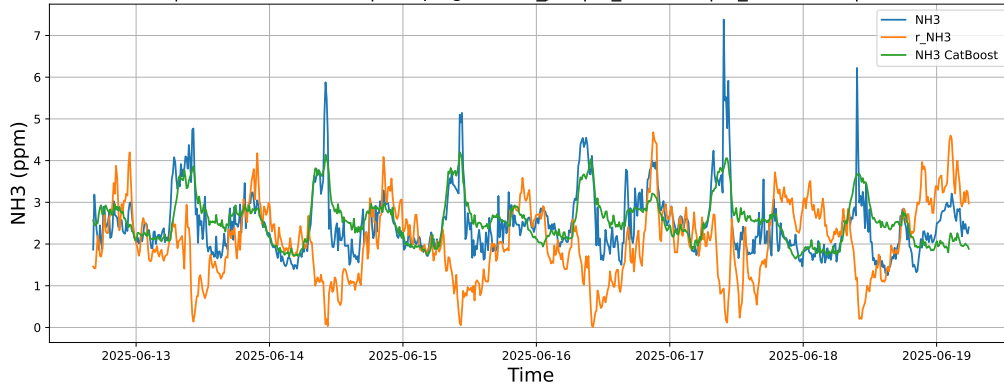
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



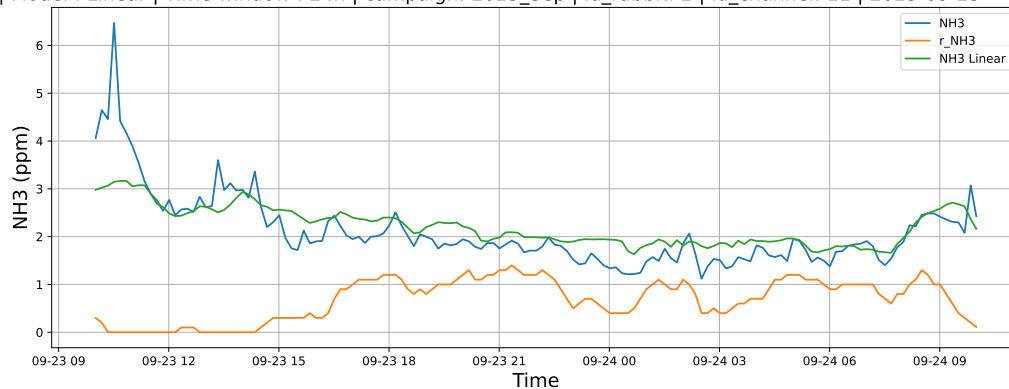
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



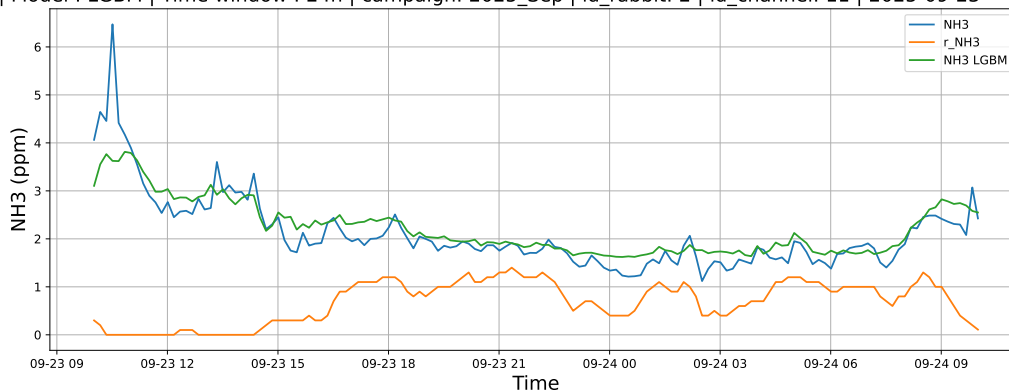
4.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

4.1.5.1 Time window : 24

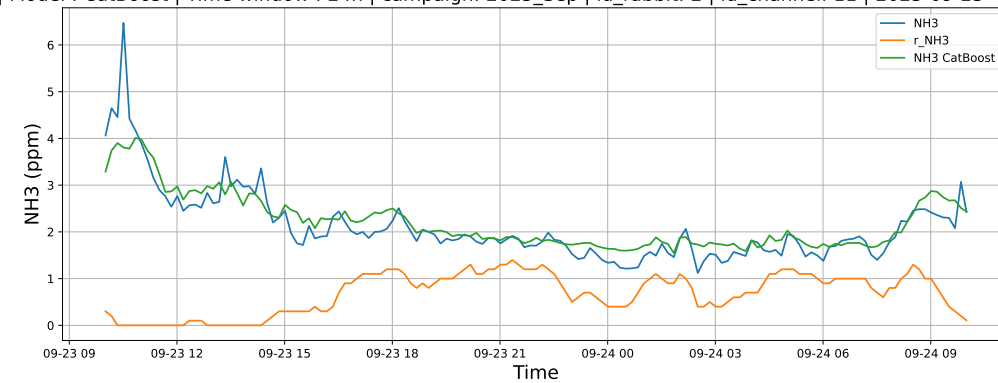
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

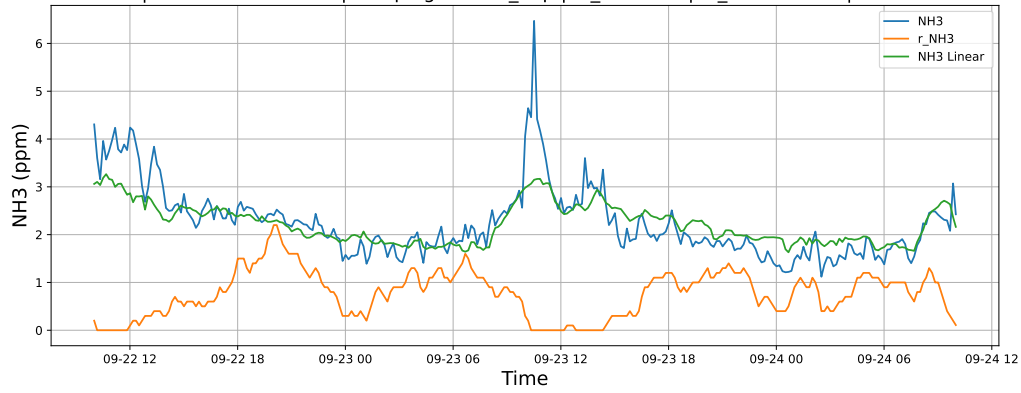


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

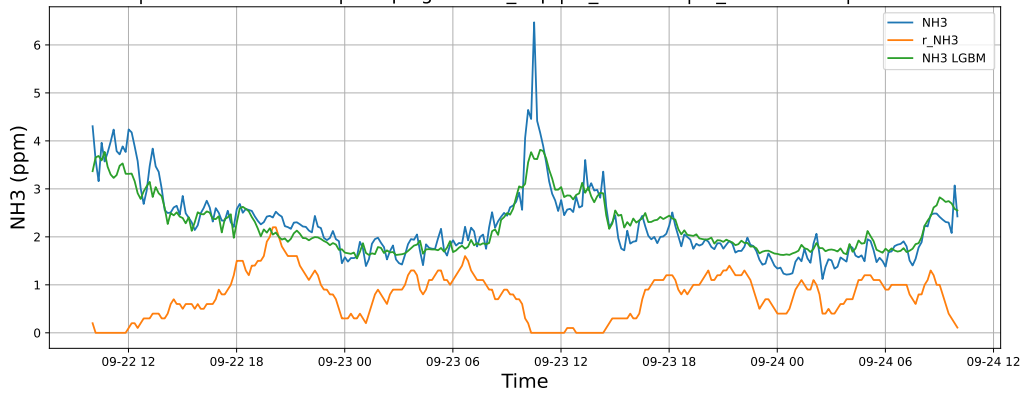


4.1.5.2 Time window : 48

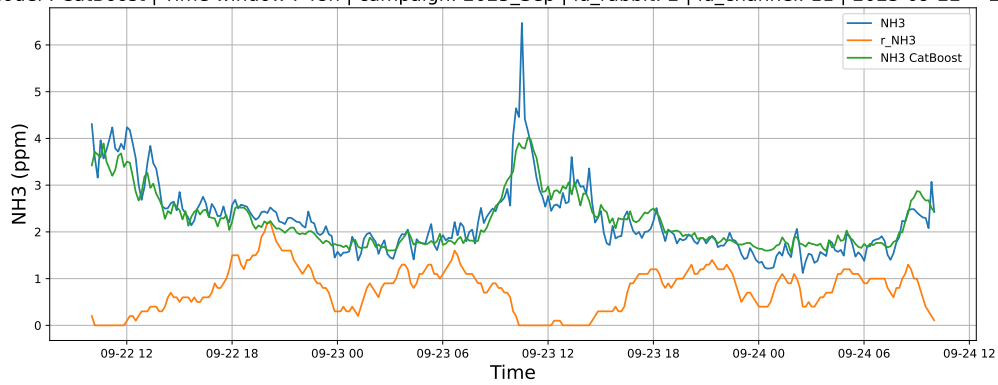
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

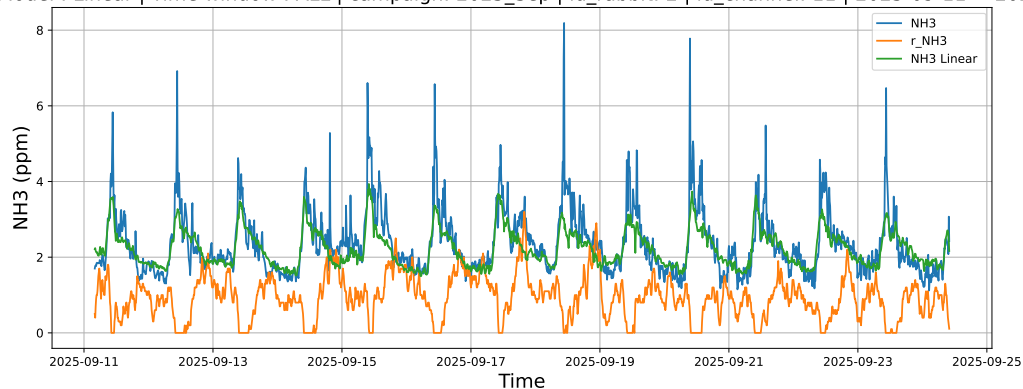


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

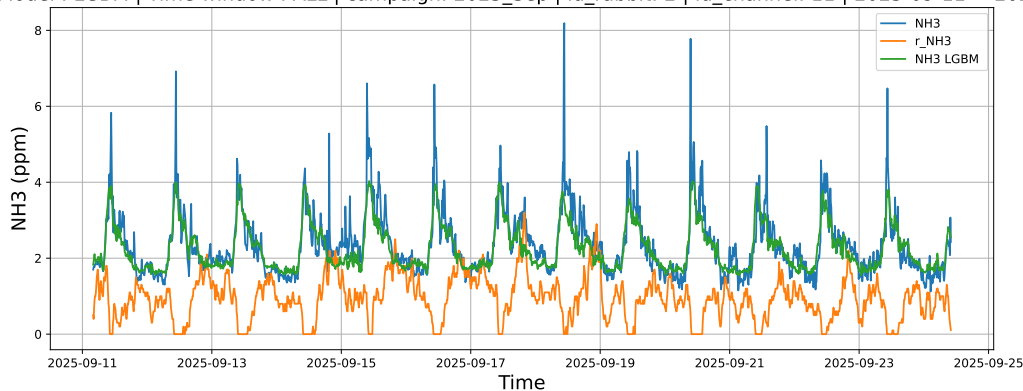


4.1.5.3 Time window : ALL

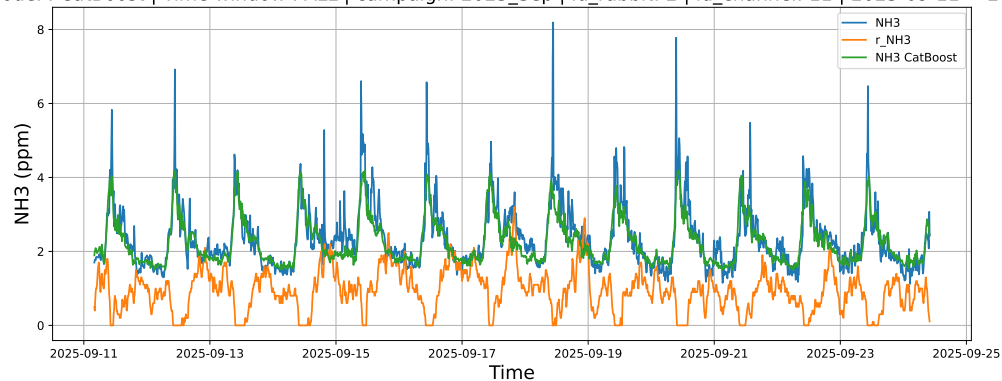
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



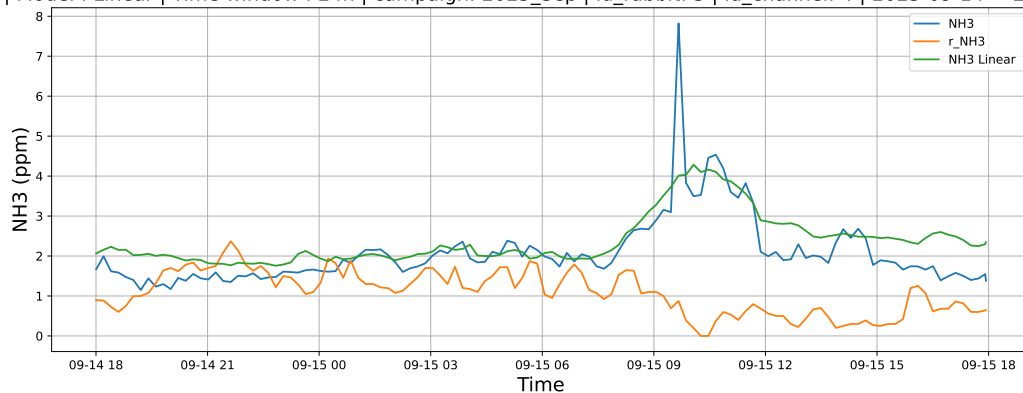
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



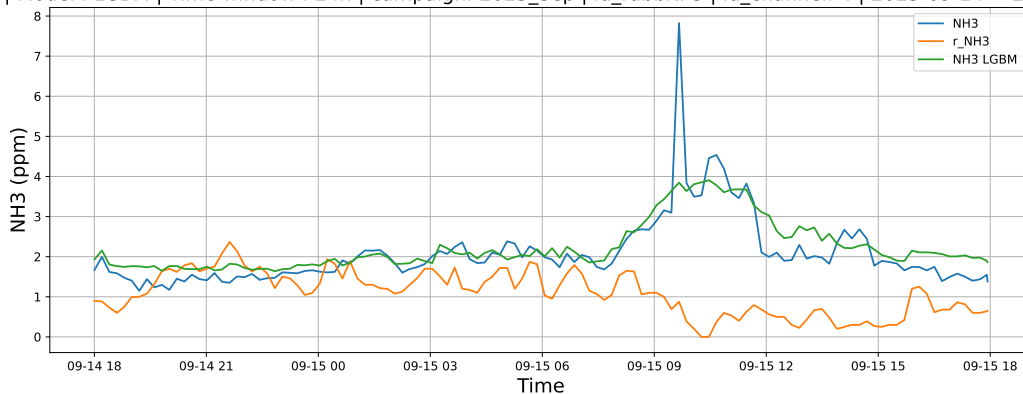
4.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

4.1.6.1 Time window : 24

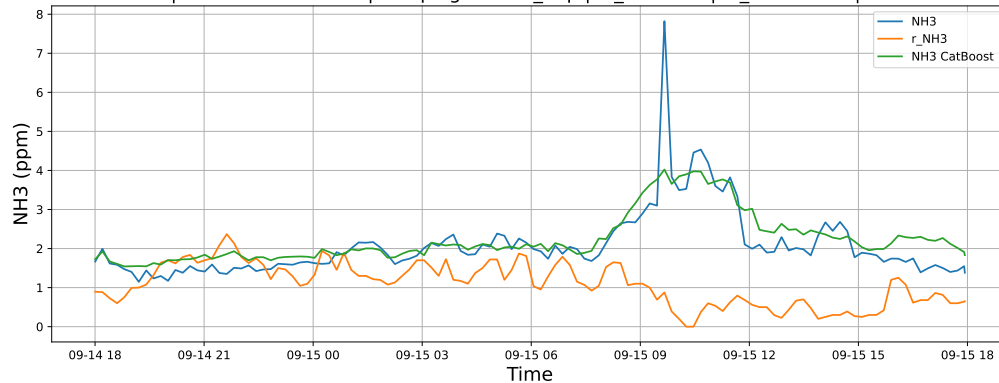
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

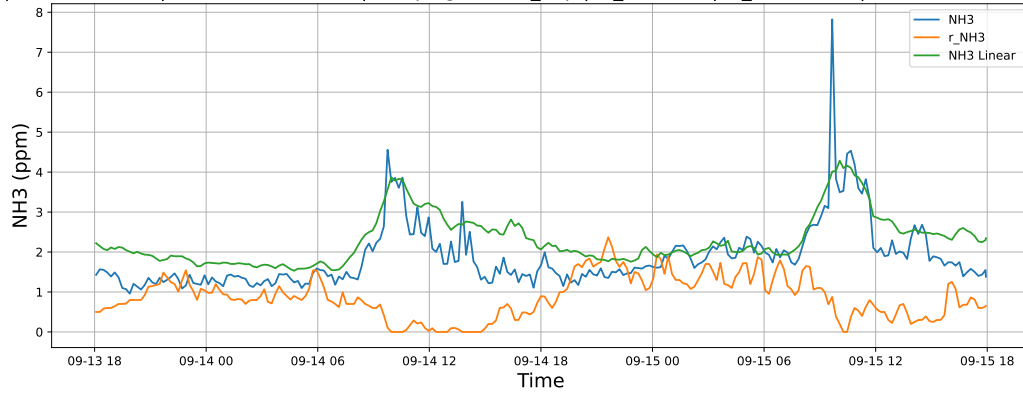


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

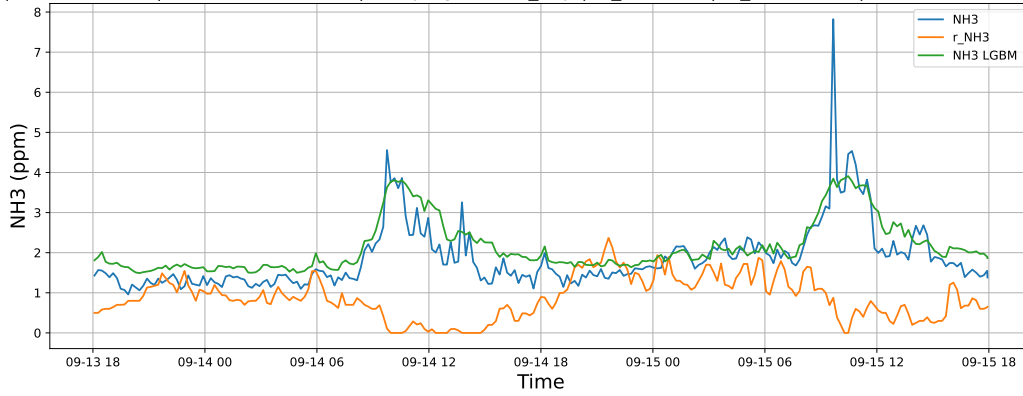


4.1.6.2 Time window : 48

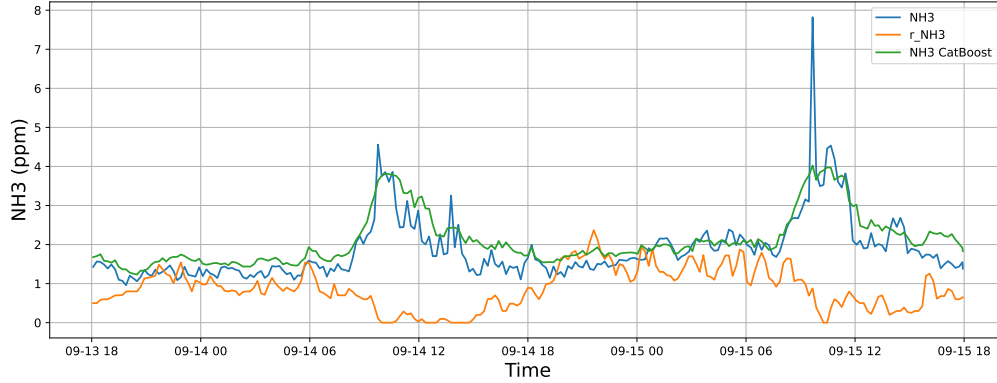
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

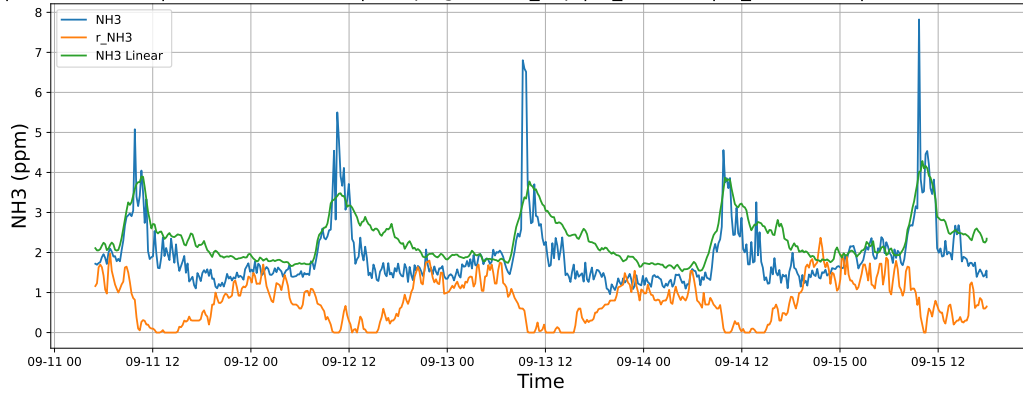


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

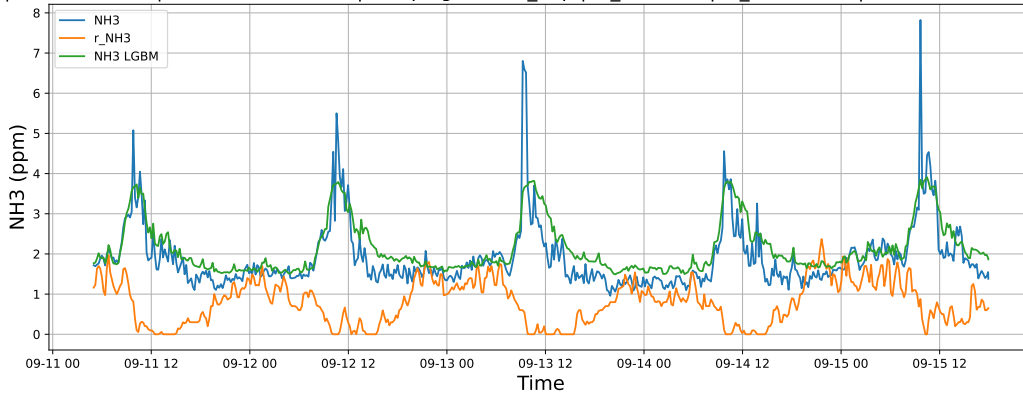


4.1.6.3 Time window : ALL

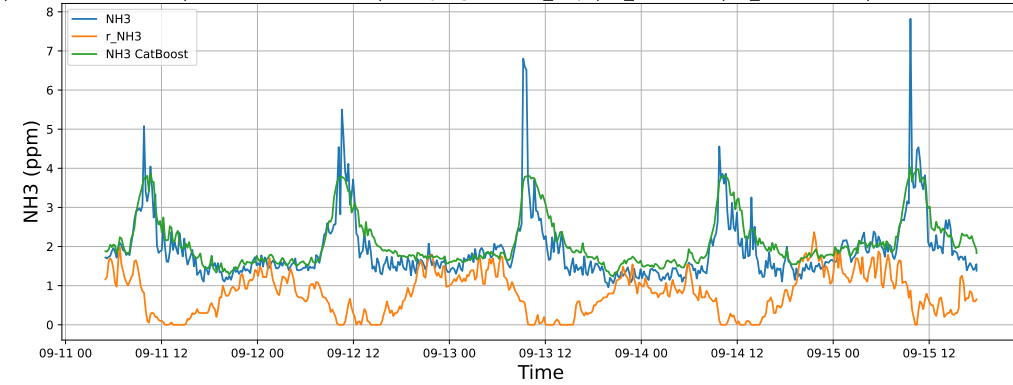
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



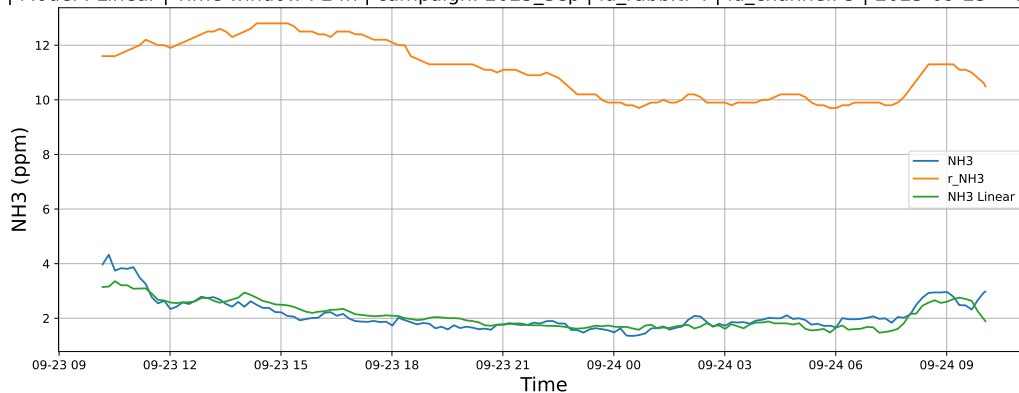
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



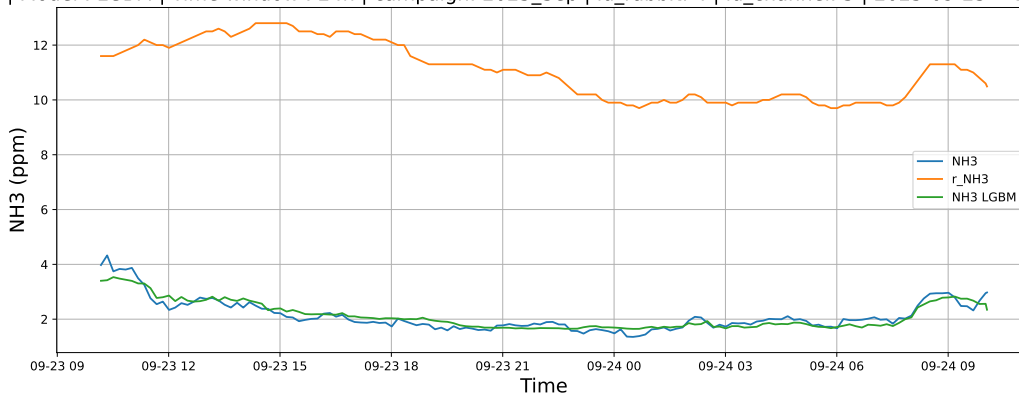
4.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

4.1.7.1 Time window : 24

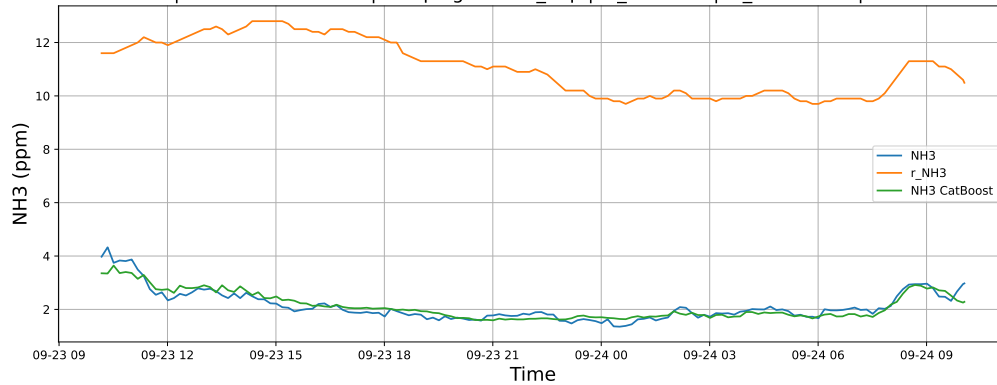
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

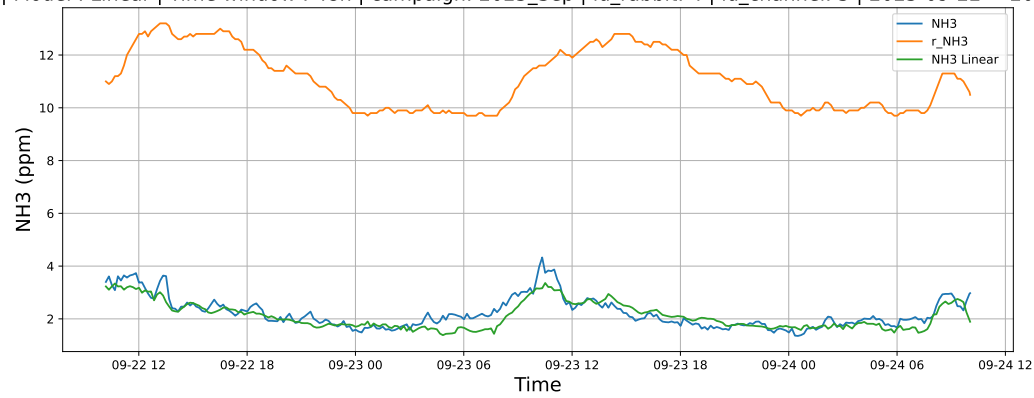


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

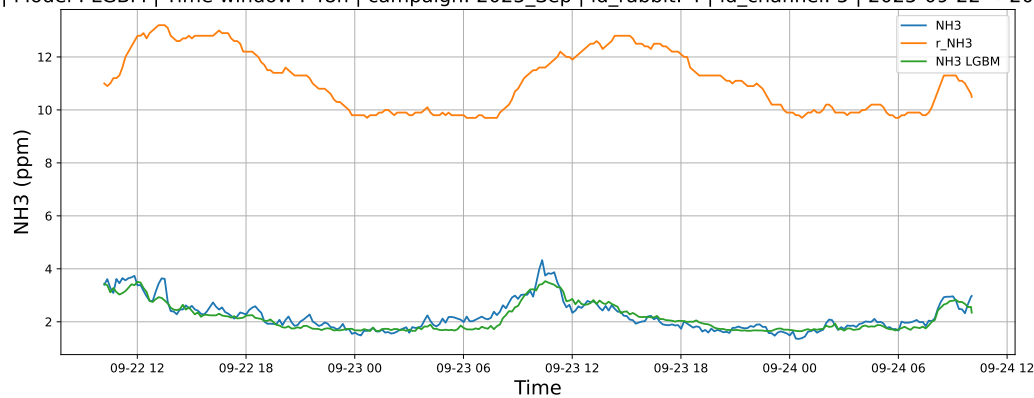


4.1.7.2 Time window : 48

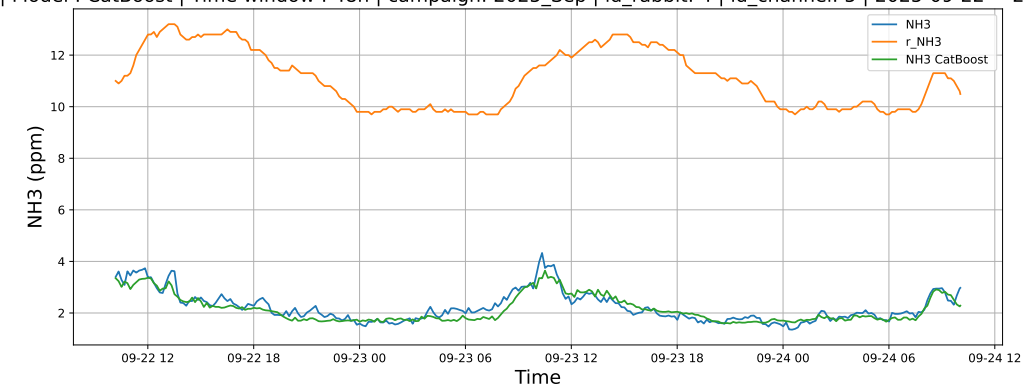
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

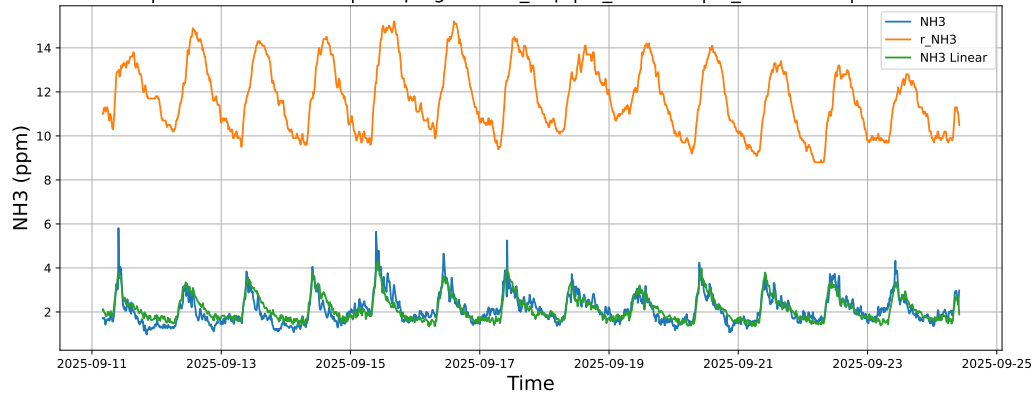


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

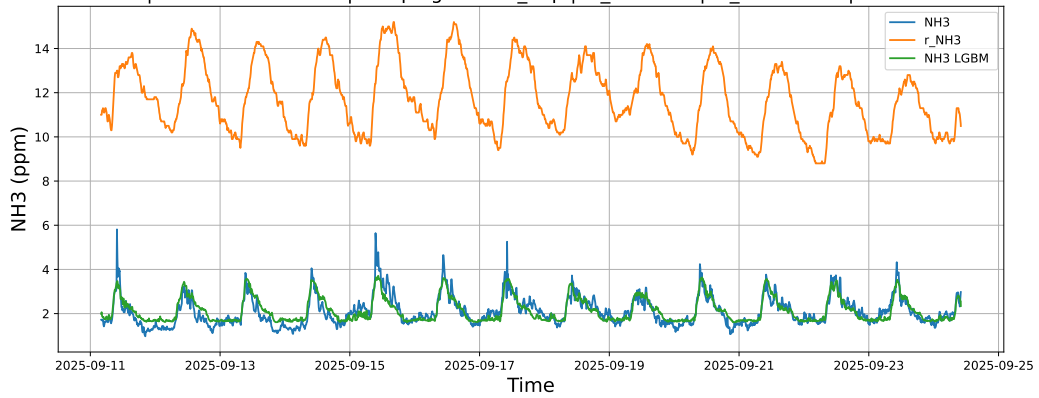


4.1.7.3 Time window : ALL

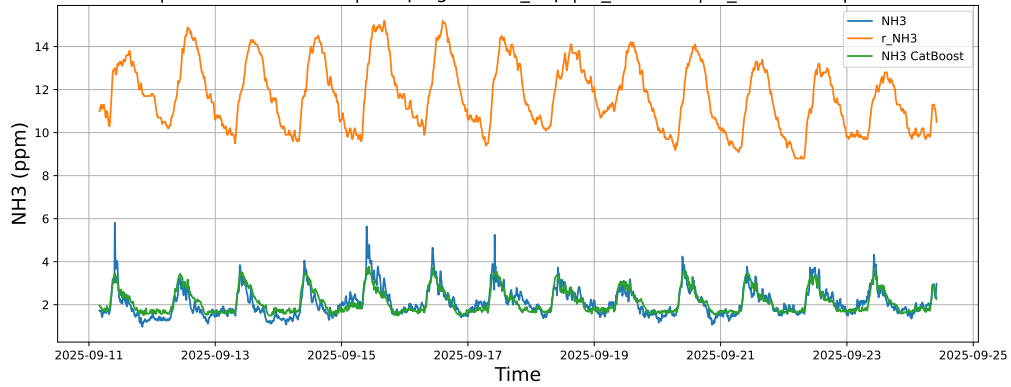
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



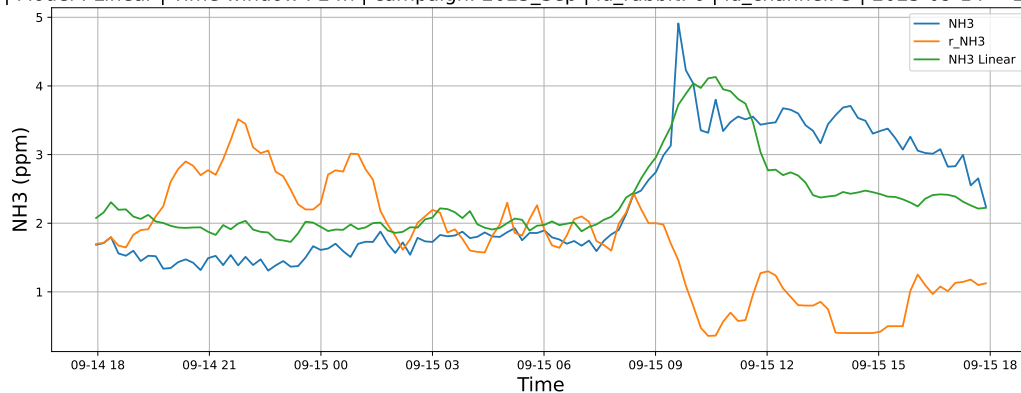
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



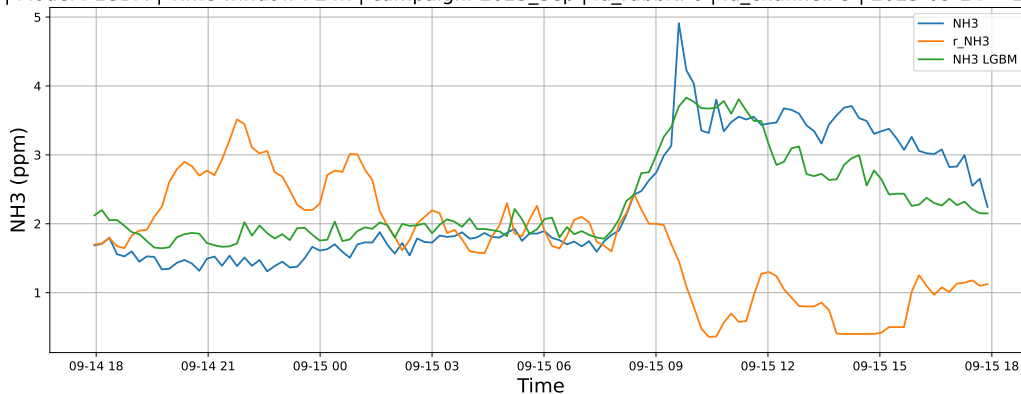
4.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

4.1.8.1 Time window : 24

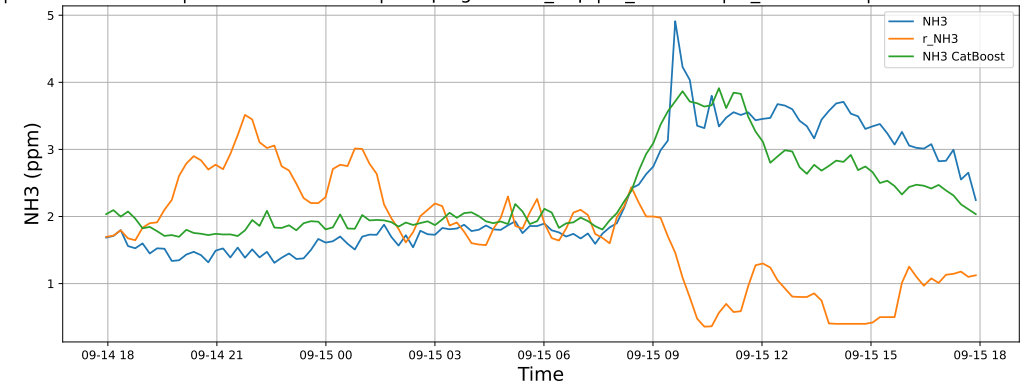
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

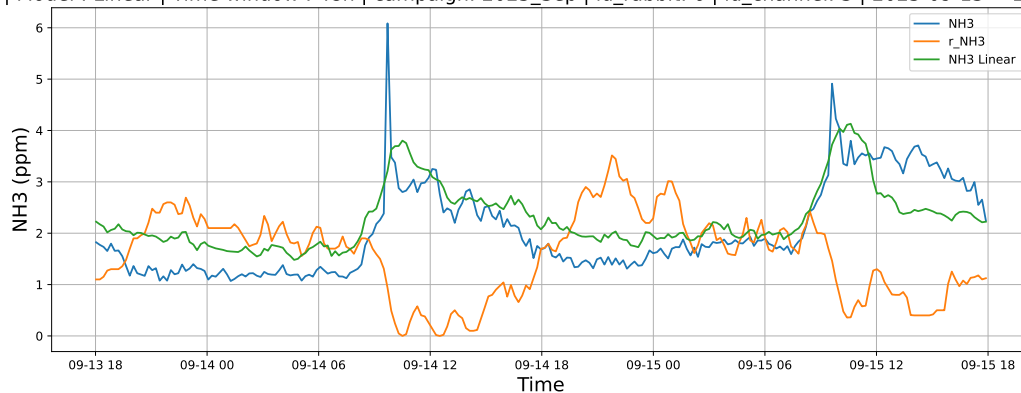


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

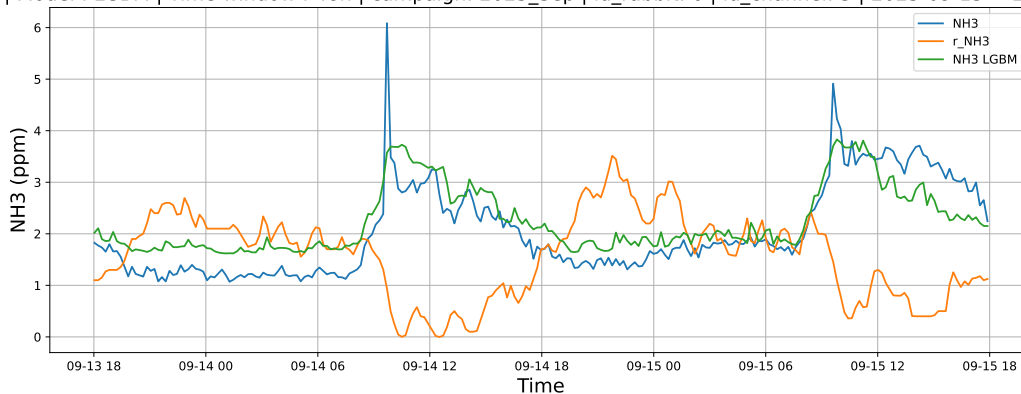


4.1.8.2 Time window : 48

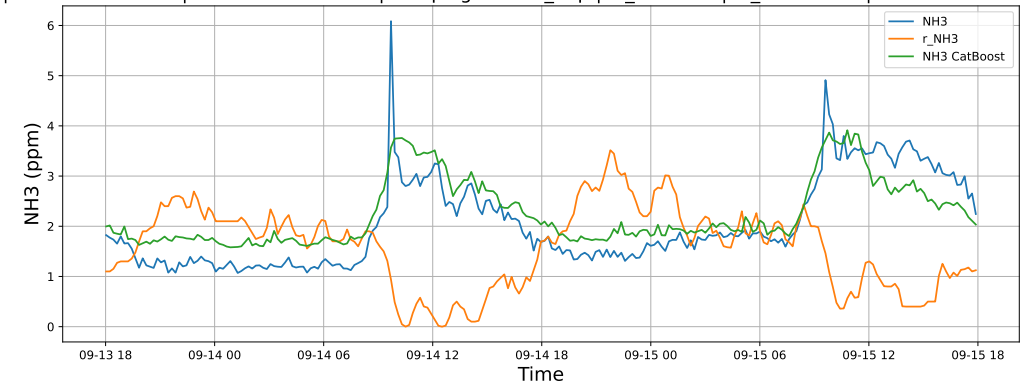
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

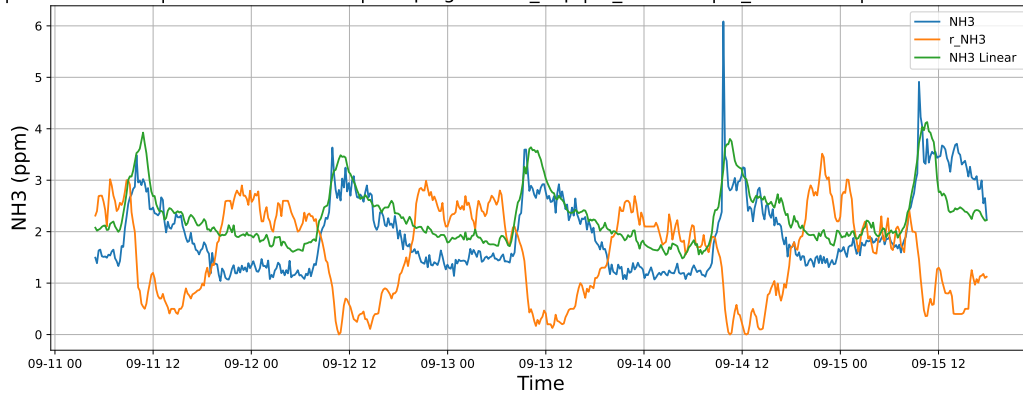


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

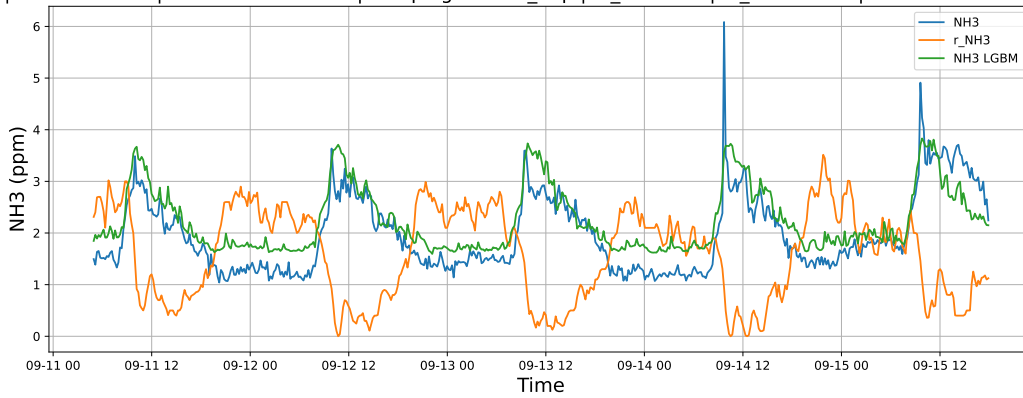


4.1.8.3 Time window : ALL

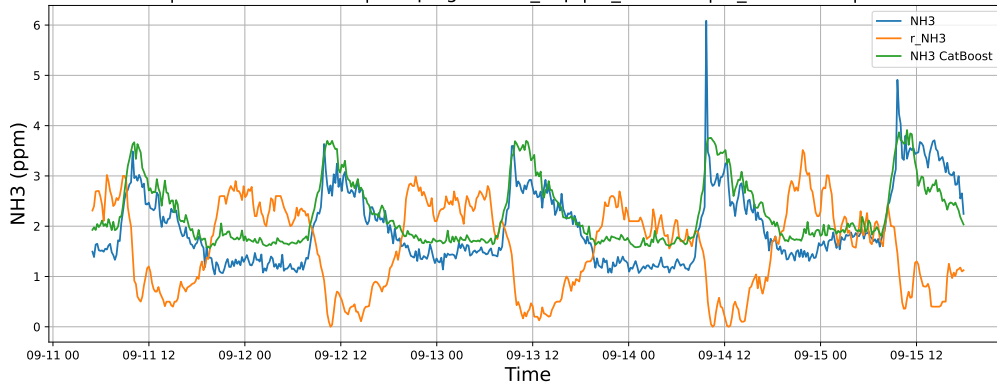
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

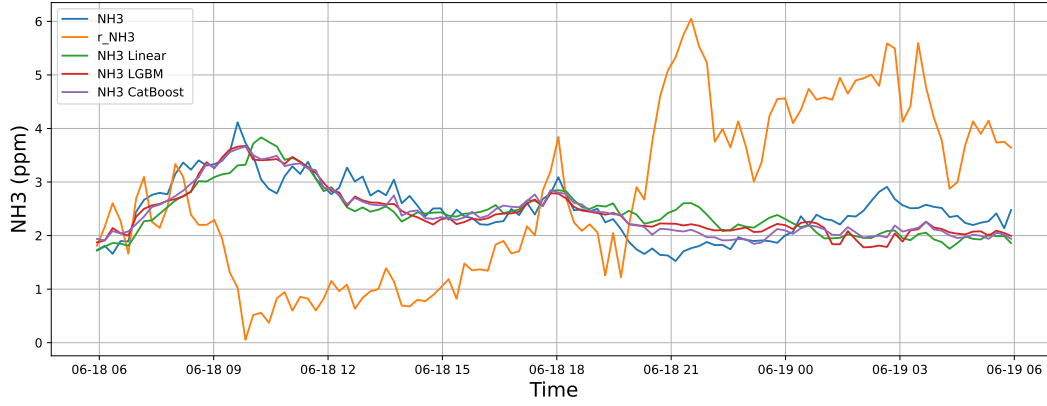


4.2 Compare plots

4.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

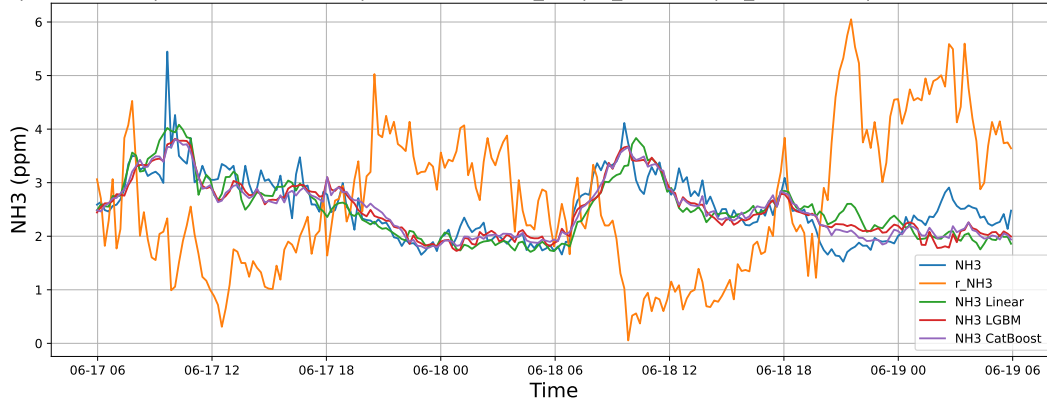
4.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



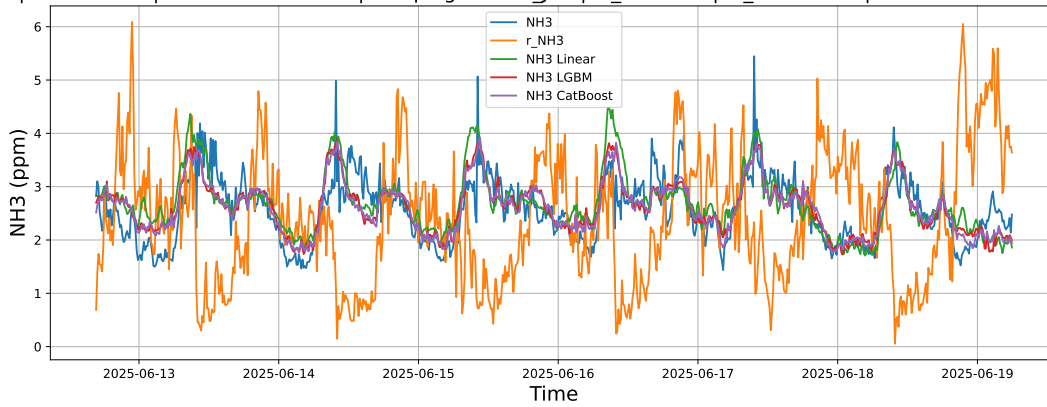
4.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



4.2.1.3 Time window : ALL

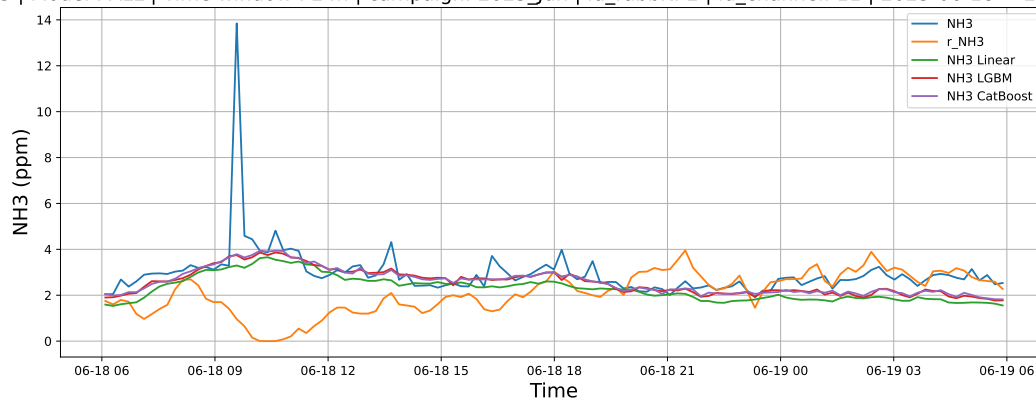
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



4.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

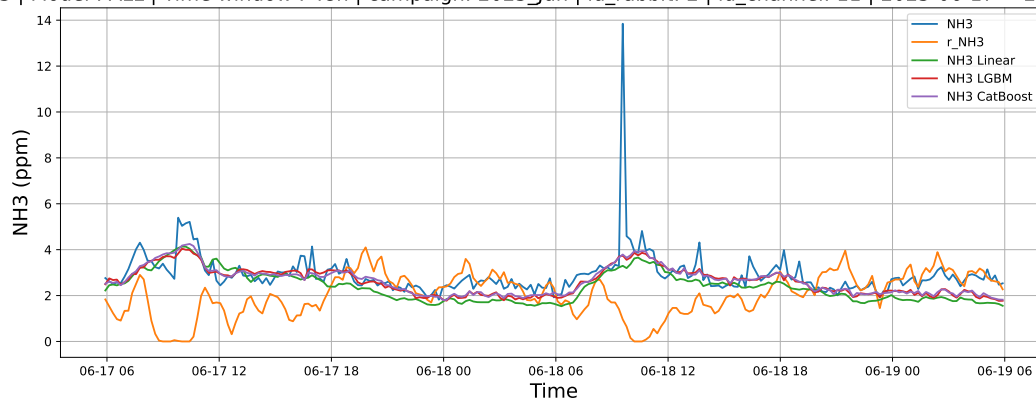
4.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



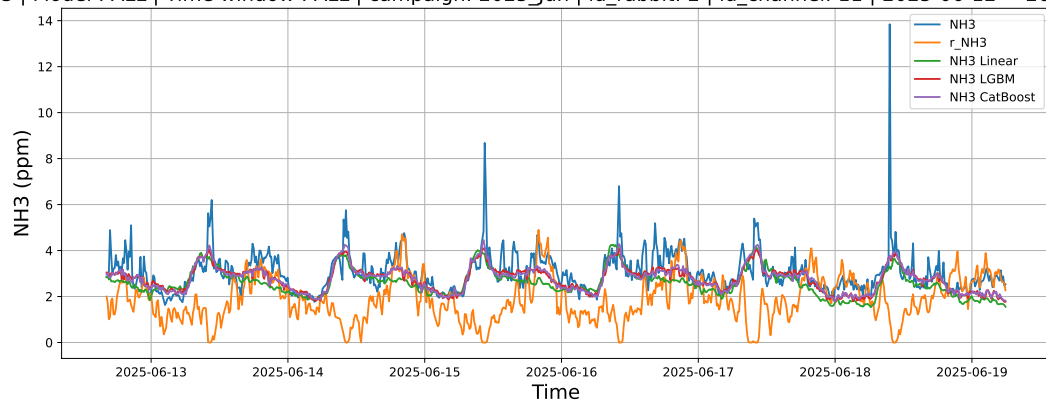
4.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



4.2.2.3 Time window : ALL

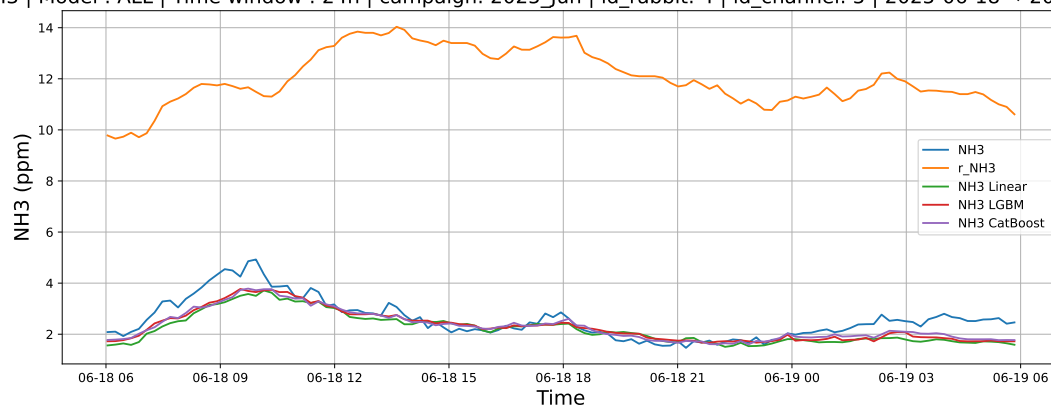
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



4.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

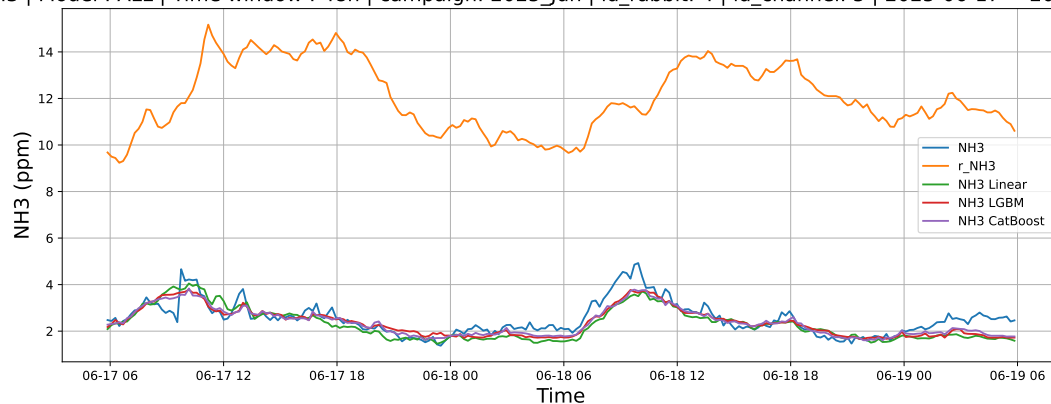
4.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



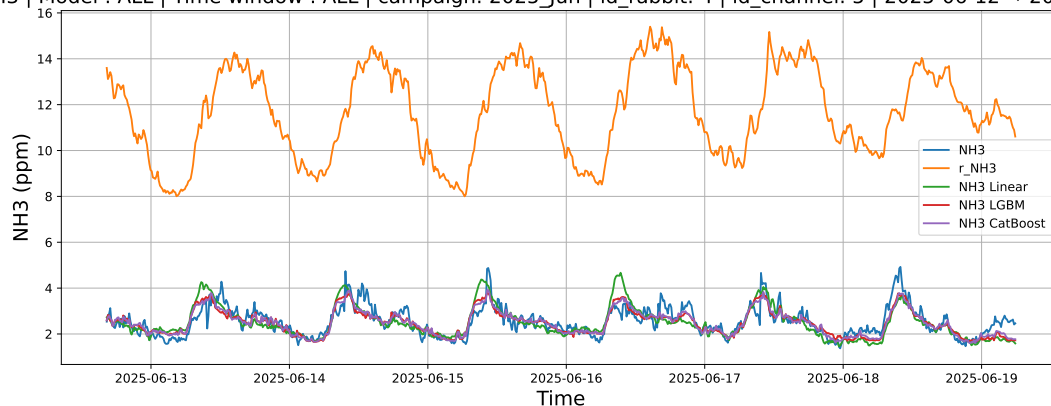
4.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



4.2.3.3 Time window : ALL

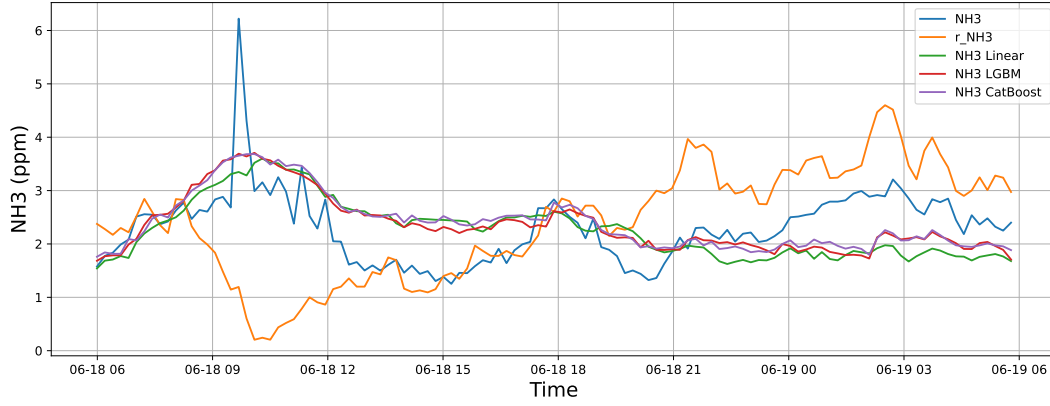
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



4.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

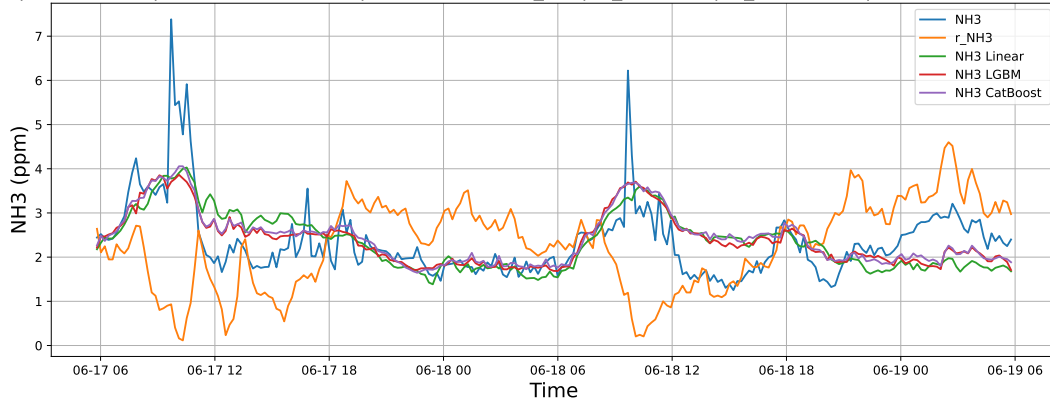
4.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



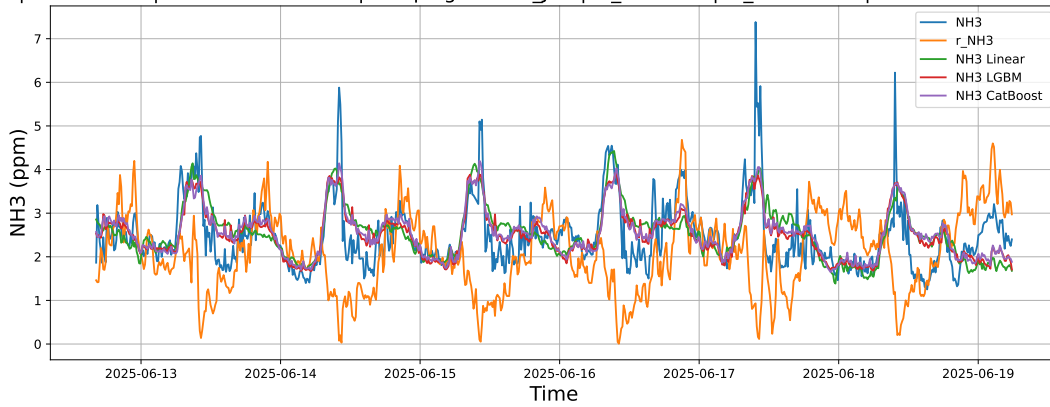
4.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



4.2.4.3 Time window : ALL

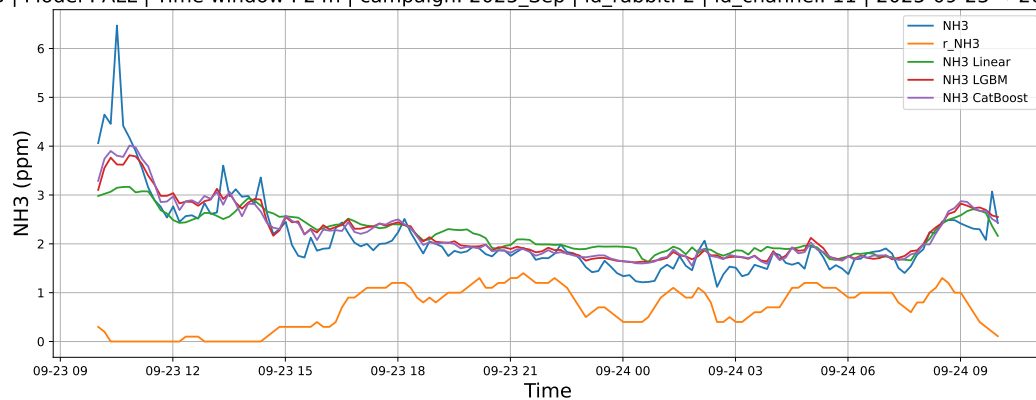
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



4.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

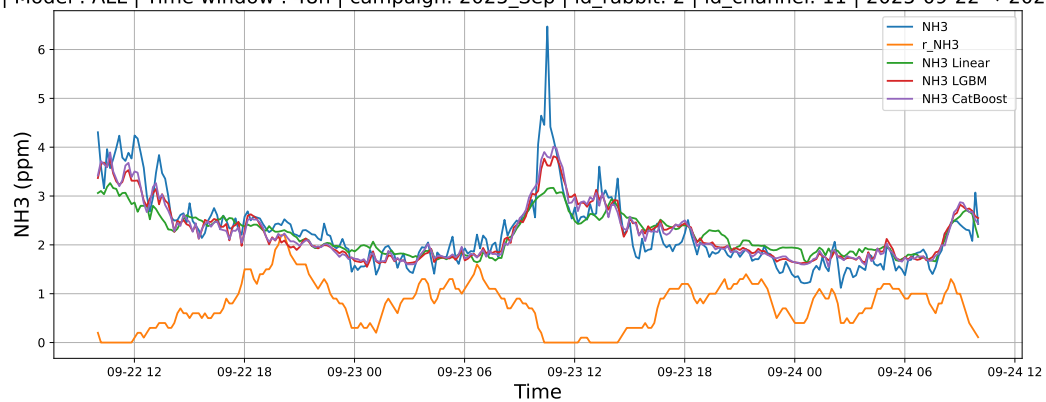
4.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



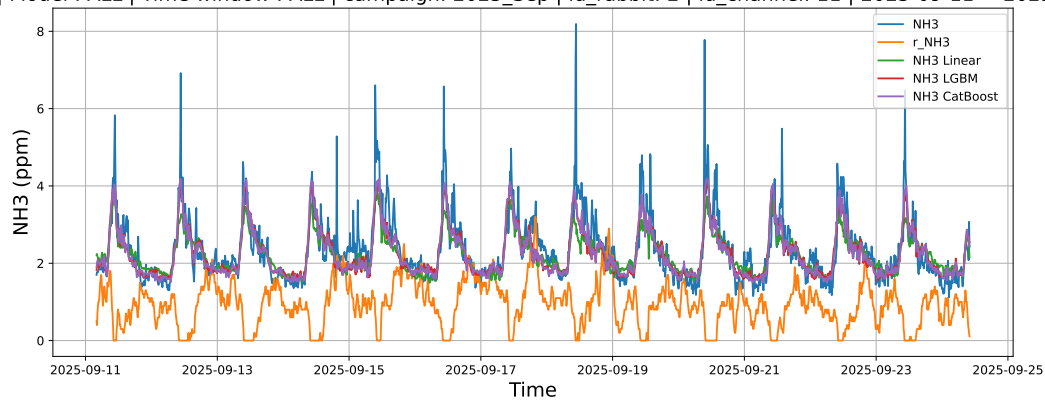
4.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



4.2.5.3 Time window : ALL

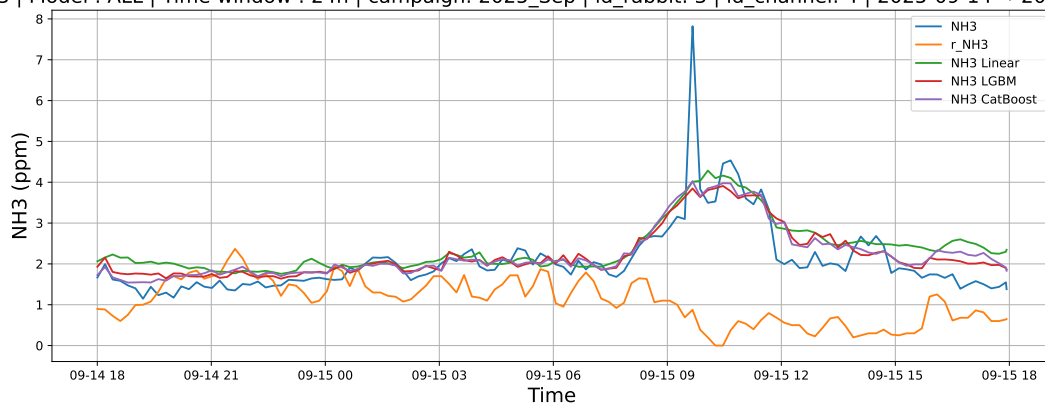
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



4.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

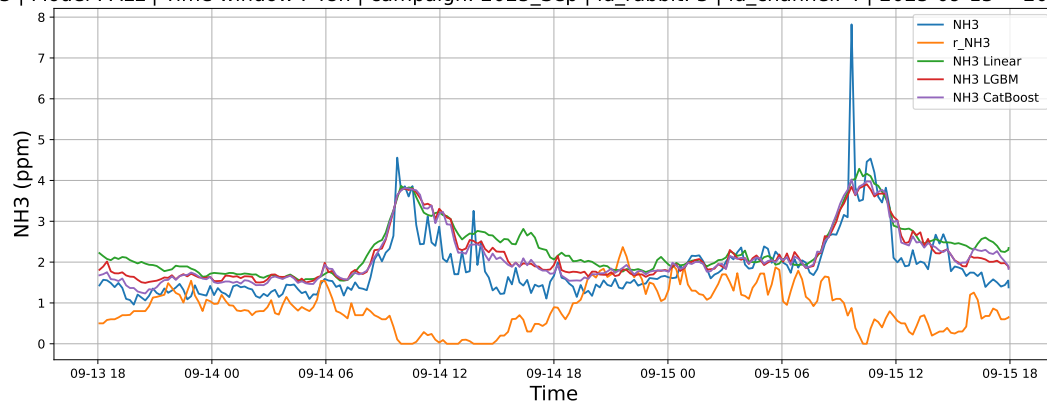
4.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



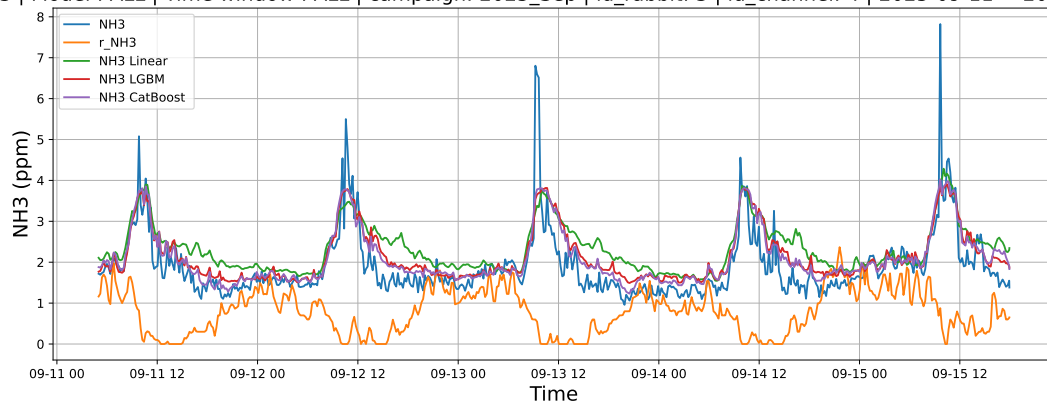
4.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



4.2.6.3 Time window : ALL

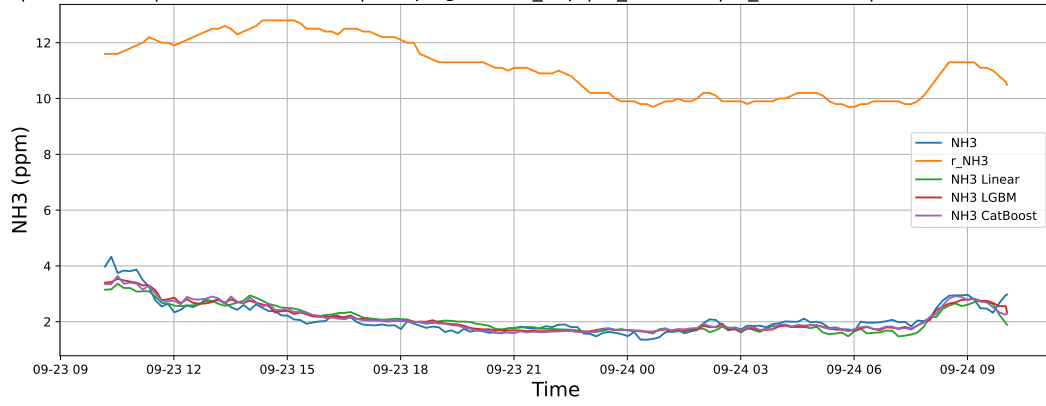
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



4.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

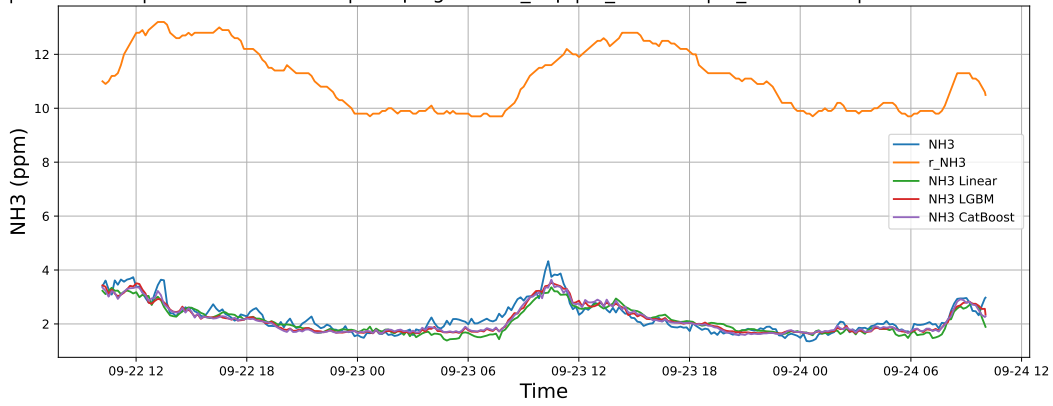
4.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



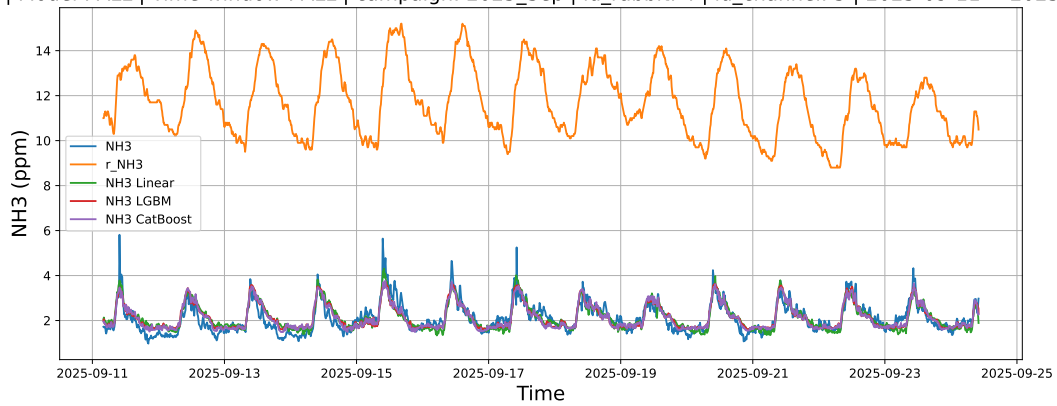
4.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



4.2.7.3 Time window : ALL

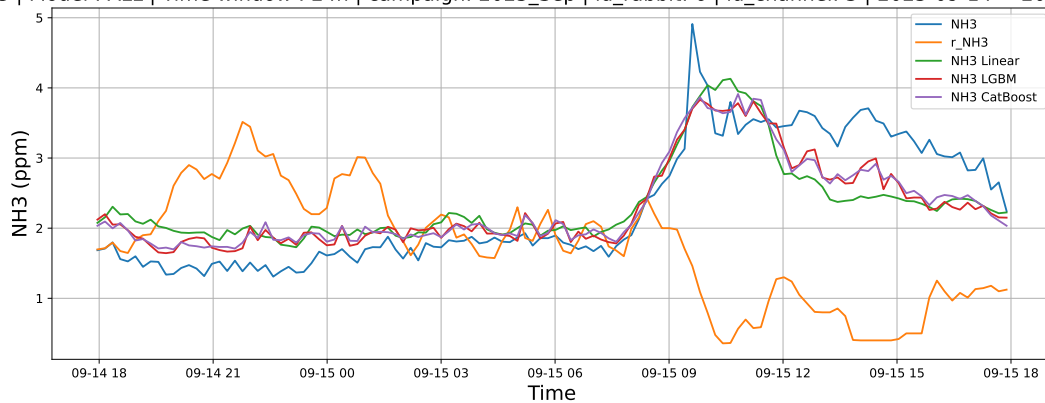
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



4.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

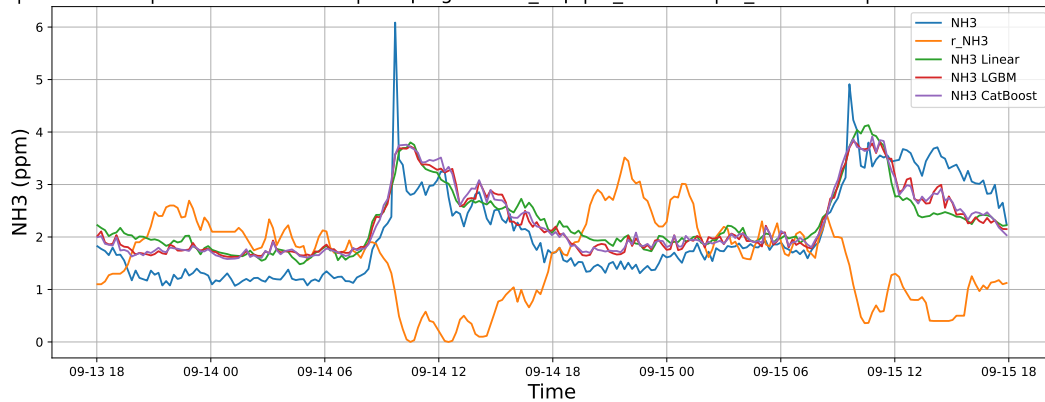
4.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



4.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



4.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

